

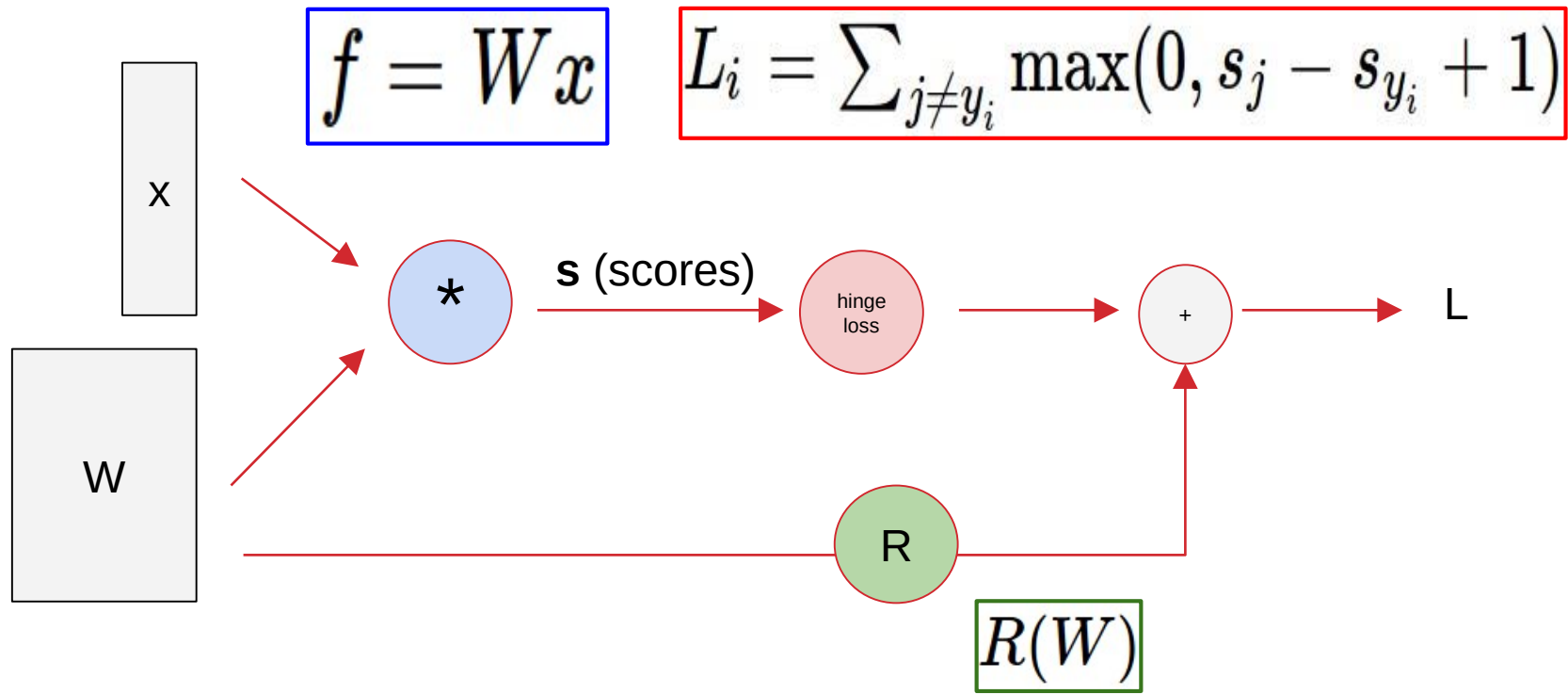
Redes Neurais e Deep Learning

PROPAGACÃO RETRÓGRADA

Zenilton K. G. Patrocínio Jr

zenilton@pucminas.br

Grafo de Computação da Função de Perda

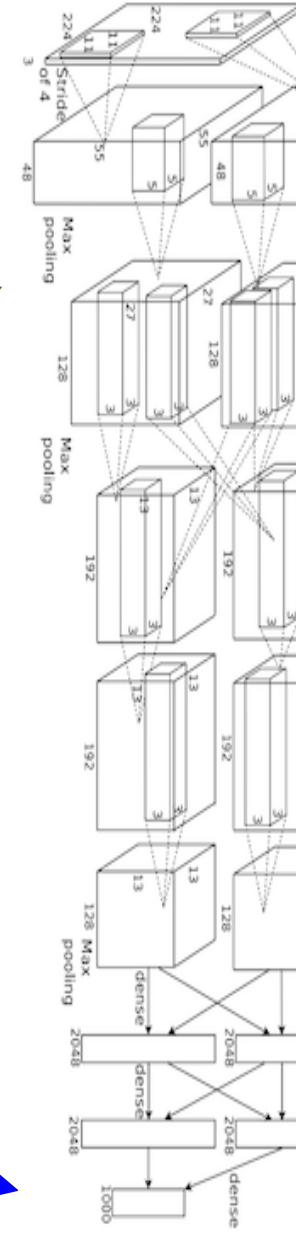


Rede Convolutucional (AlexNet)

Imagem de entrada

Pesos

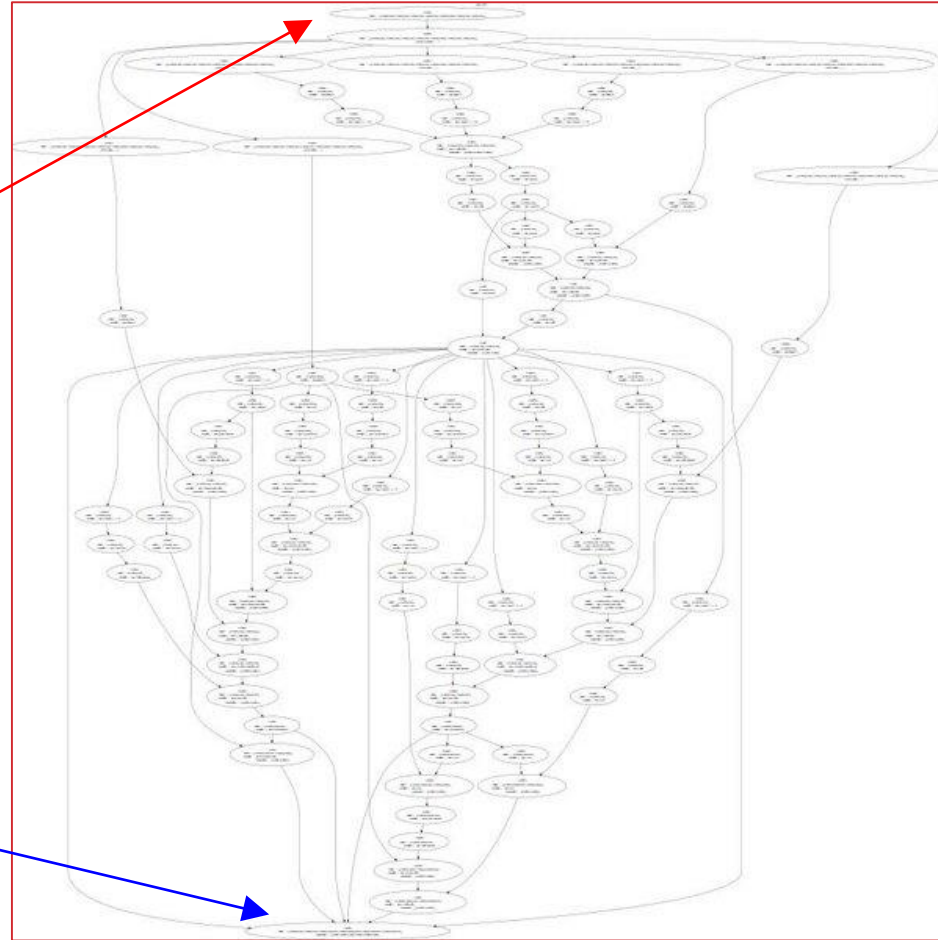
Perda



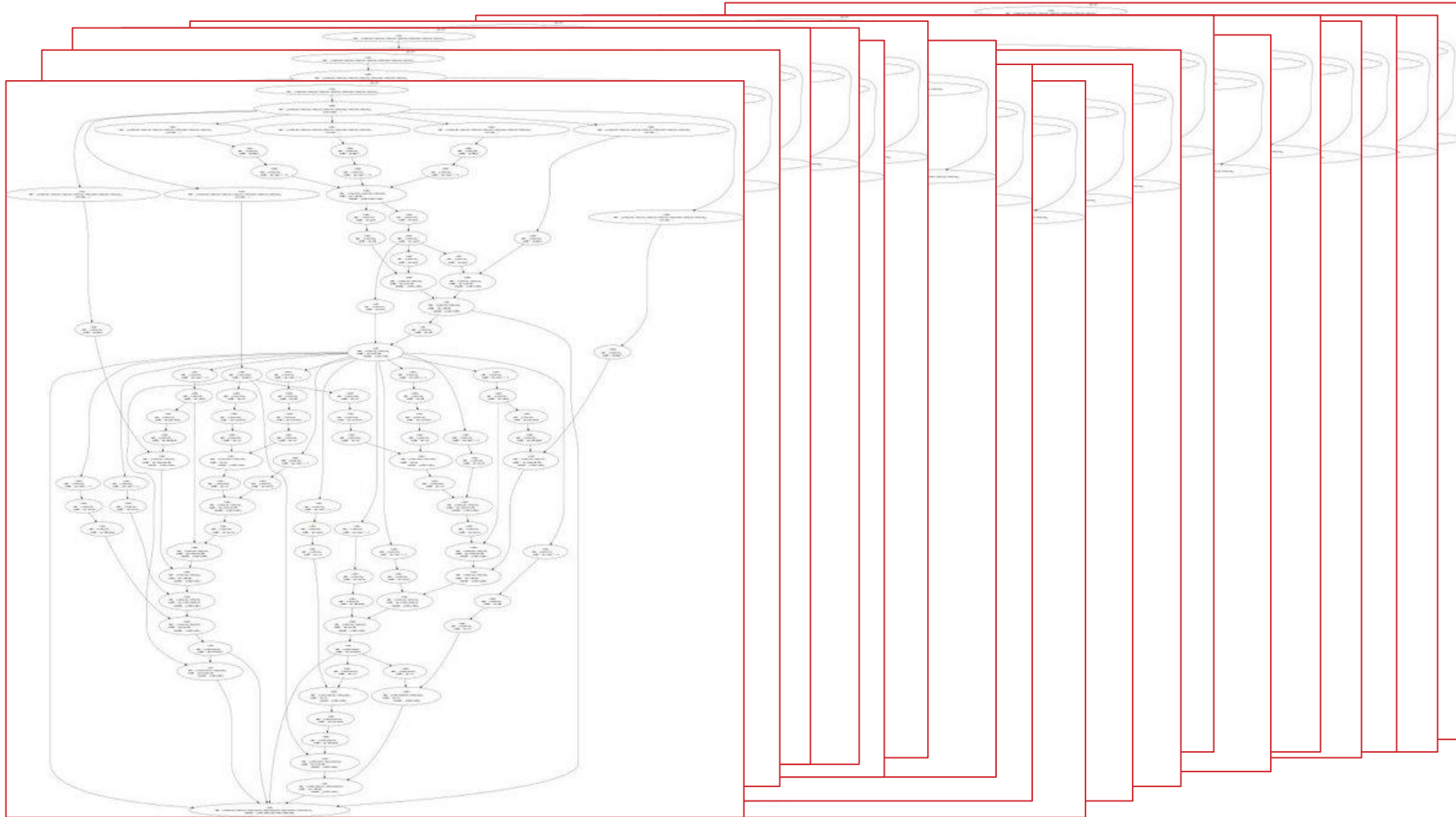
Máquina Neural de Turing

Fita de entrada

Perda

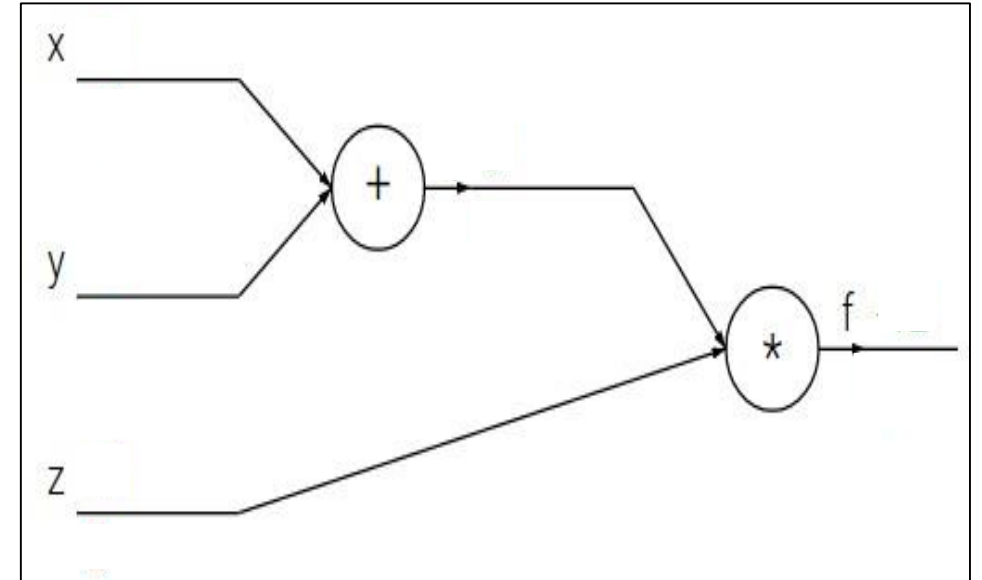


Neural Turing Machine



Diferenciação de um Grafo de Computação

Seja $f(x, y, z) = (x + y) \times z$

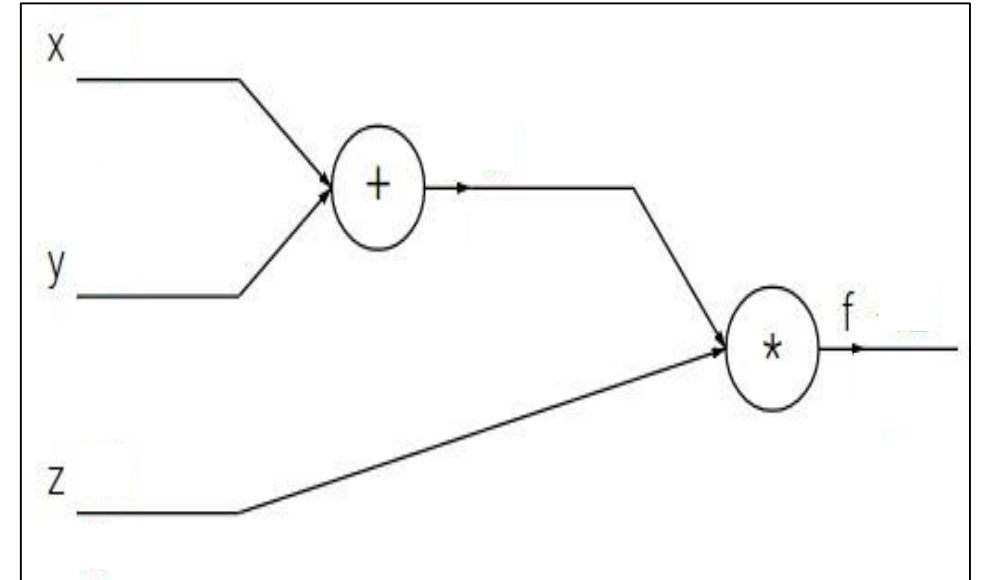


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Então, pode-se dizer que

$$f(x, y, z) = g(h(x, y), z)$$



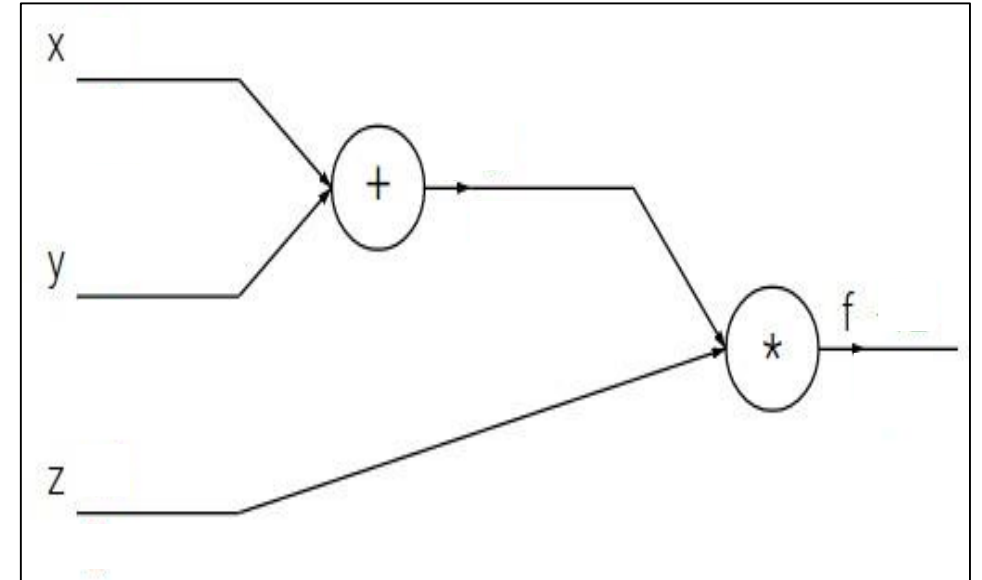
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em $h(x, y) = x + y$



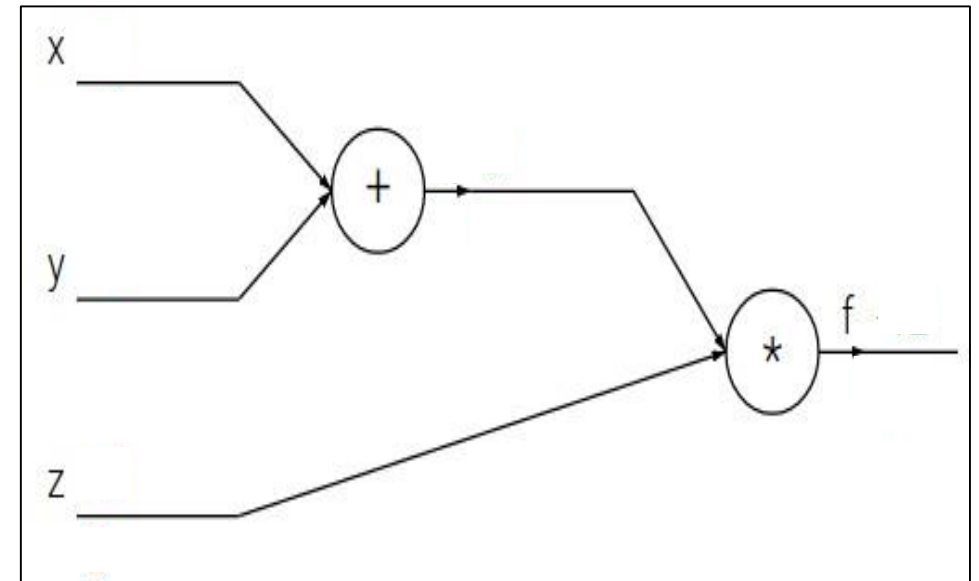
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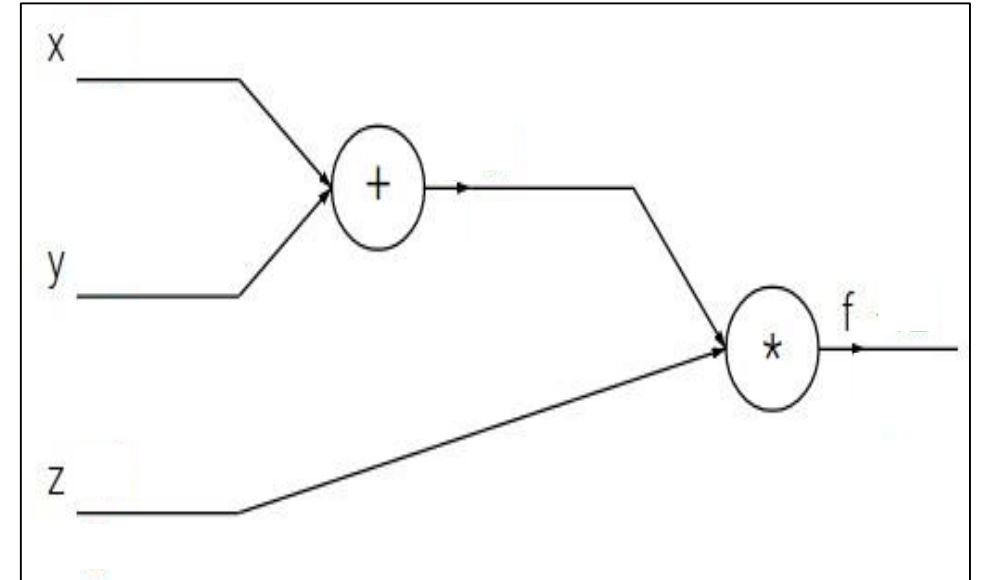
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$$\frac{df}{dx} = \frac{dg}{dh} \times \frac{dh}{dx}$$



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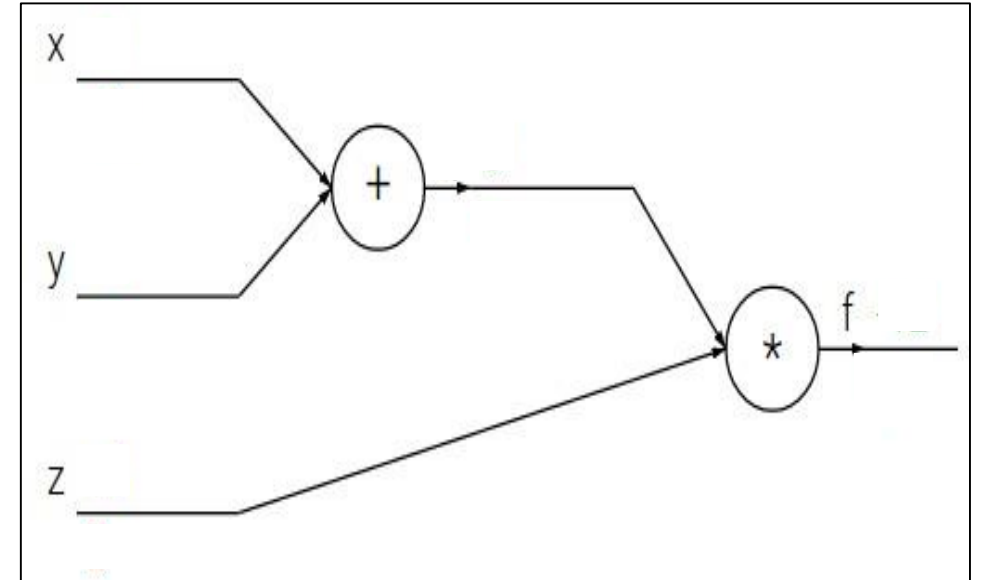
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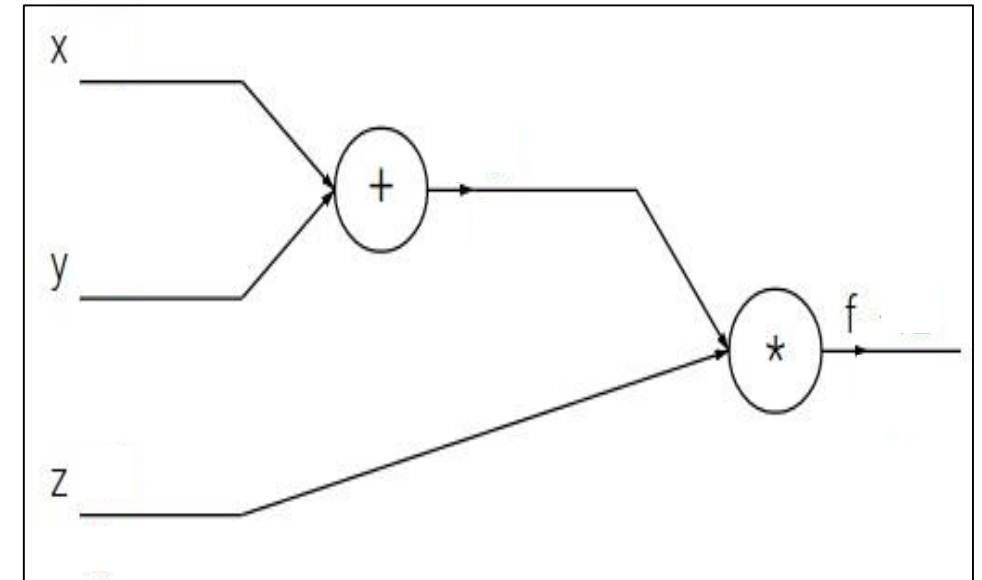
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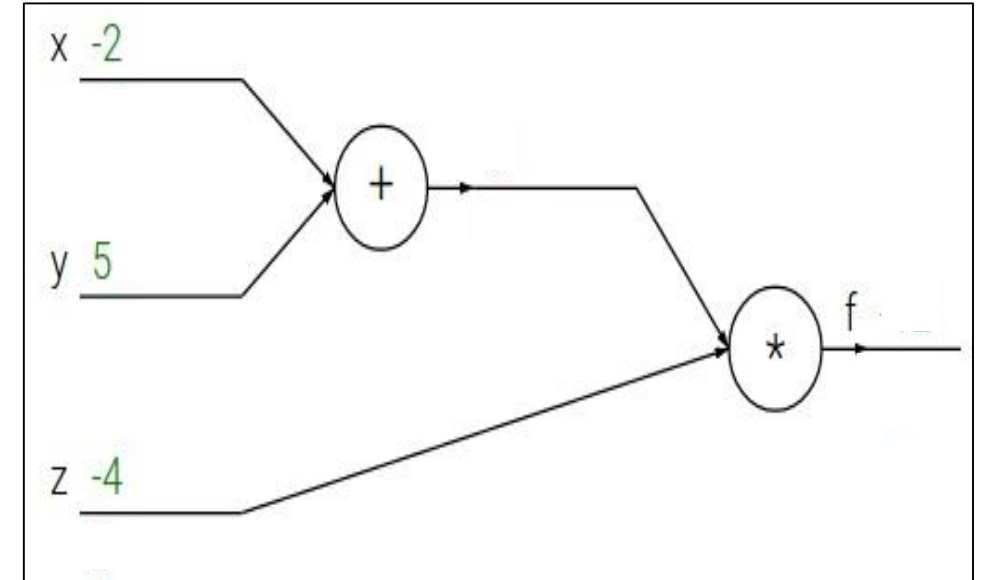
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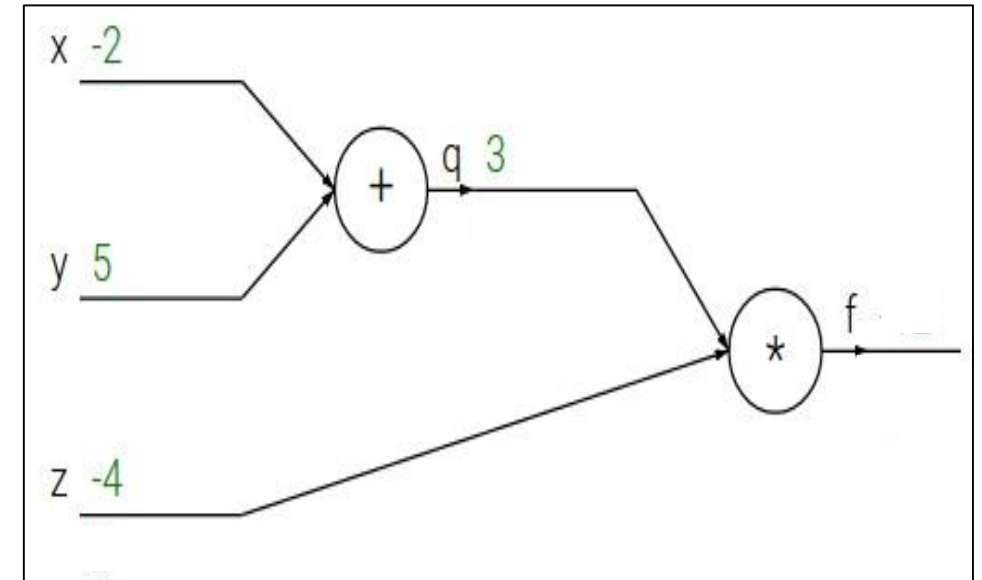


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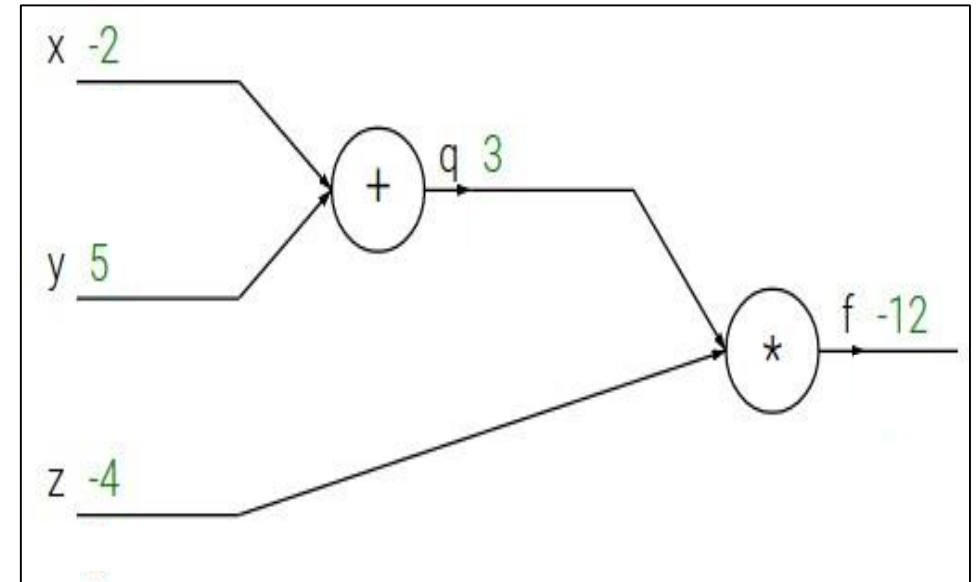
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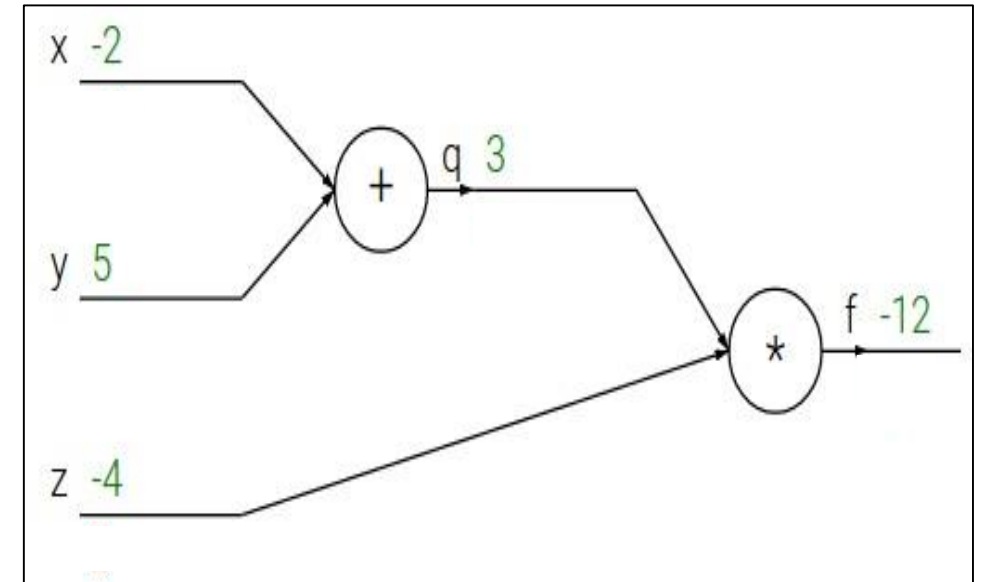
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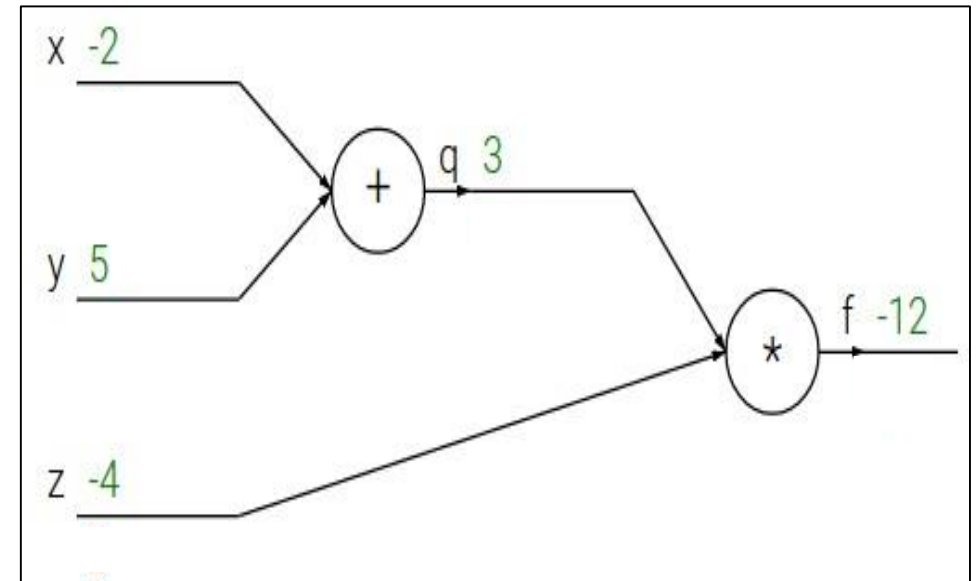
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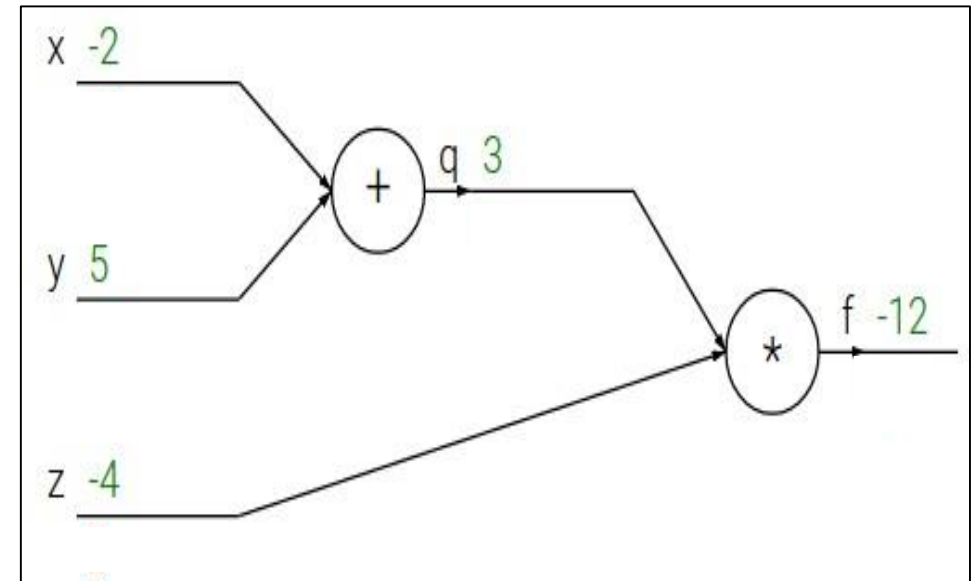
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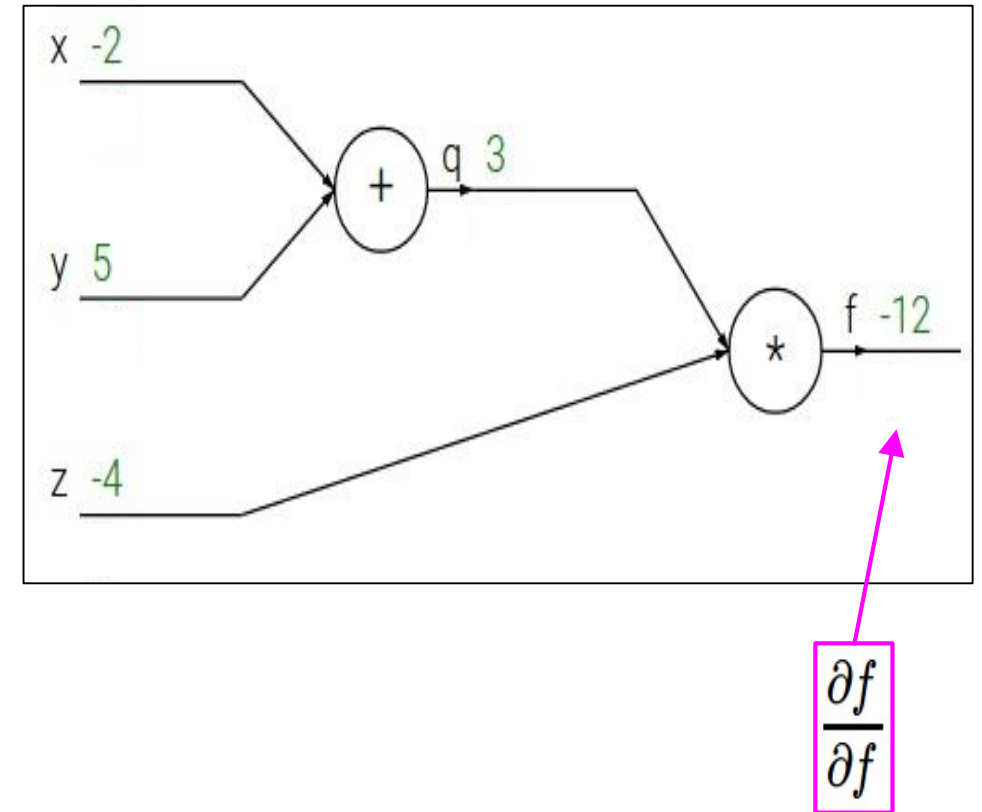
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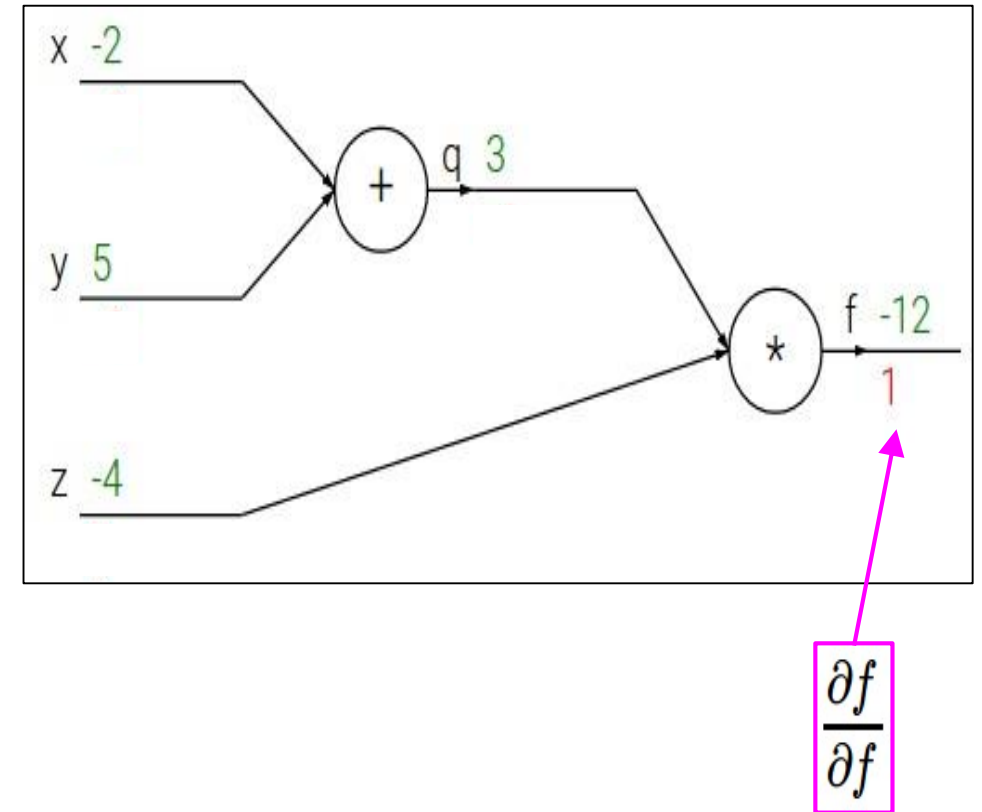
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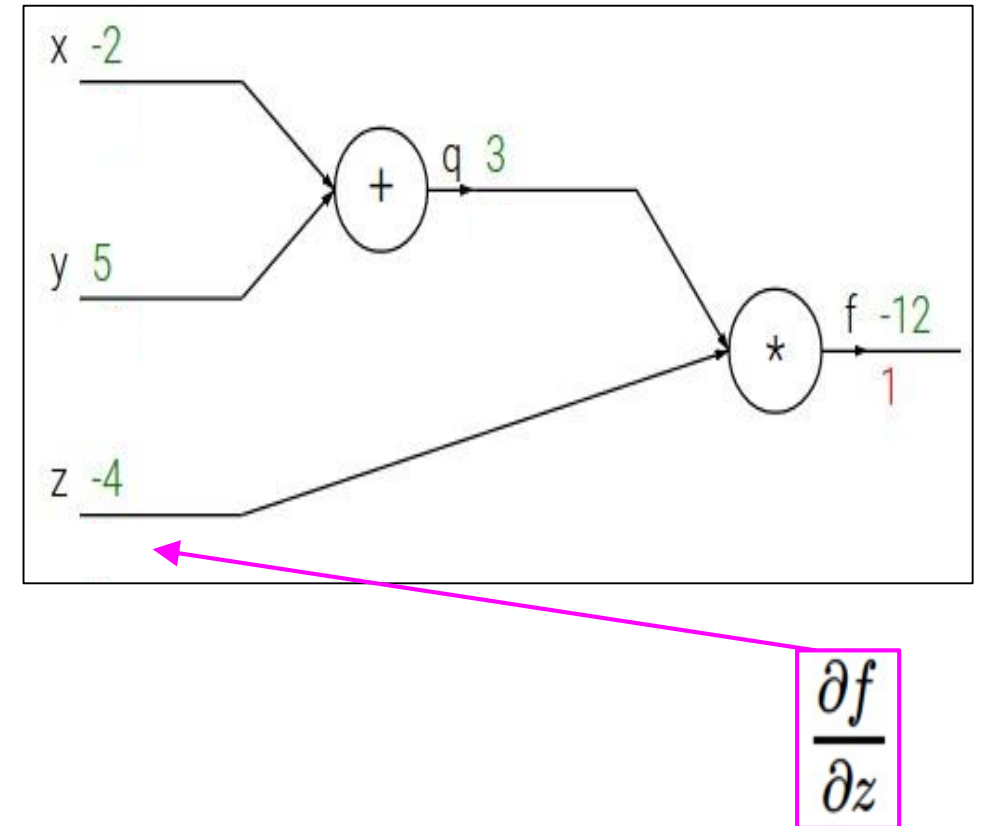
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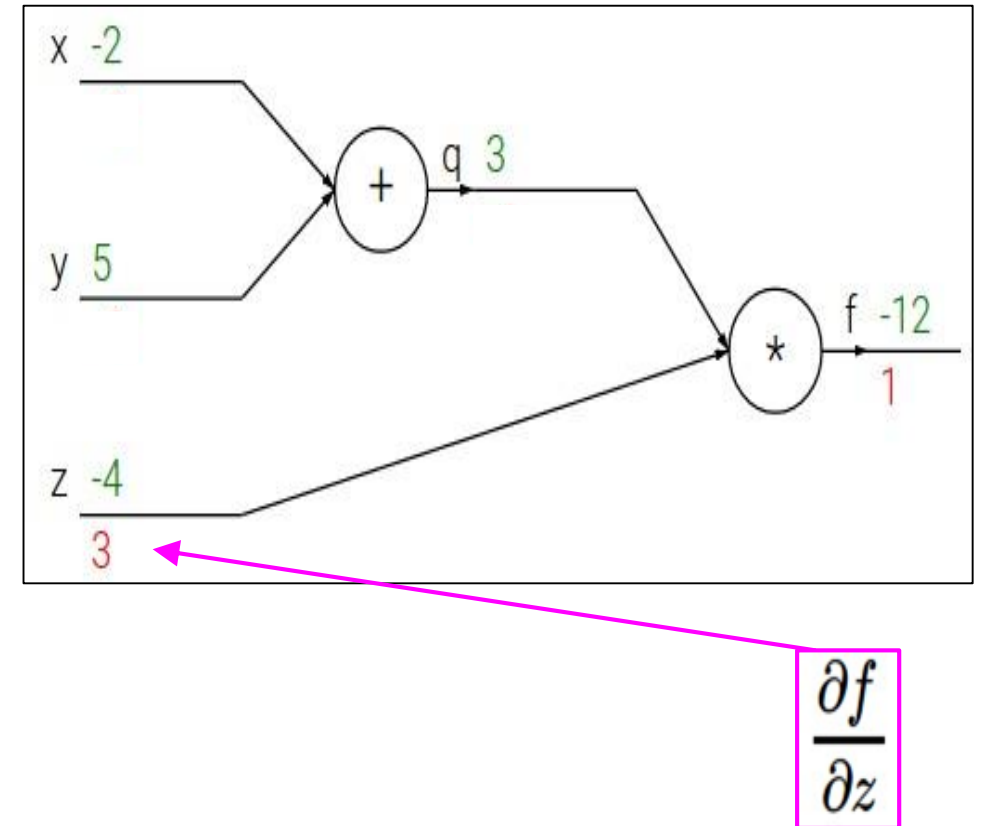
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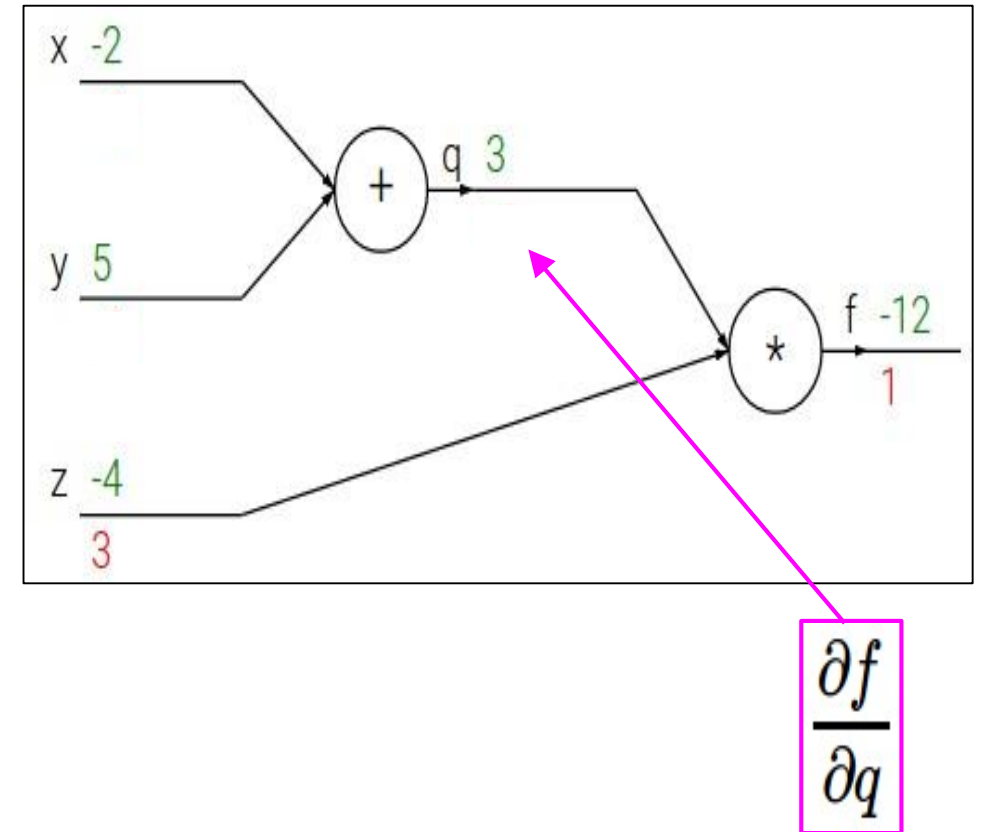
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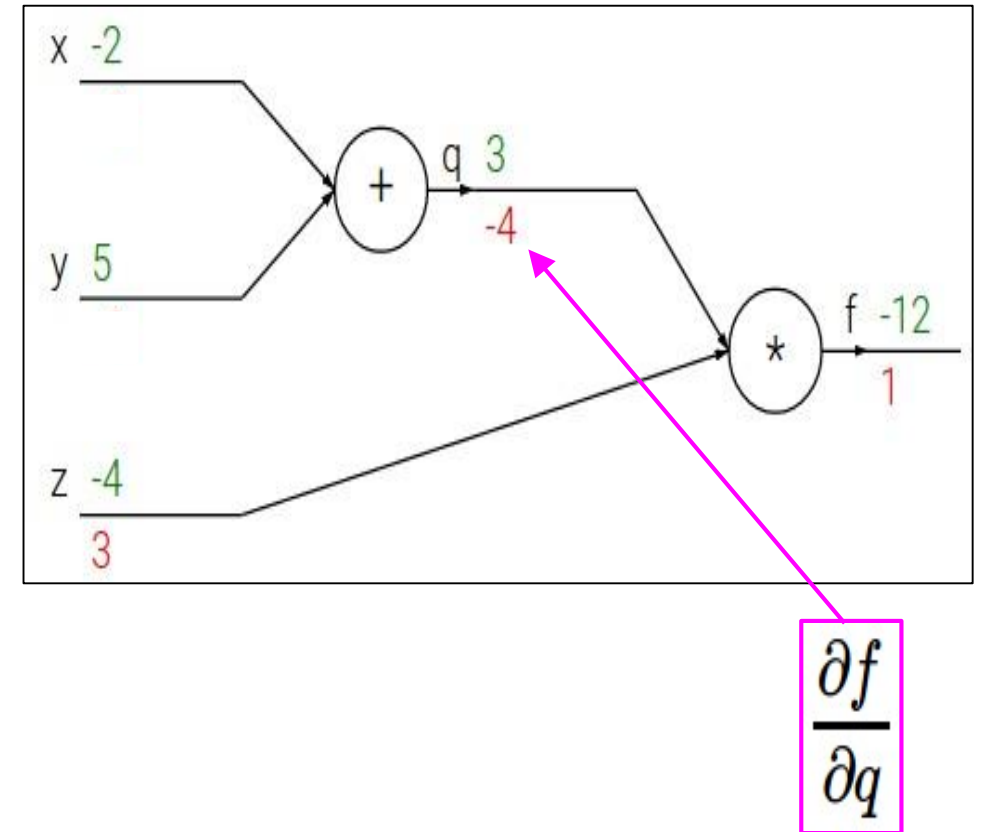
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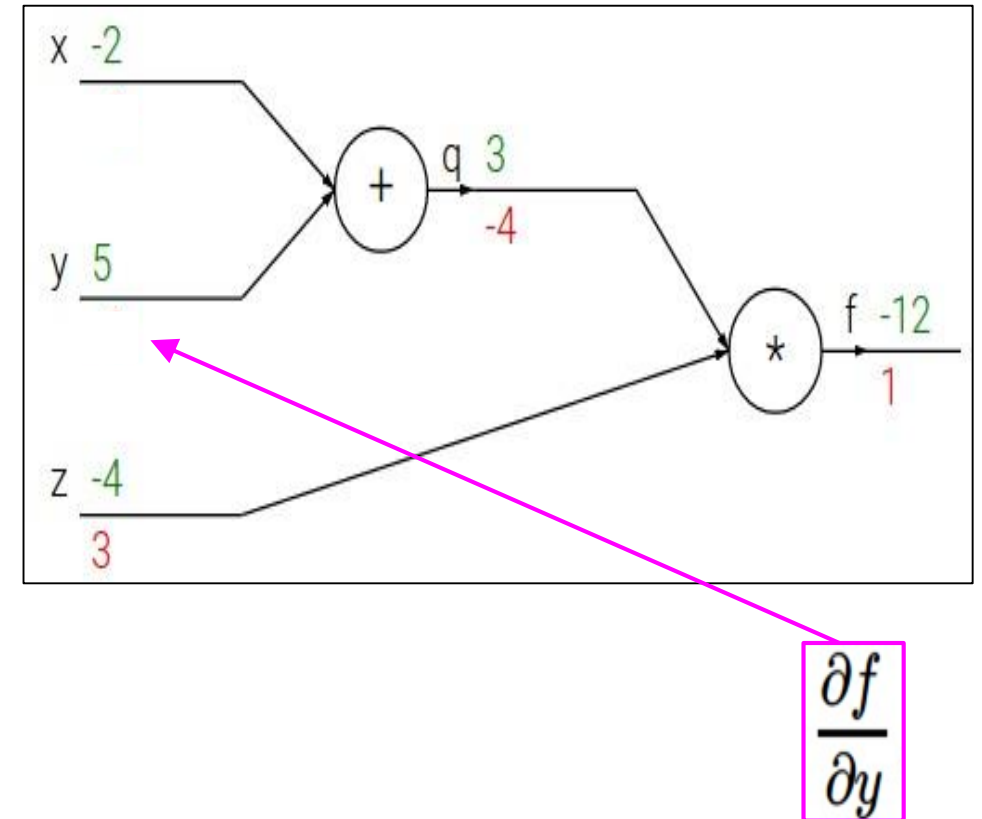
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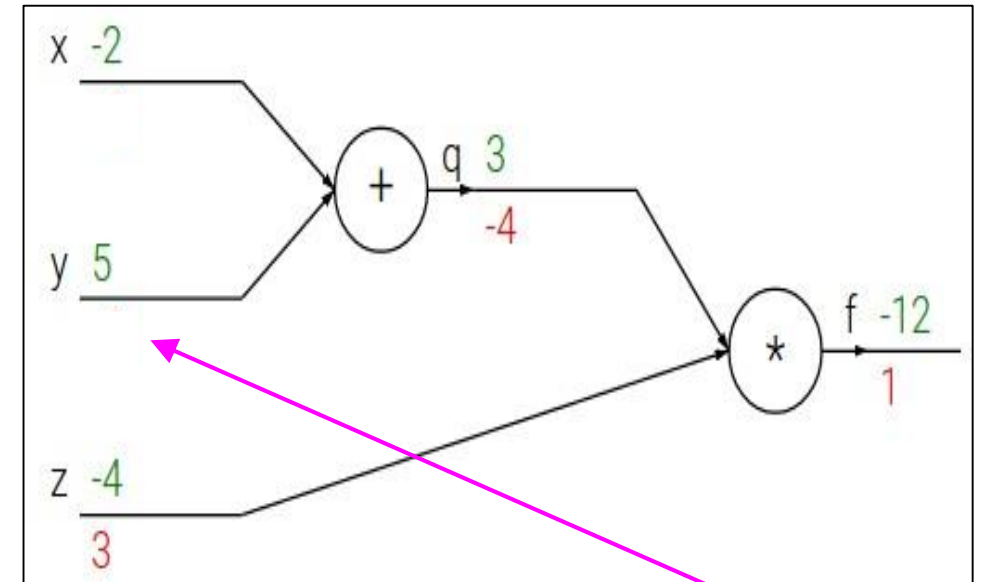
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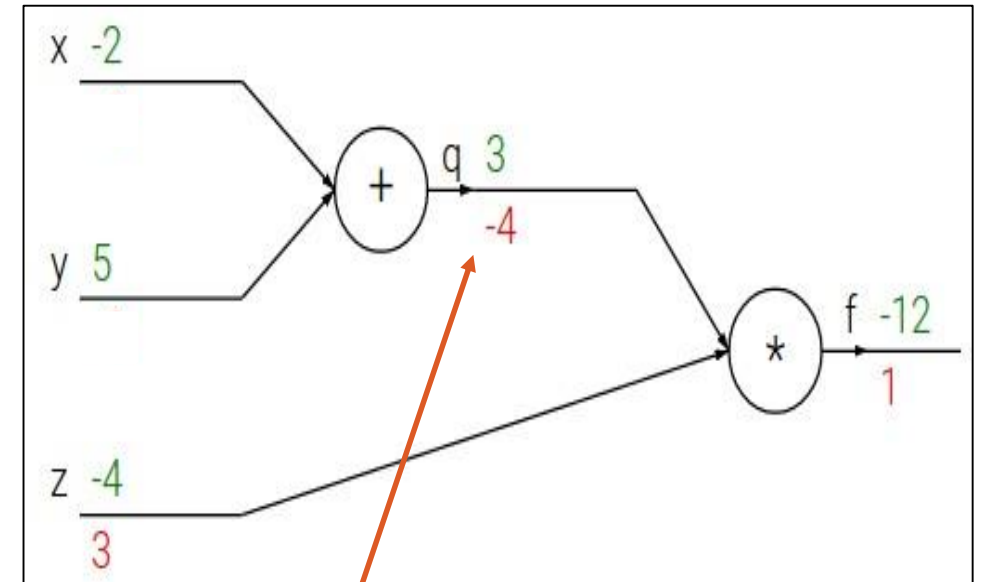
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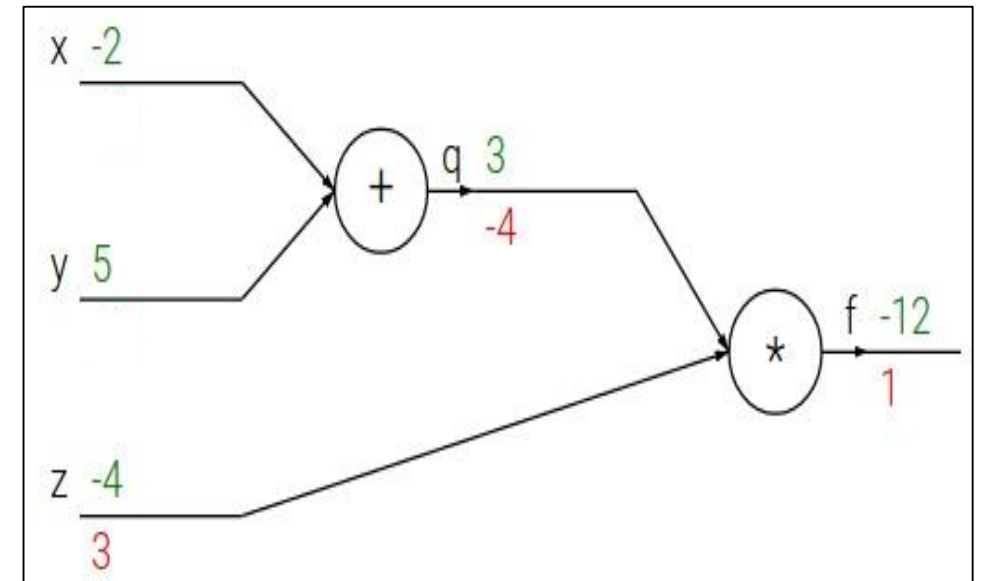
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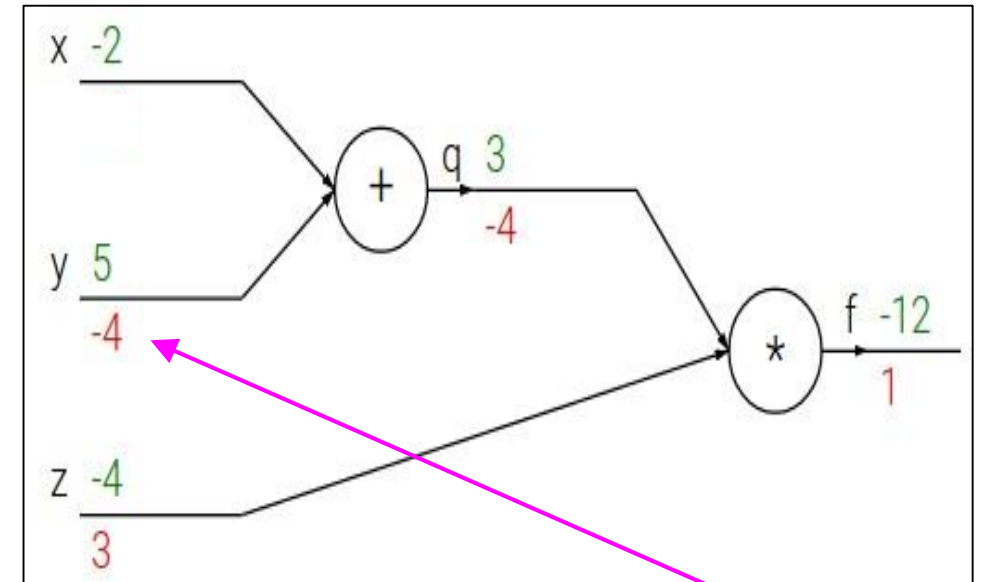
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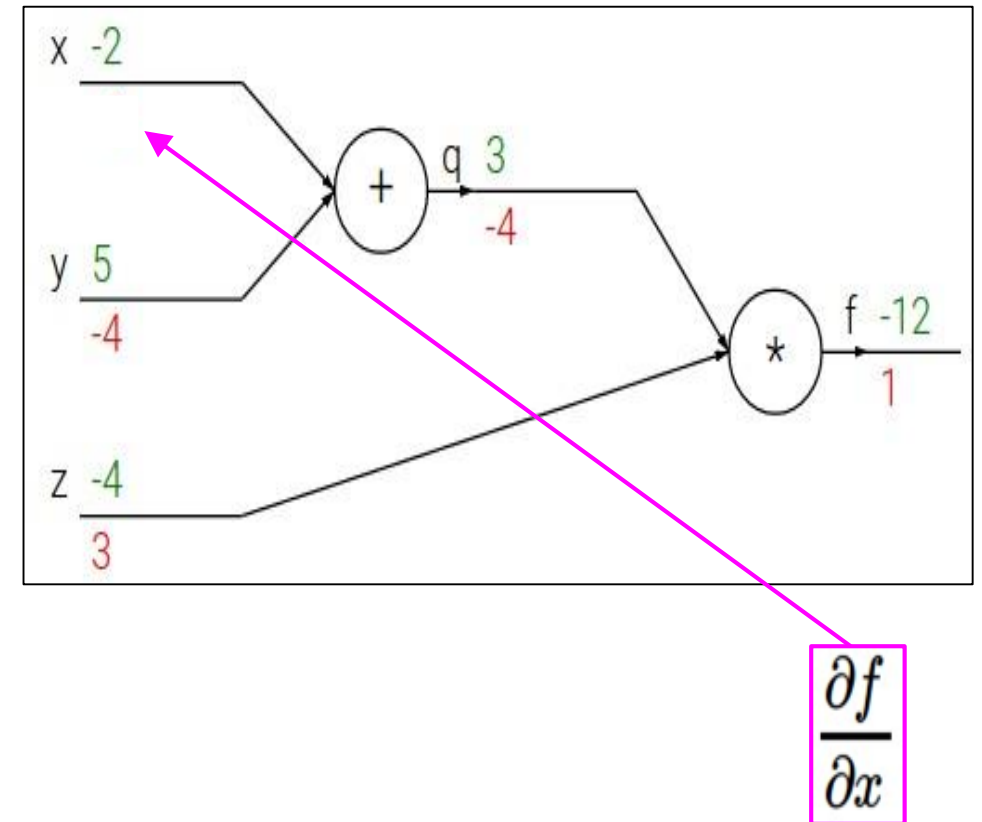
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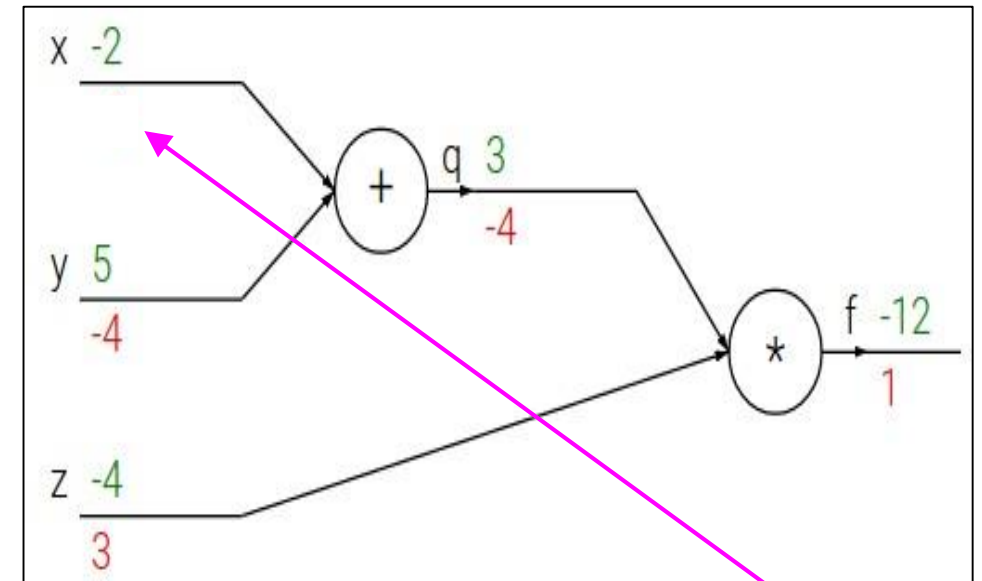
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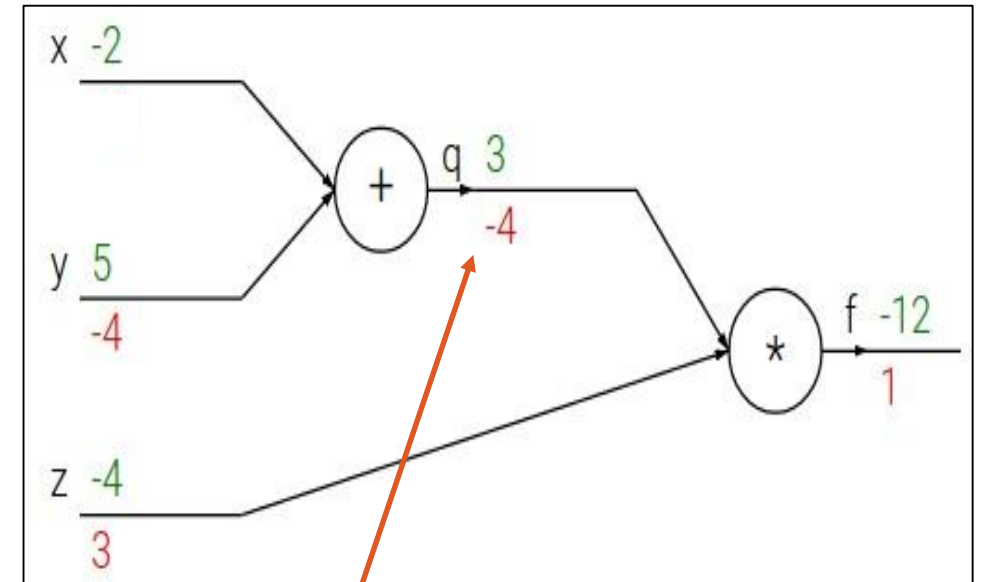
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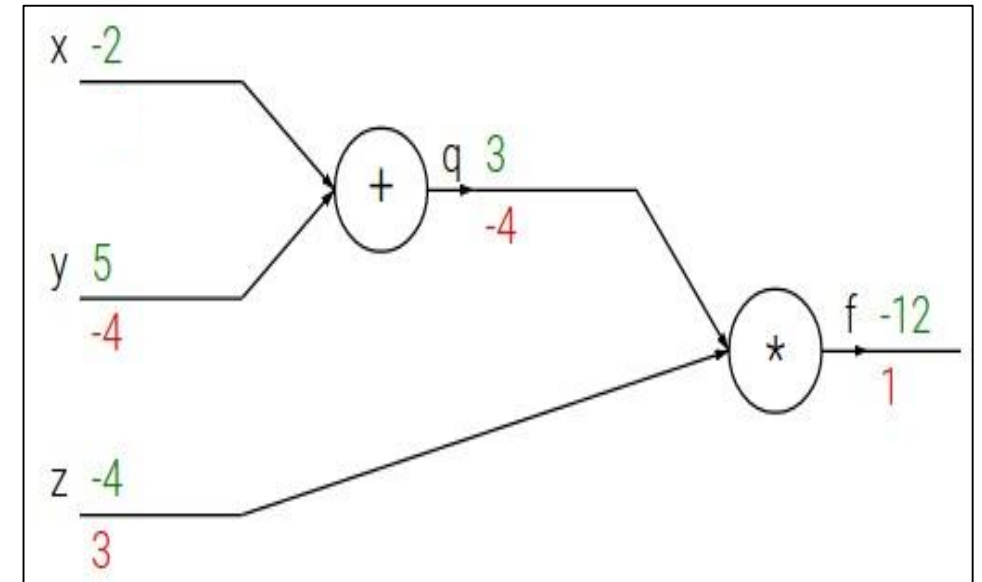
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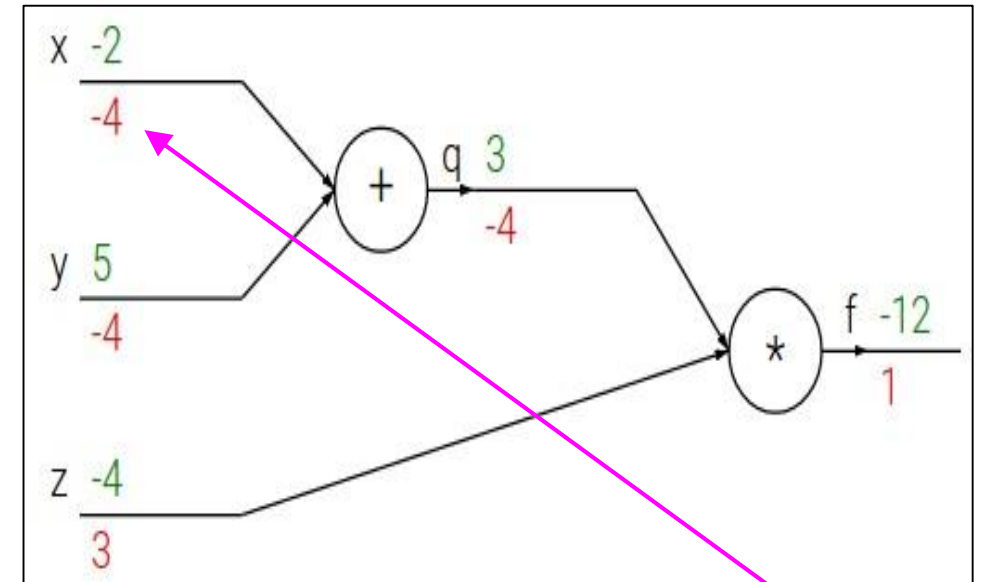
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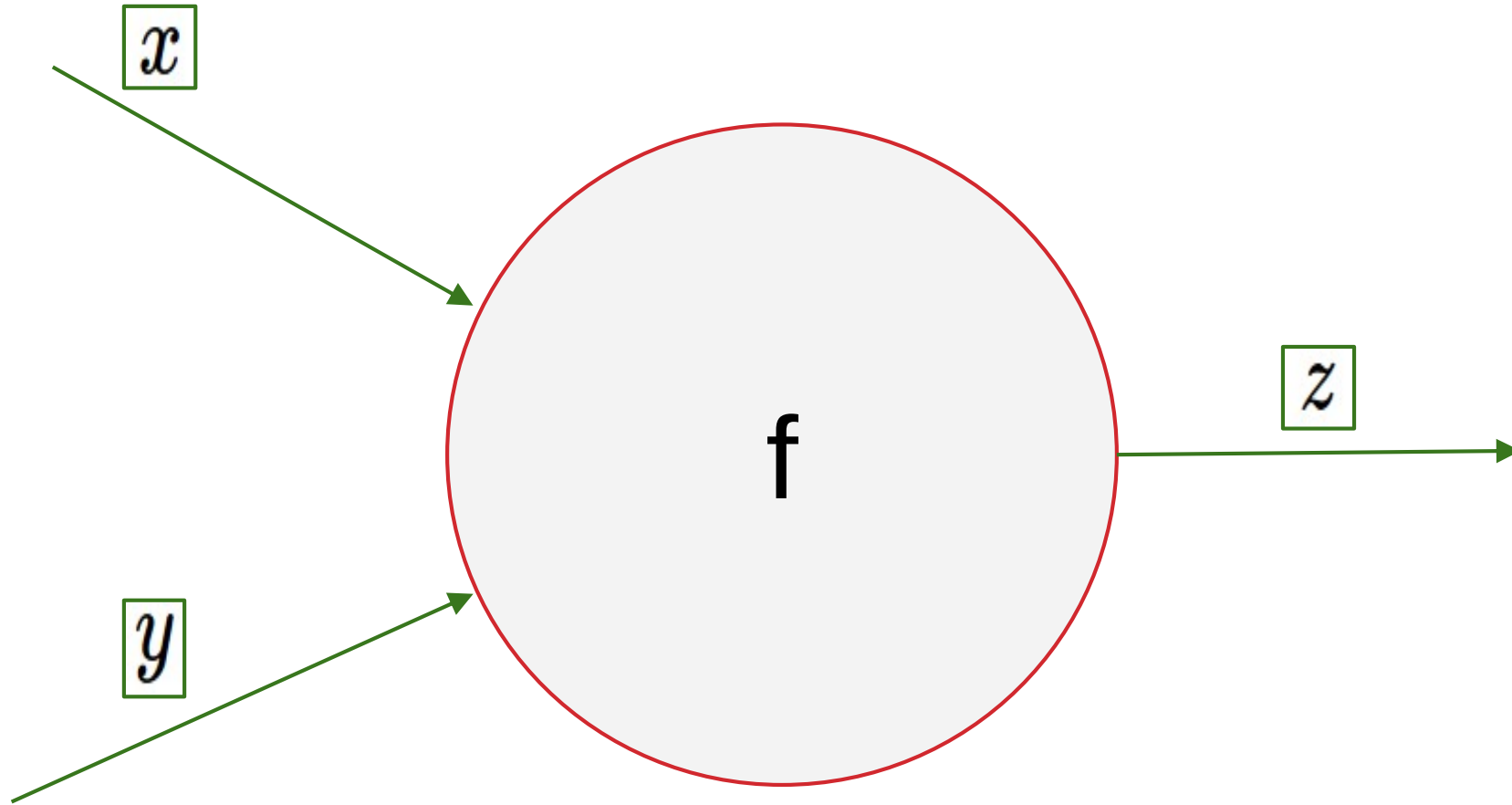


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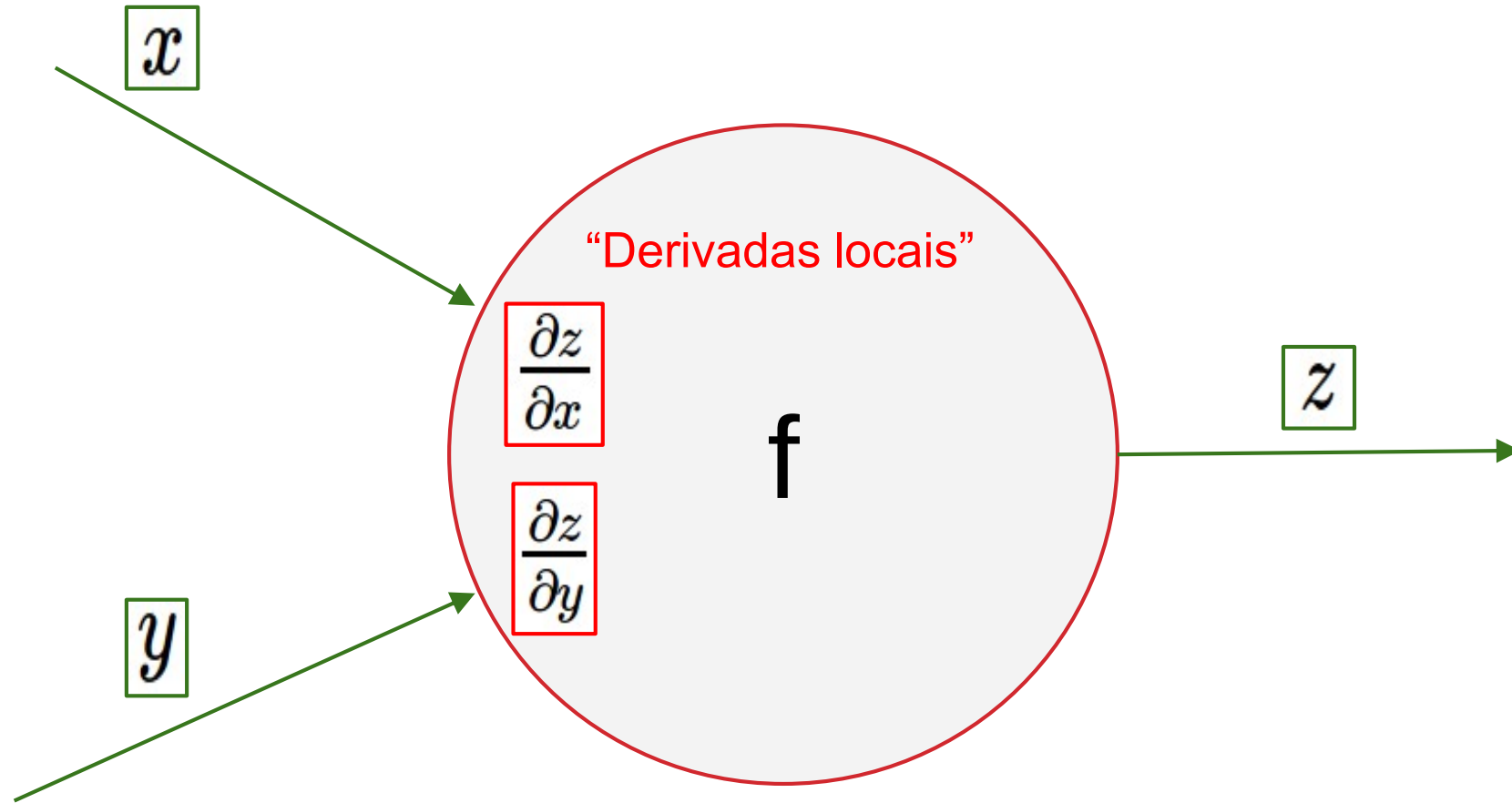
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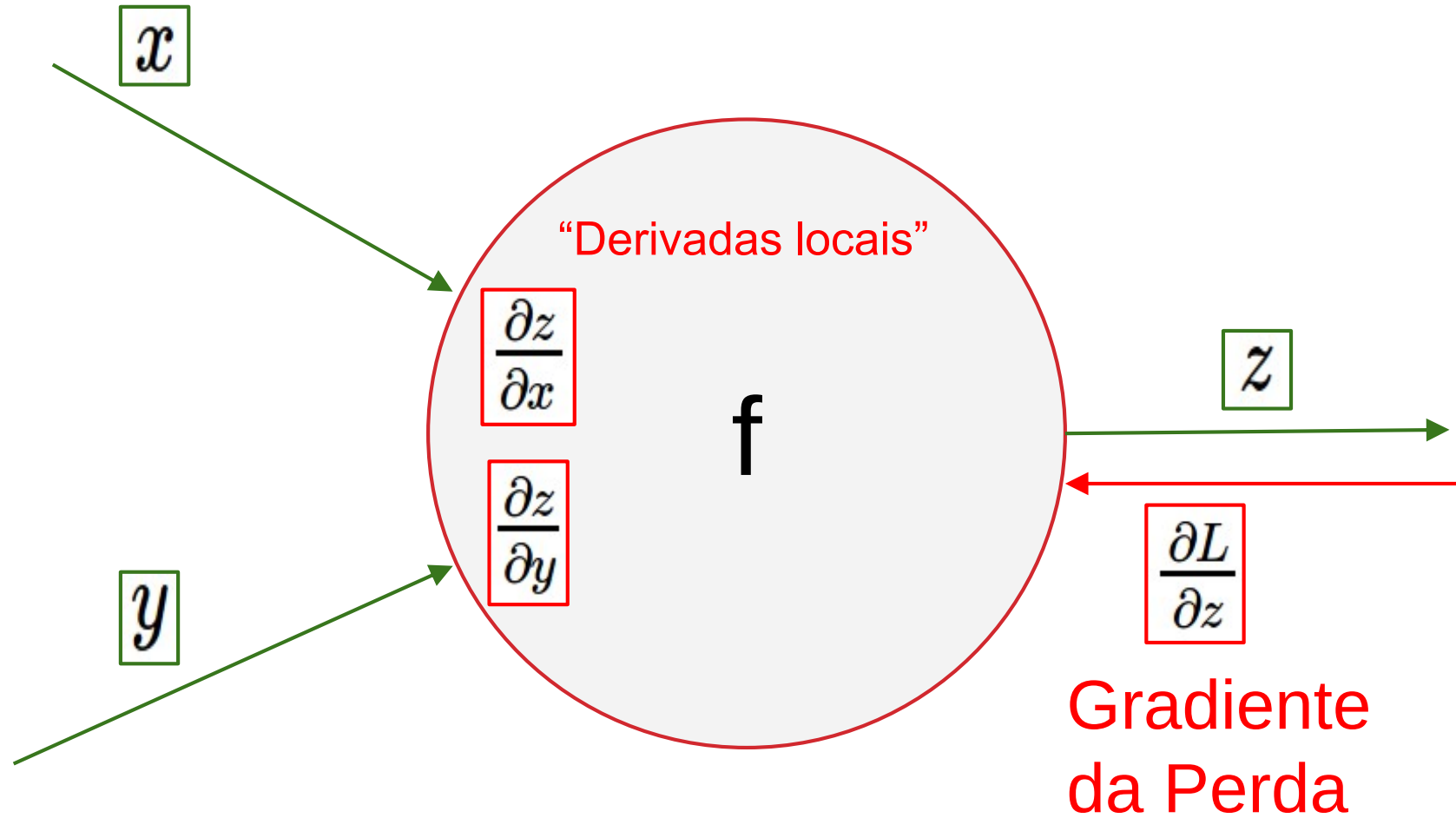
Passo Retrógrado (*Backward Pass*)



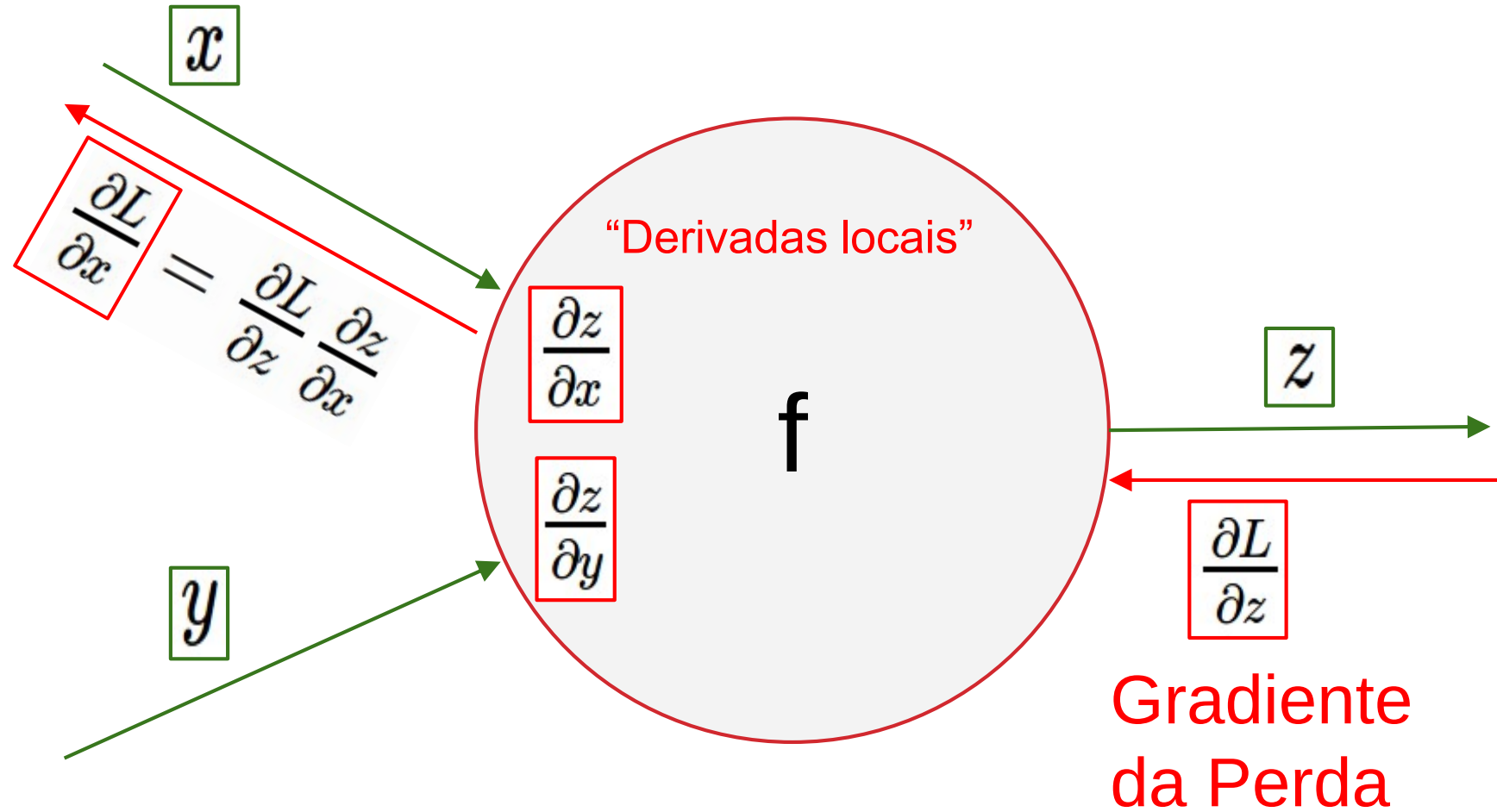
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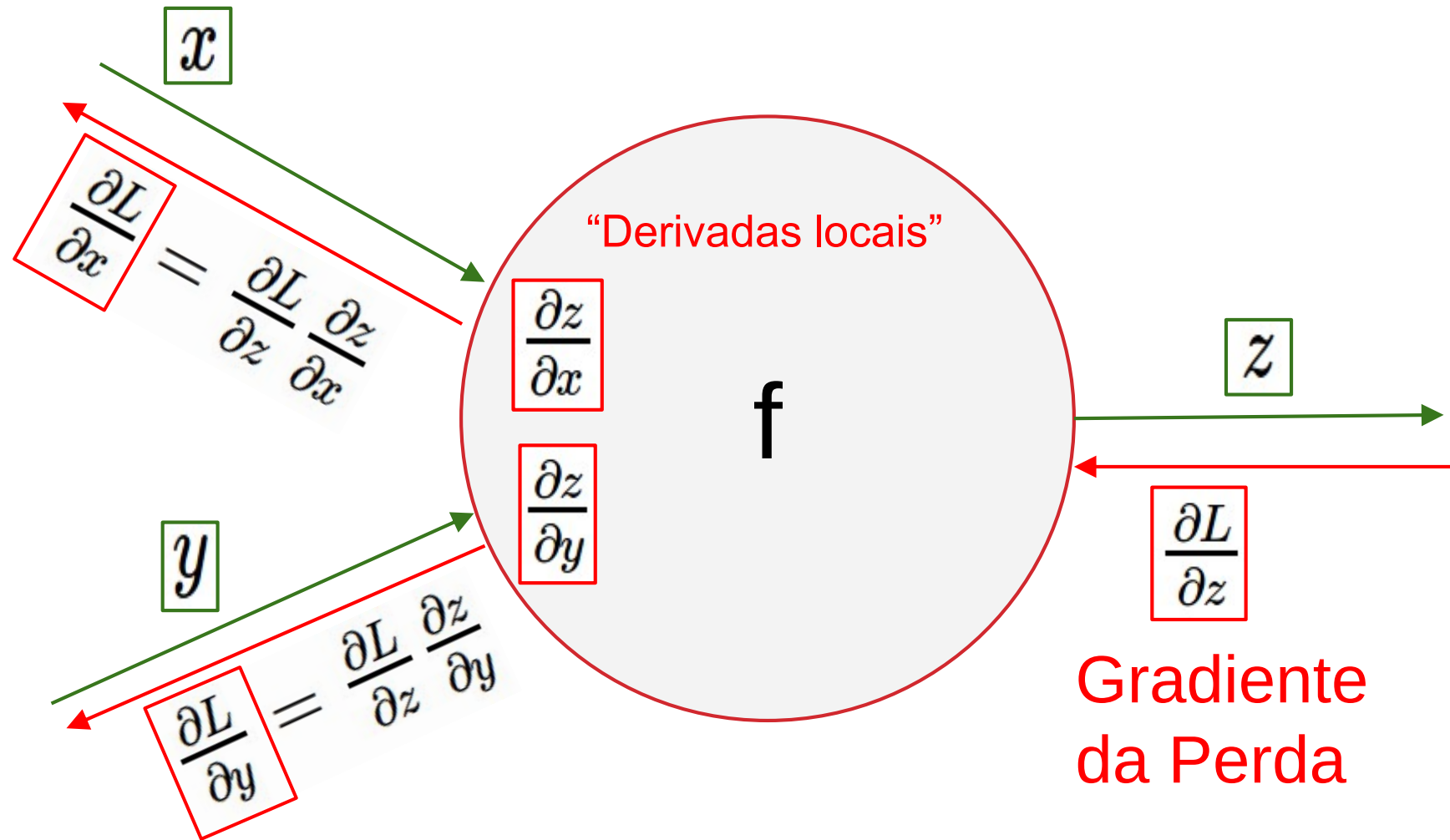
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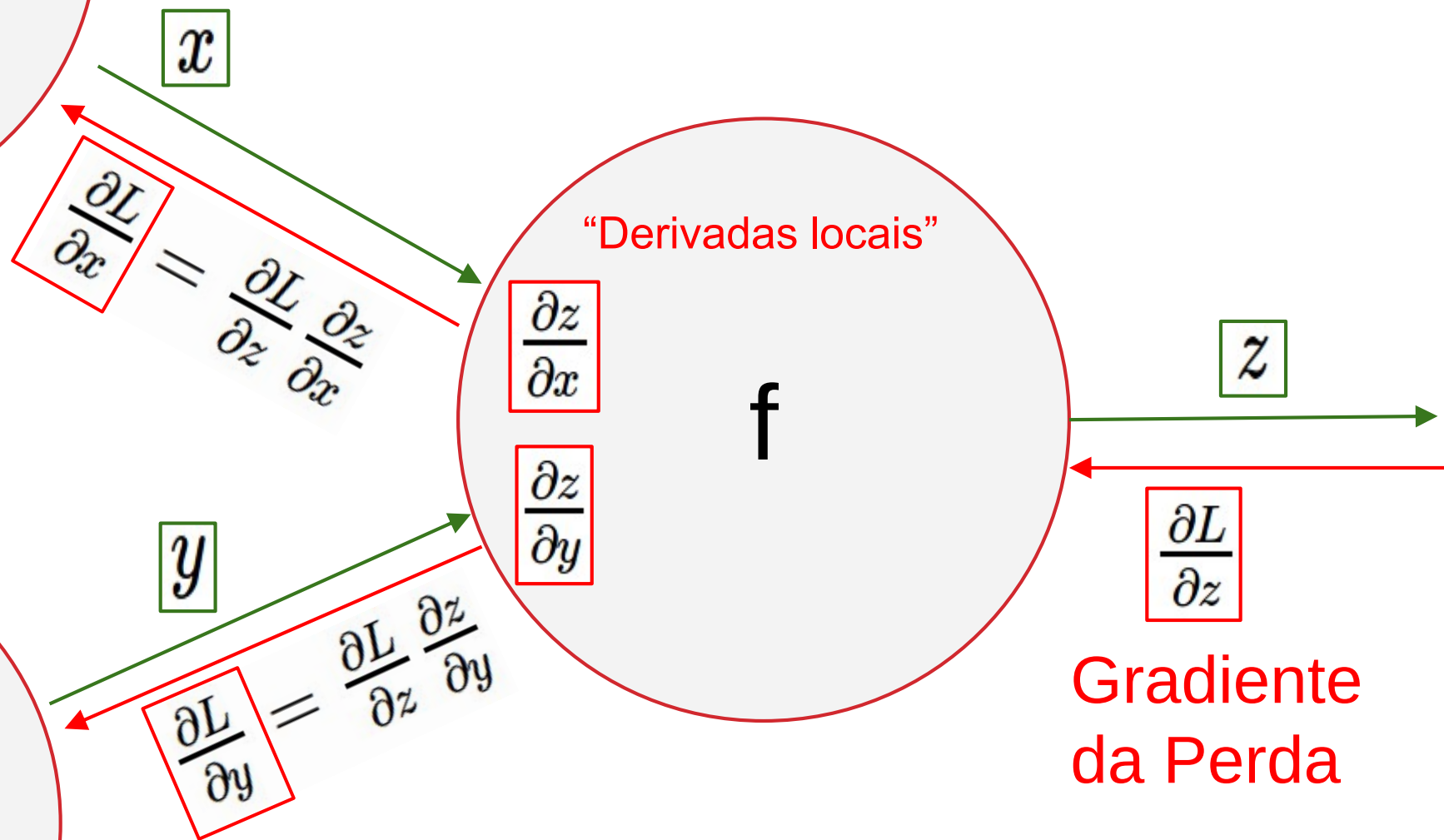
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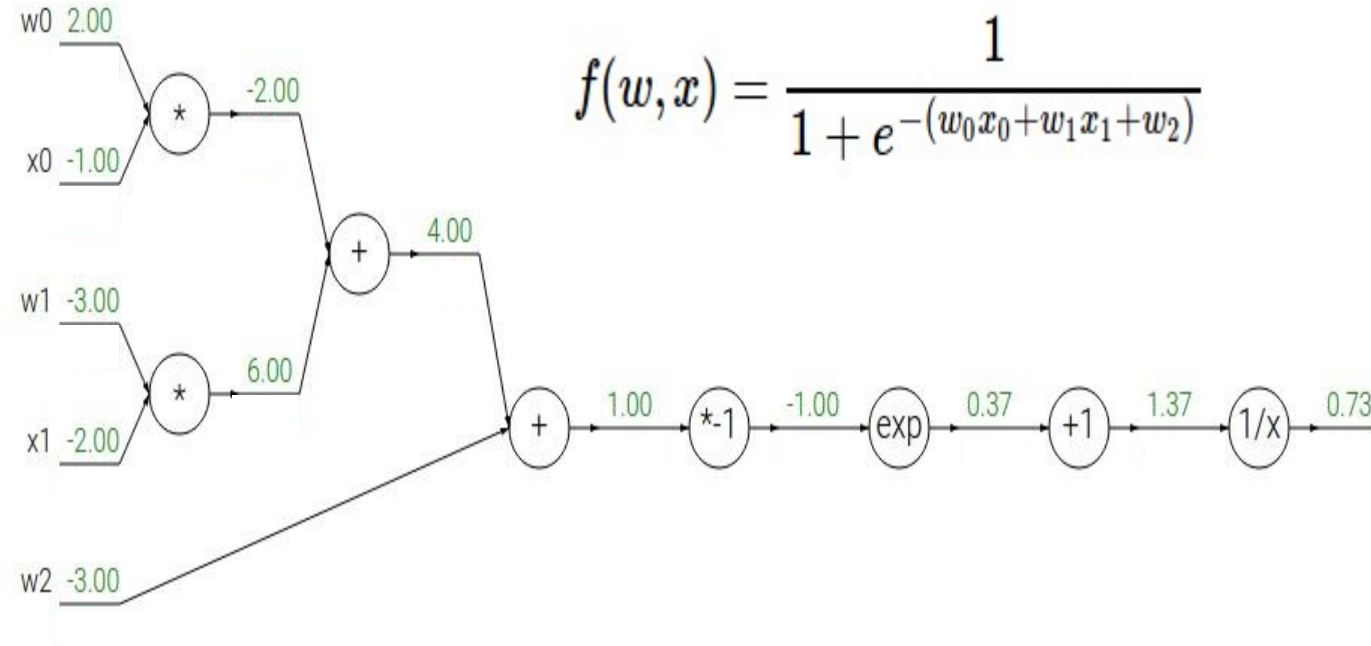
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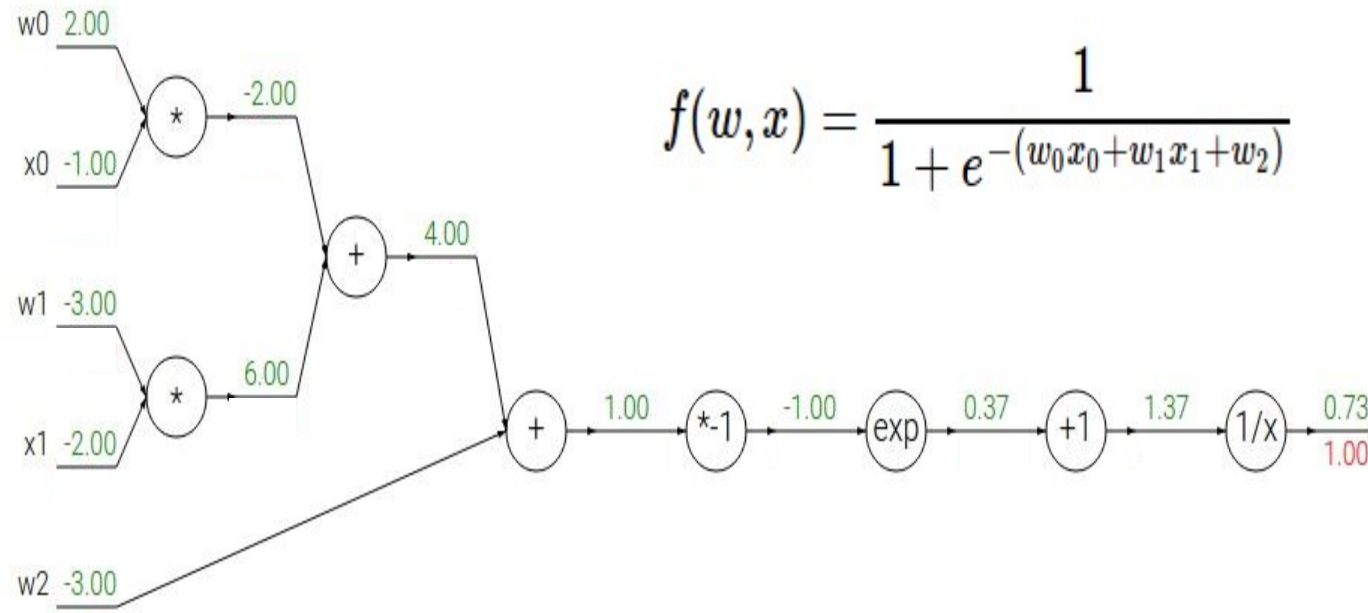
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Outro Exemplo – Passo Retrógrado

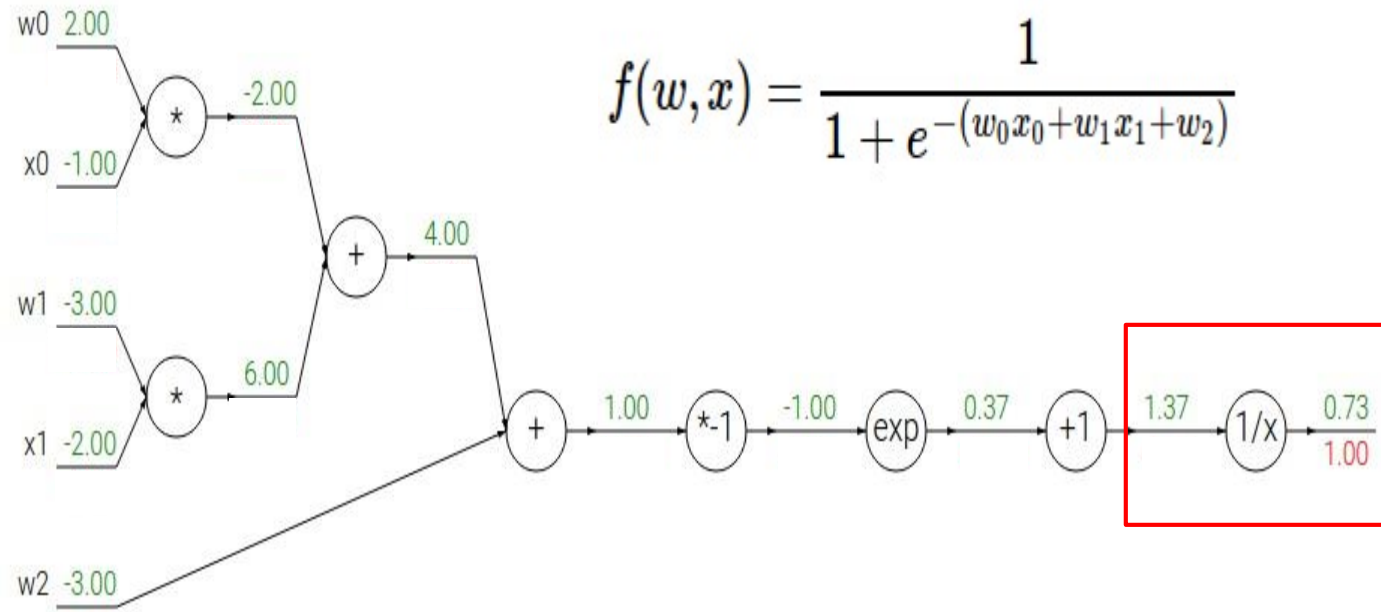


Outro Exemplo – Passo Retrógrado



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Outro Exemplo – Passo Retrógrado



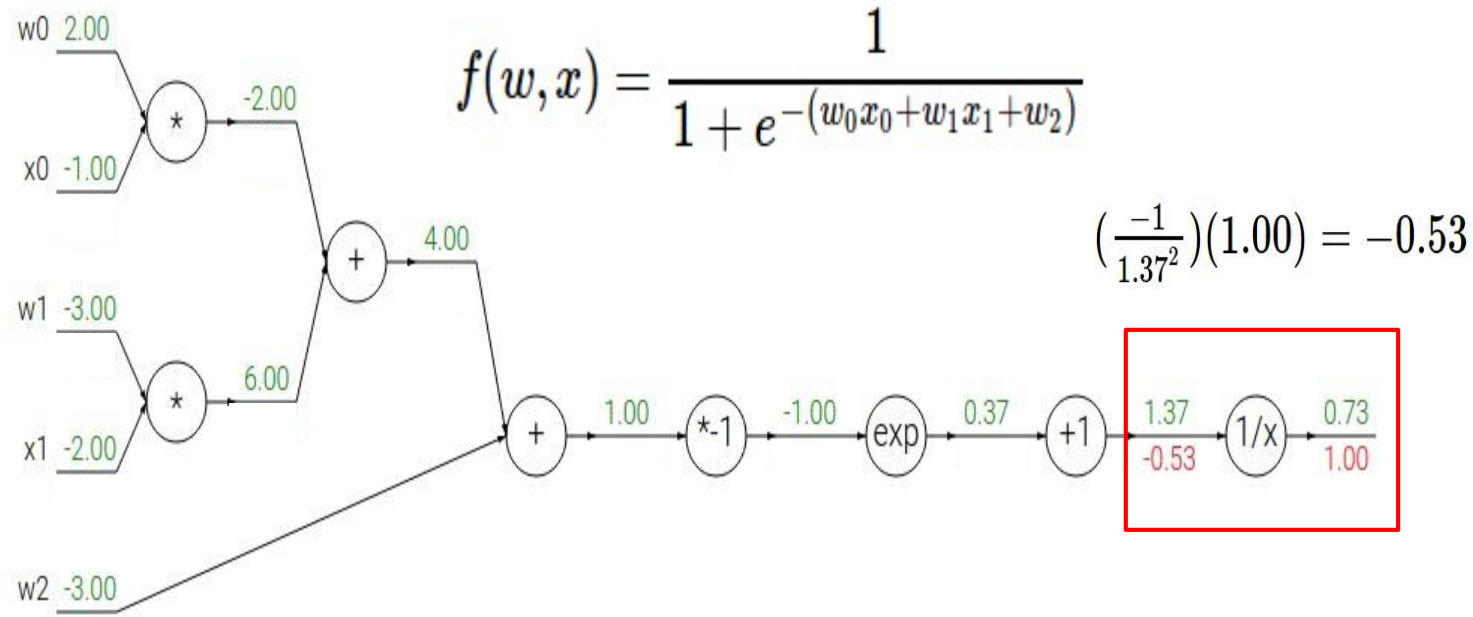
$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

$$f_a(x) = ax \rightarrow \frac{df}{dx} = a$$

$$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$$

$$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$$

Outro Exemplo – Passo Retrógrado



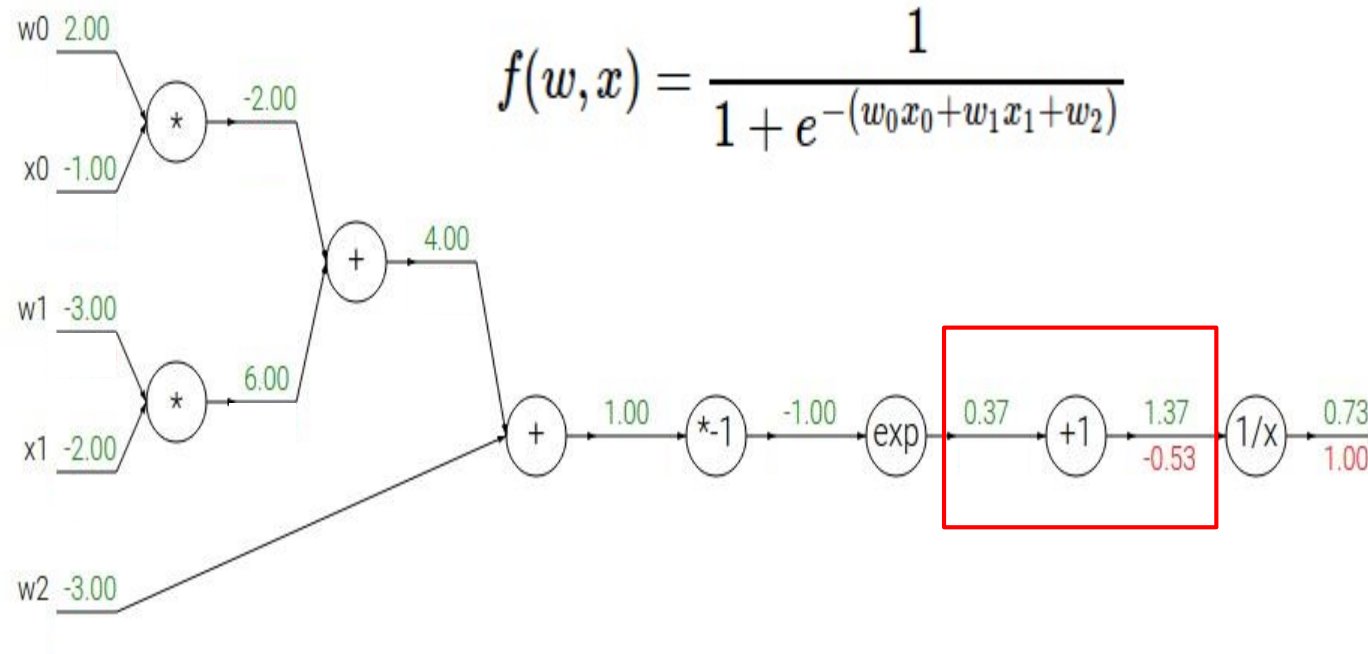
$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

$$f_a(x) = ax \rightarrow \frac{df}{dx} = a$$

$$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$$

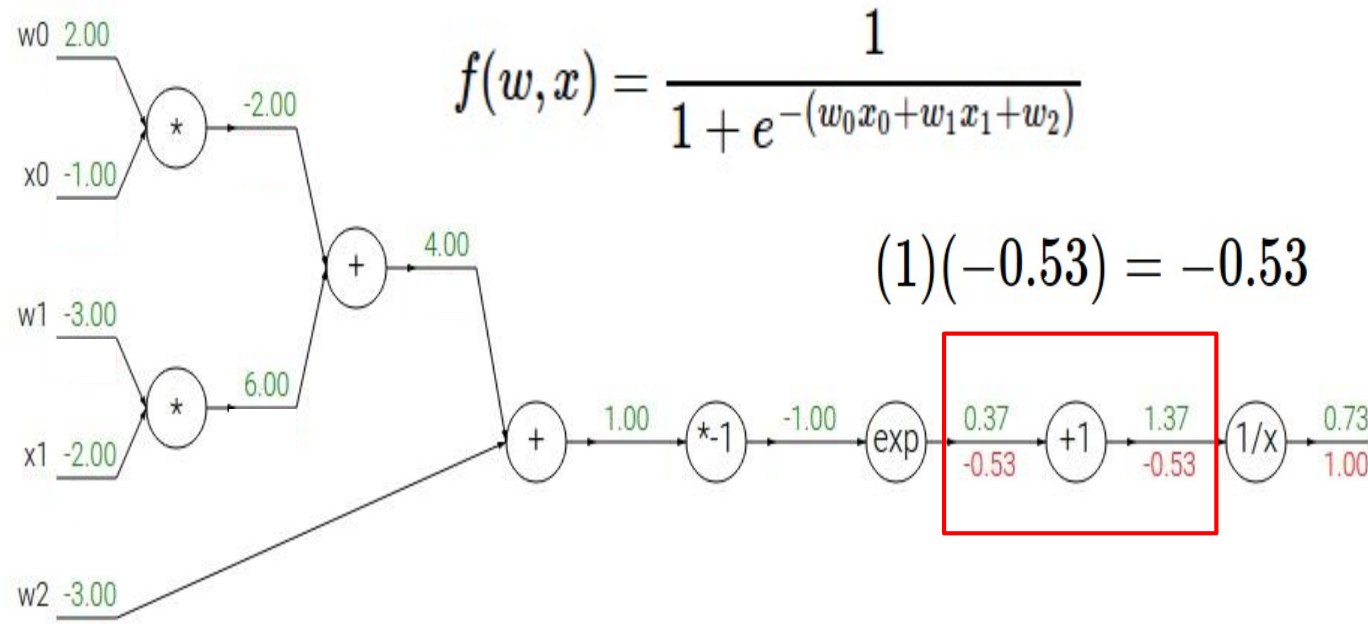
$$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$$

Outro Exemplo – Passo Retrógrado



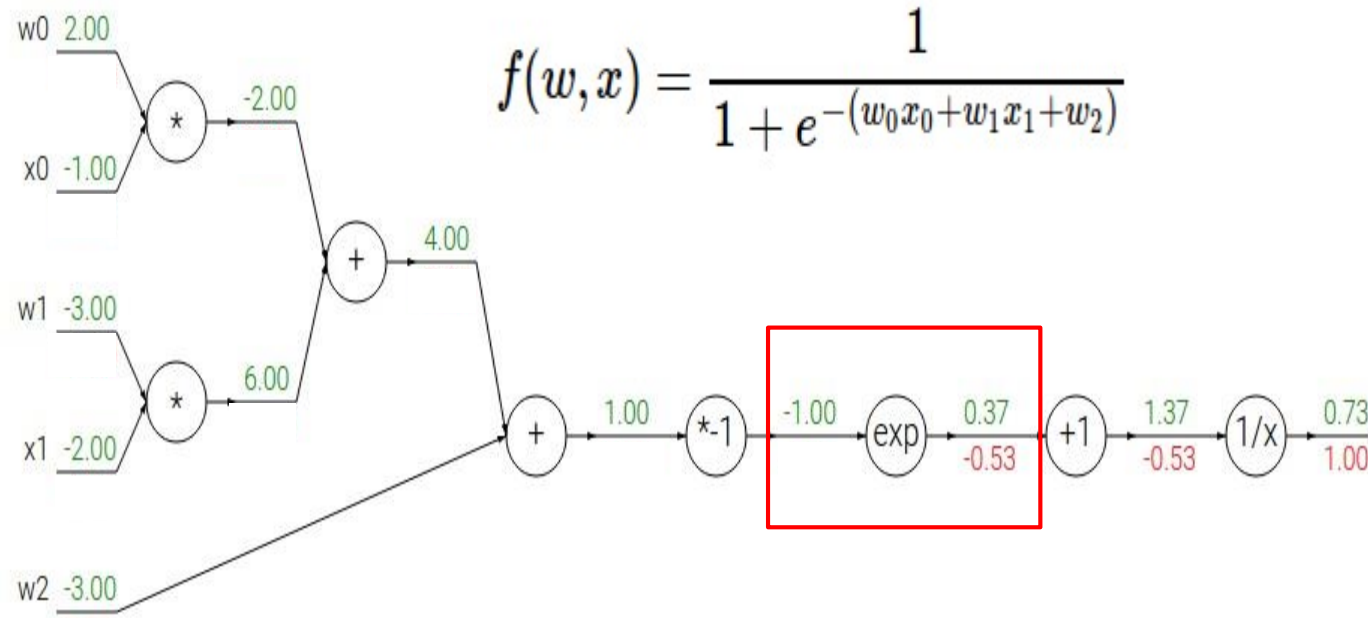
$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Outro Exemplo – Passo Retrógrado



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Outro Exemplo – Passo Retrógrado



$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

$$f_a(x) = ax \rightarrow \frac{df}{dx} = a$$

$$f(x) = \frac{1}{x}$$

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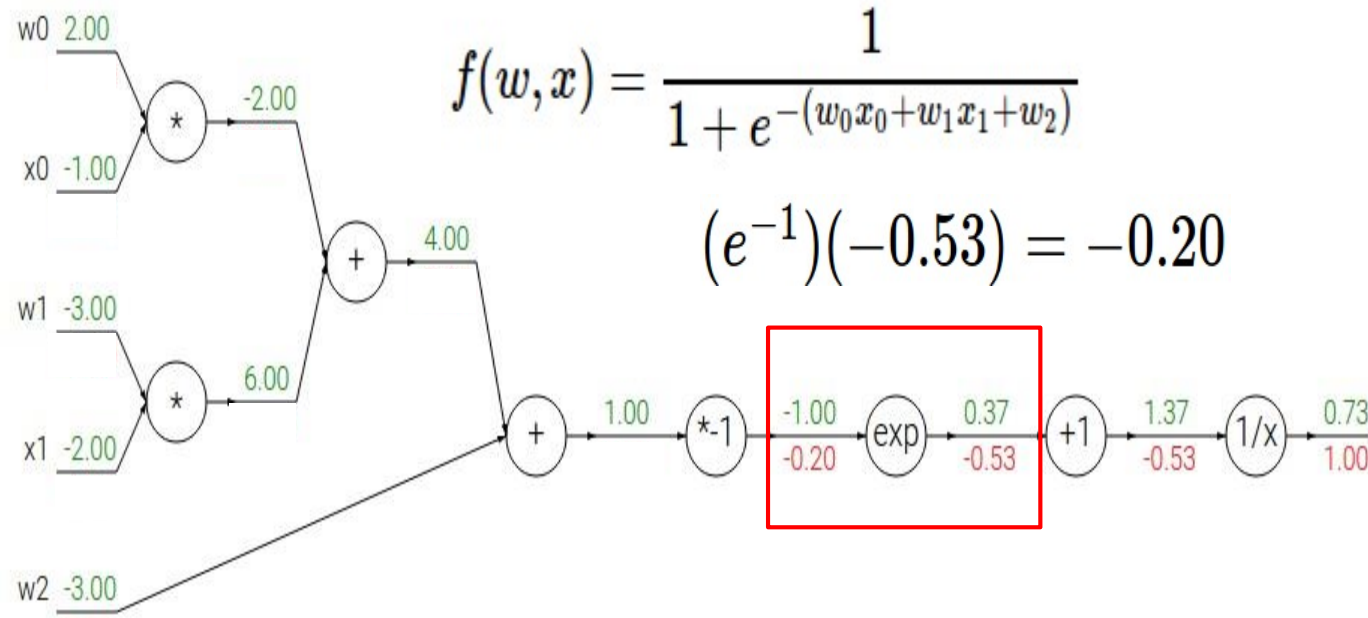
→

$$\frac{df}{dx} = -1/x^2$$

→

$$\frac{df}{dx} = 1$$

Outro Exemplo – Passo Retrógrado



$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

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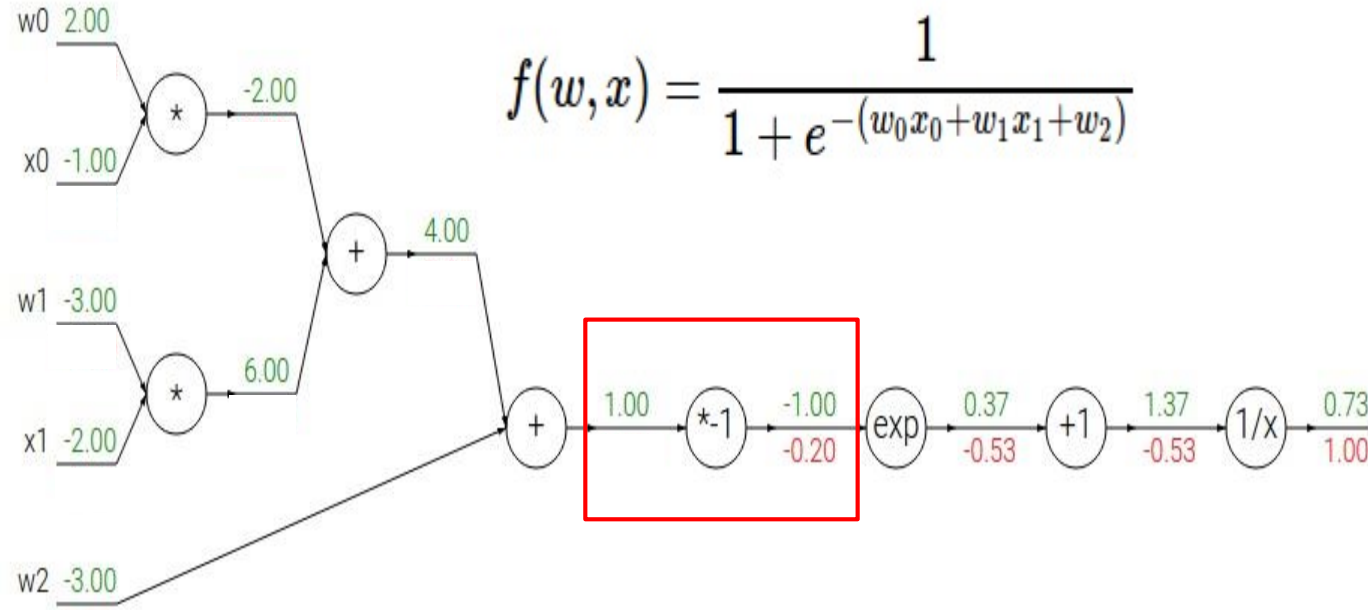
→

$$\frac{df}{dx} = -1/x^2$$

→

$$\frac{df}{dx} = 1$$

Outro Exemplo – Passo Retrógrado



$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

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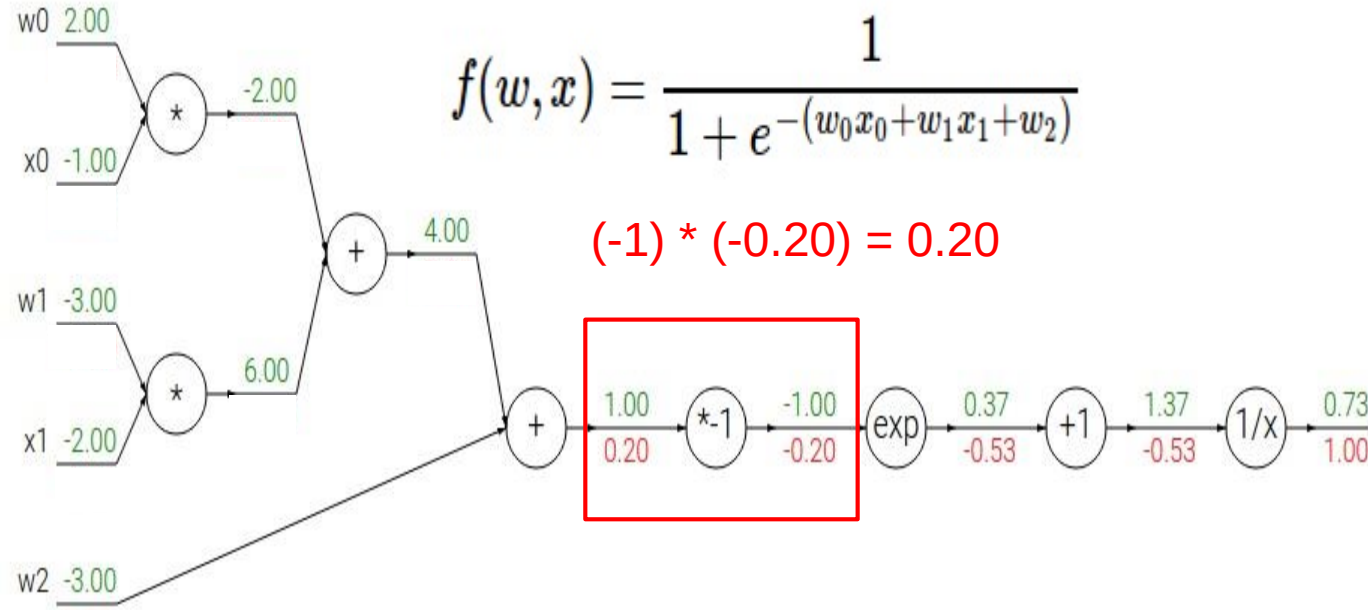
$\rightarrow$

$$\frac{df}{dx} = -1/x^2$$

$\rightarrow$

$$\frac{df}{dx} = 1$$

Outro Exemplo – Passo Retrógrado



$$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$$

$$f_a(x) = ax \rightarrow \frac{df}{dx} = a$$

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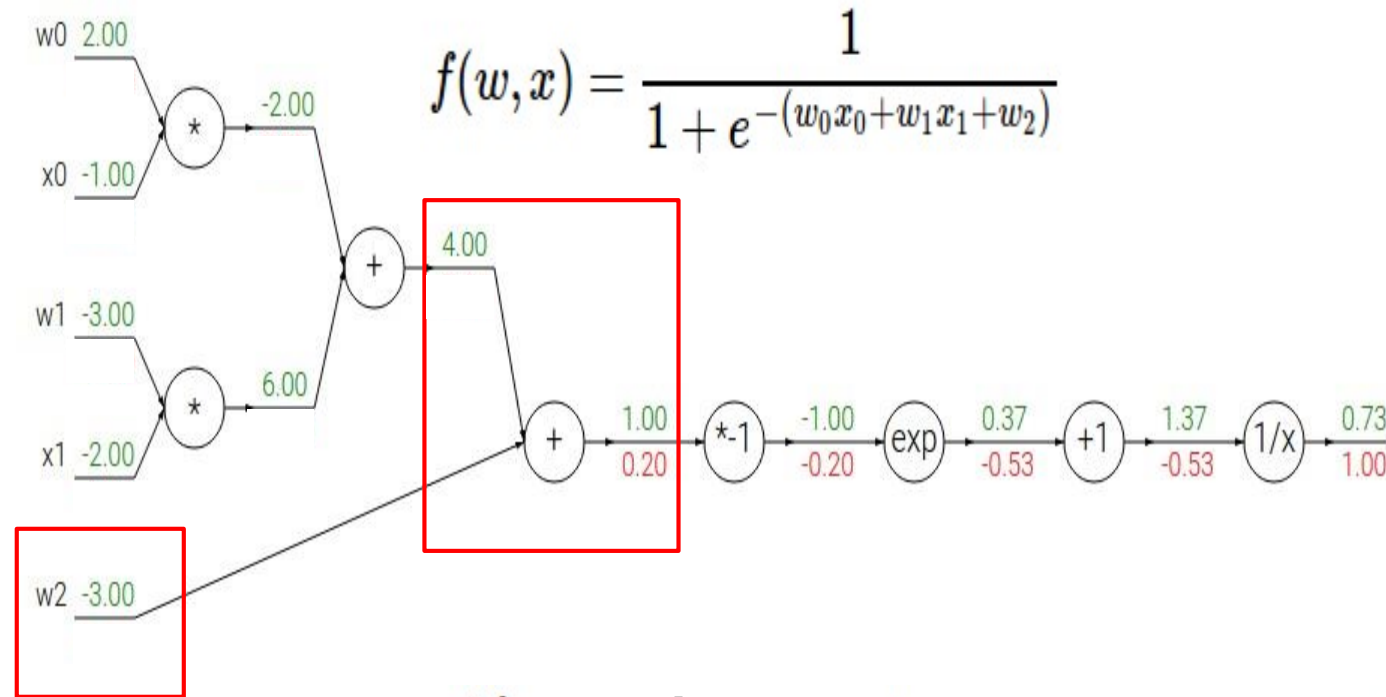
→

$$\frac{df}{dx} = -1/x^2$$

→

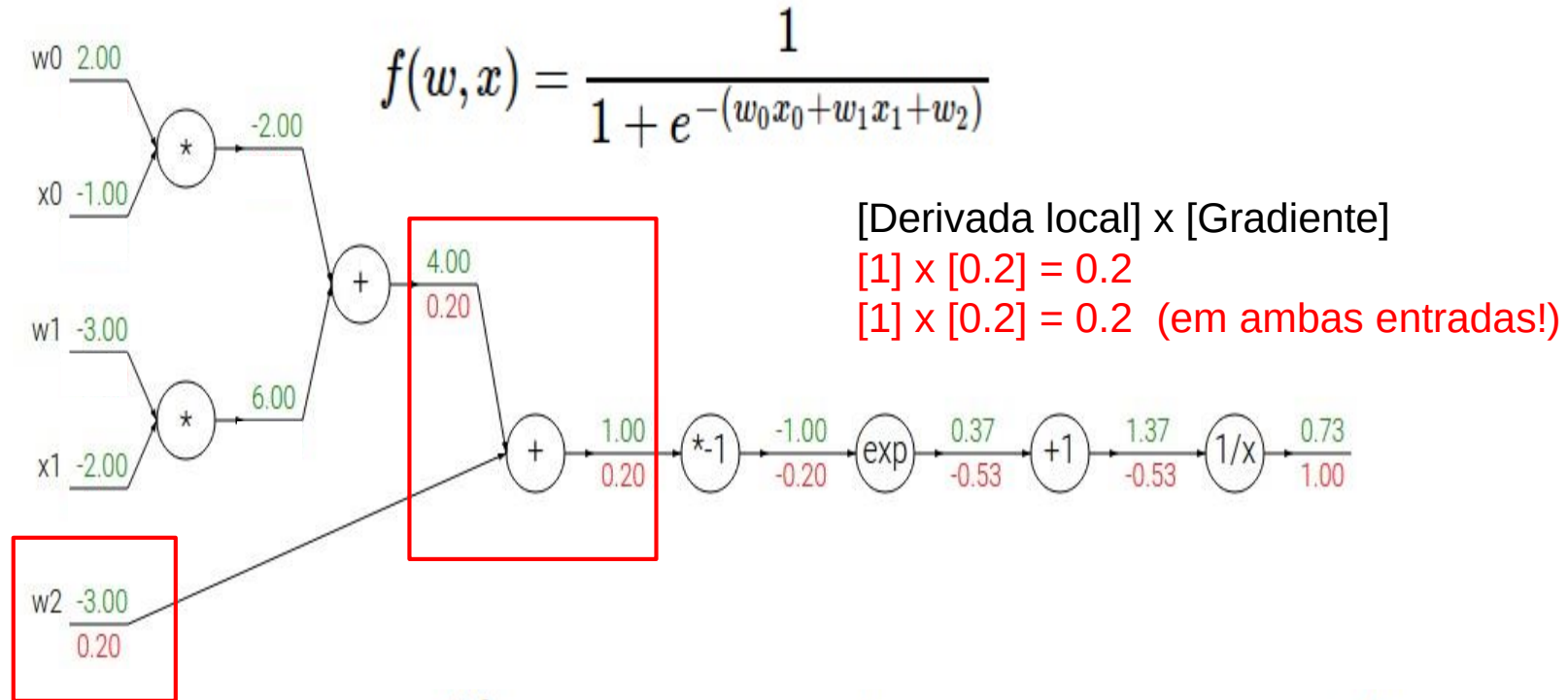
$$\frac{df}{dx} = 1$$

Outro Exemplo – Passo Retrógrado



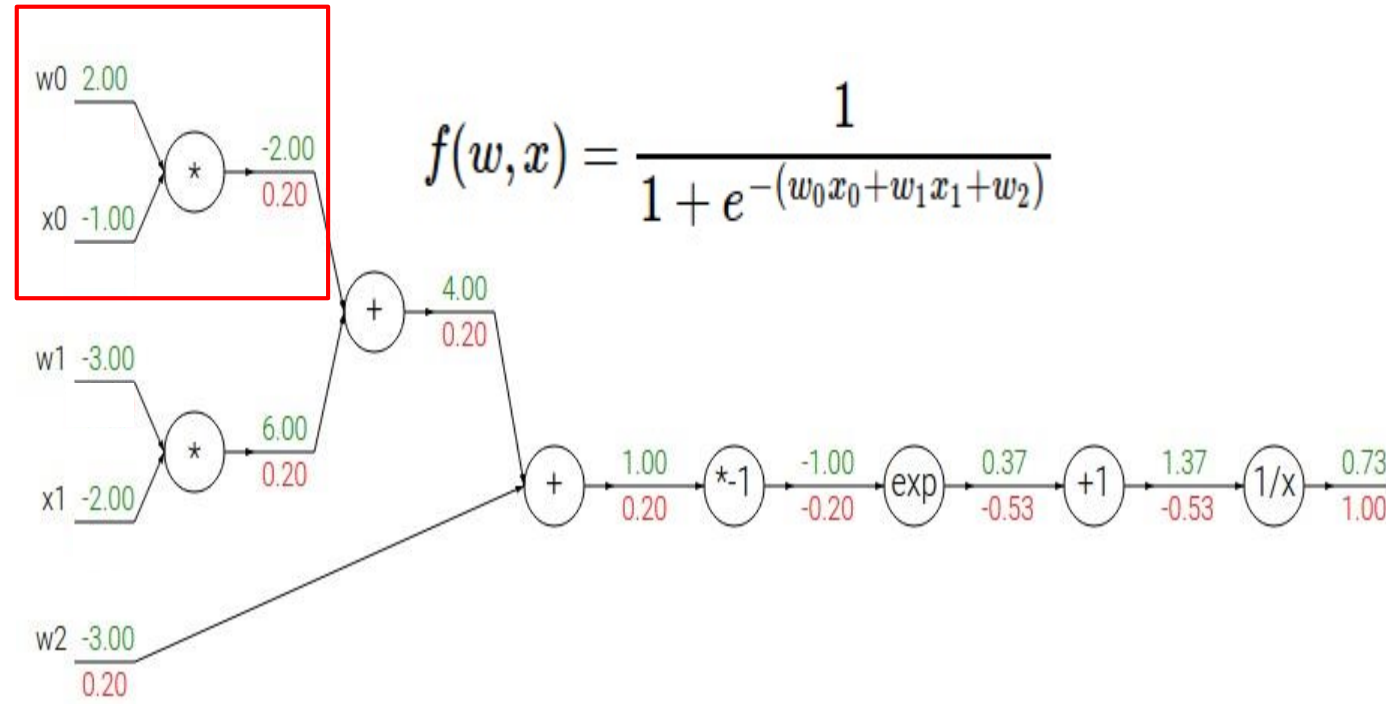
$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$		$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Outro Exemplo – Passo Retrógrado



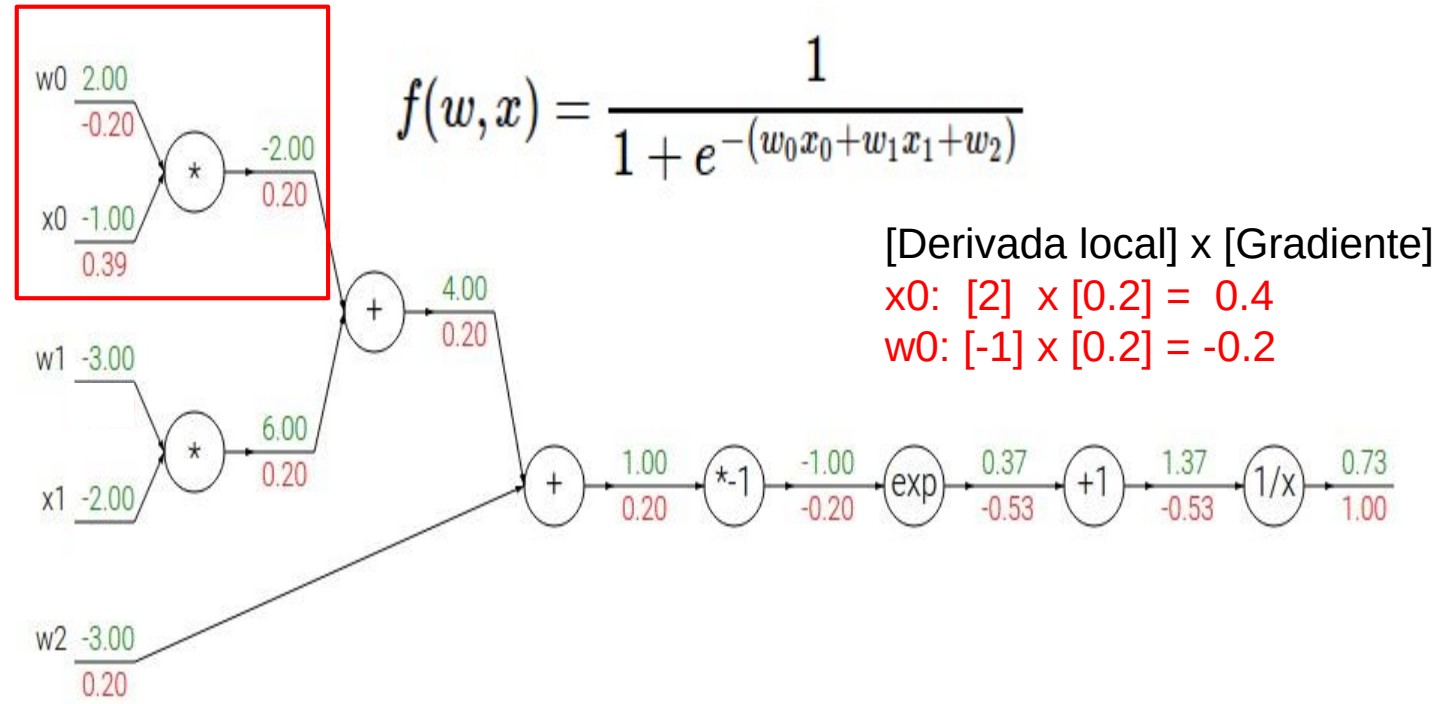
$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
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Outro Exemplo – Passo Retrógrado



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Outro Exemplo – Passo Retrógrado



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$		$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
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Outro Exemplo – Passo Retrógrado

$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2)}}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Função sigmoide

$$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1 + e^{-x})^2} = \left(\frac{1 + e^{-x} - 1}{1 + e^{-x}} \right) \left(\frac{1}{1 + e^{-x}} \right) = (1 - \sigma(x))\sigma(x)$$

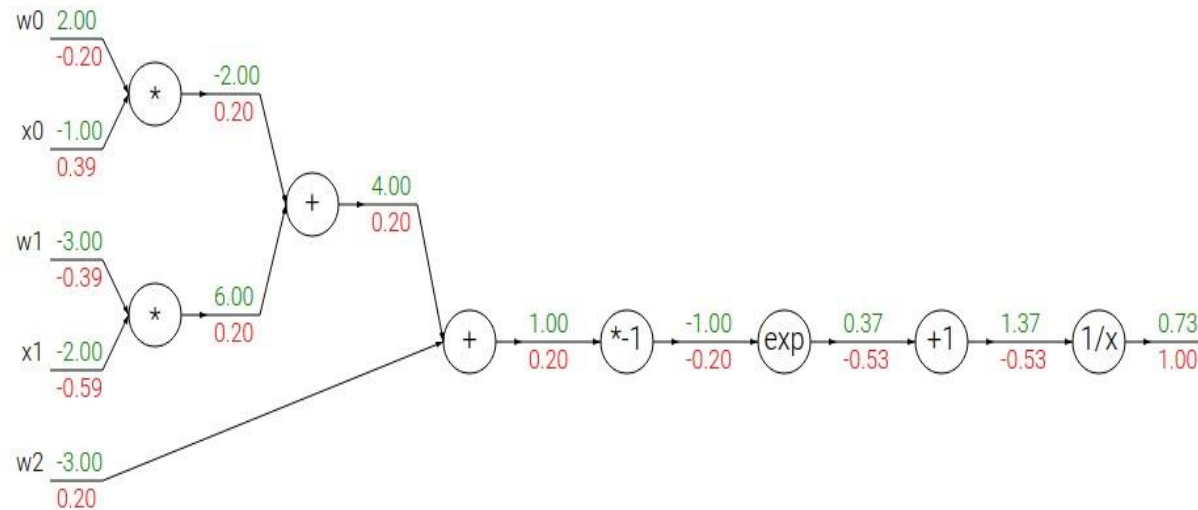
Outro Exemplo – Passo Retrógrado

$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

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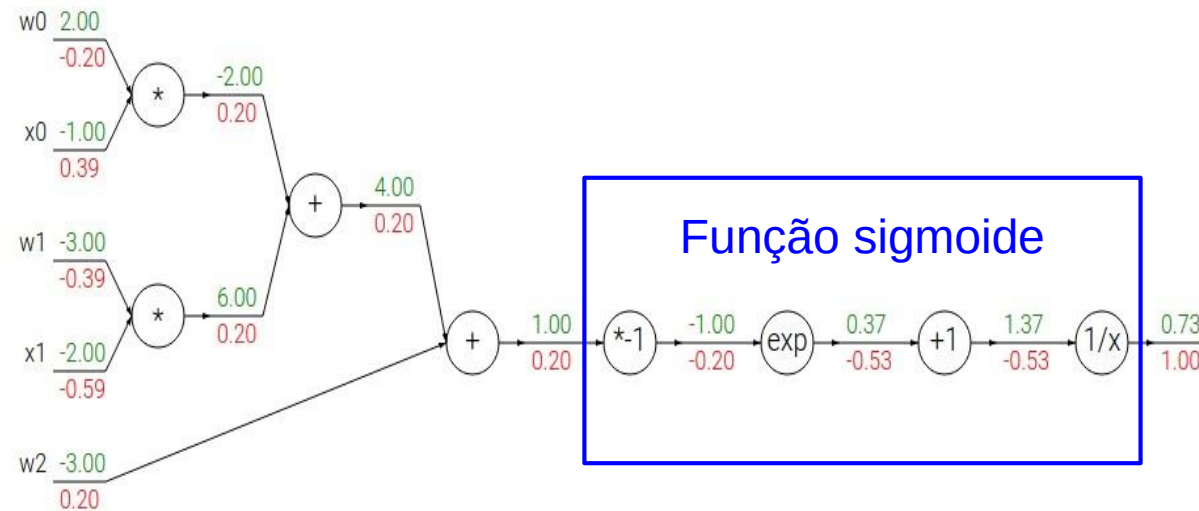
Outro Exemplo – Passo Retrógrado

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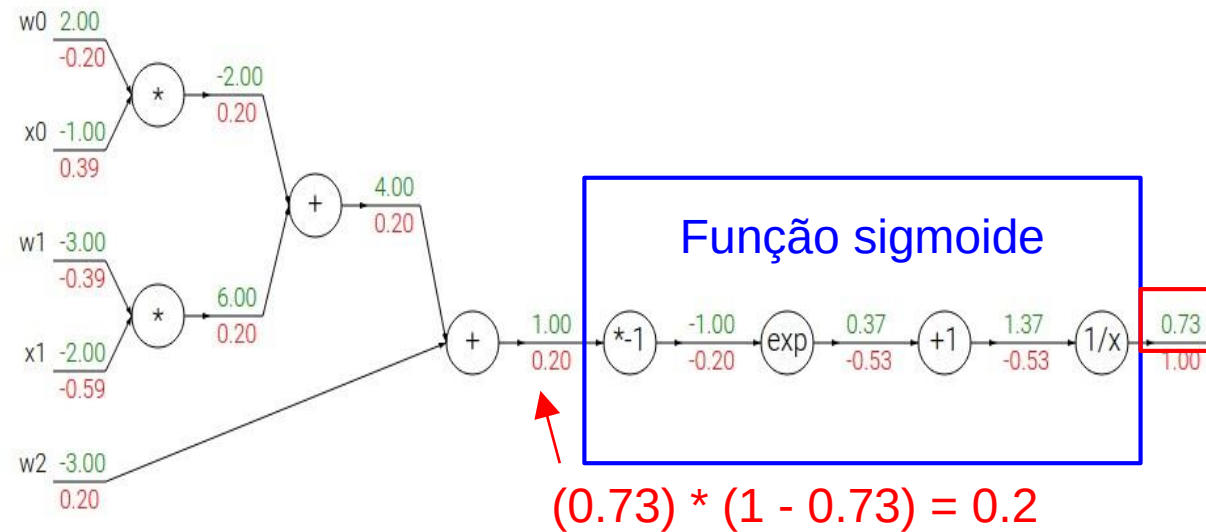
Outro Exemplo – Passo Retrógrado

$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$$

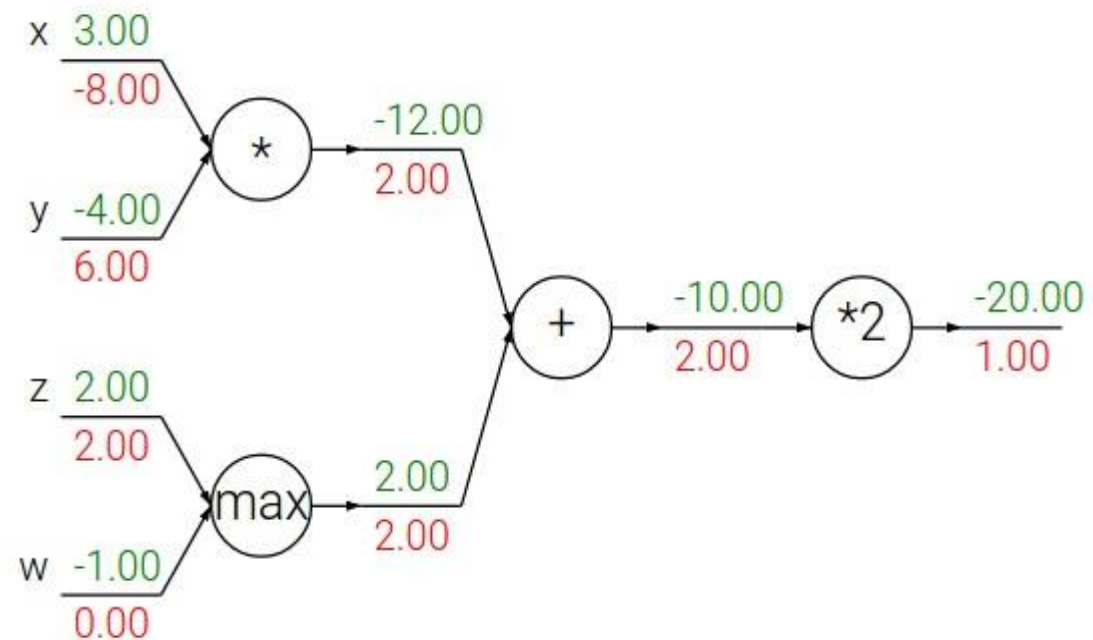
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Função sigmoide

$$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1 + e^{-x})^2} = \left(\frac{1 + e^{-x} - 1}{1 + e^{-x}} \right) \left(\frac{1}{1 + e^{-x}} \right) = (1 - \sigma(x))\sigma(x)$$

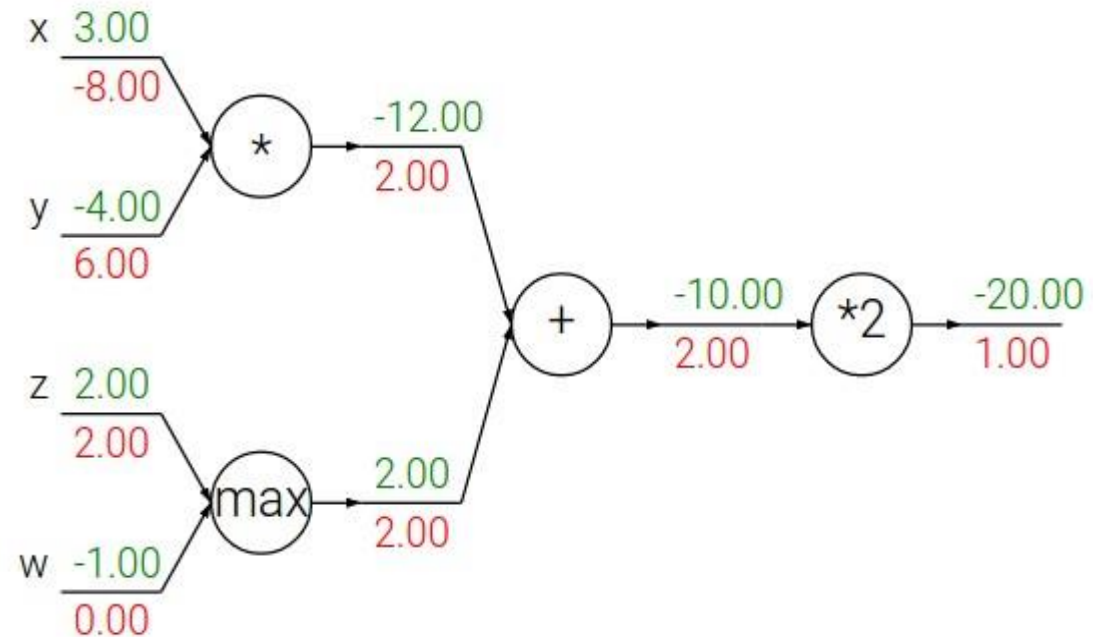


Padrões no Fluxo Reverso



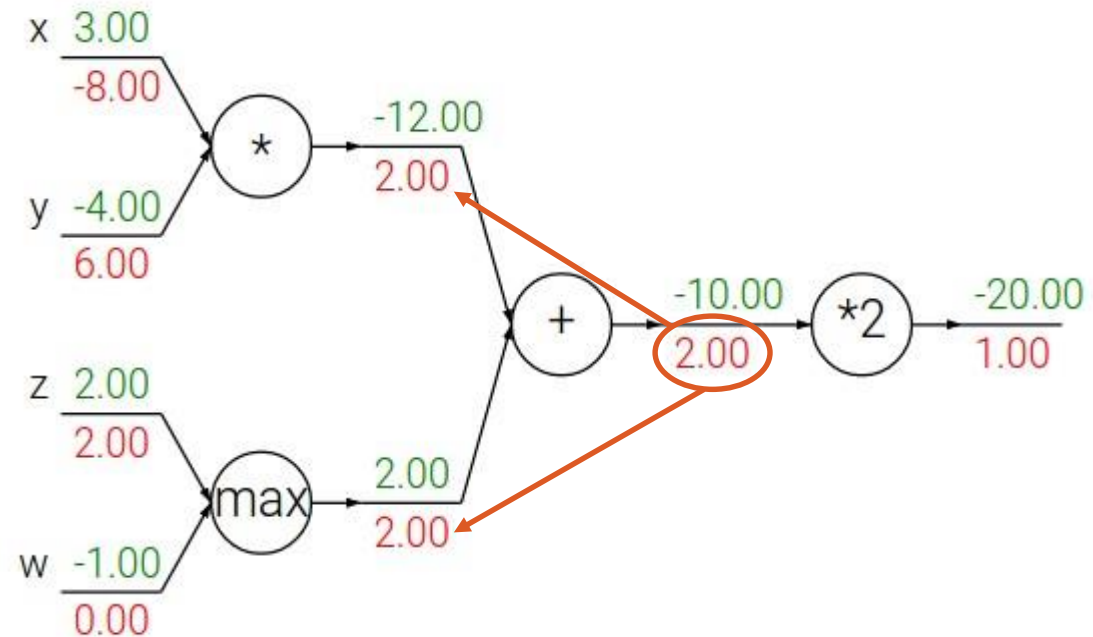
Padrões no Fluxo Reverso

Adição : distribuidor de gradiente



Padrões no Fluxo Reverso

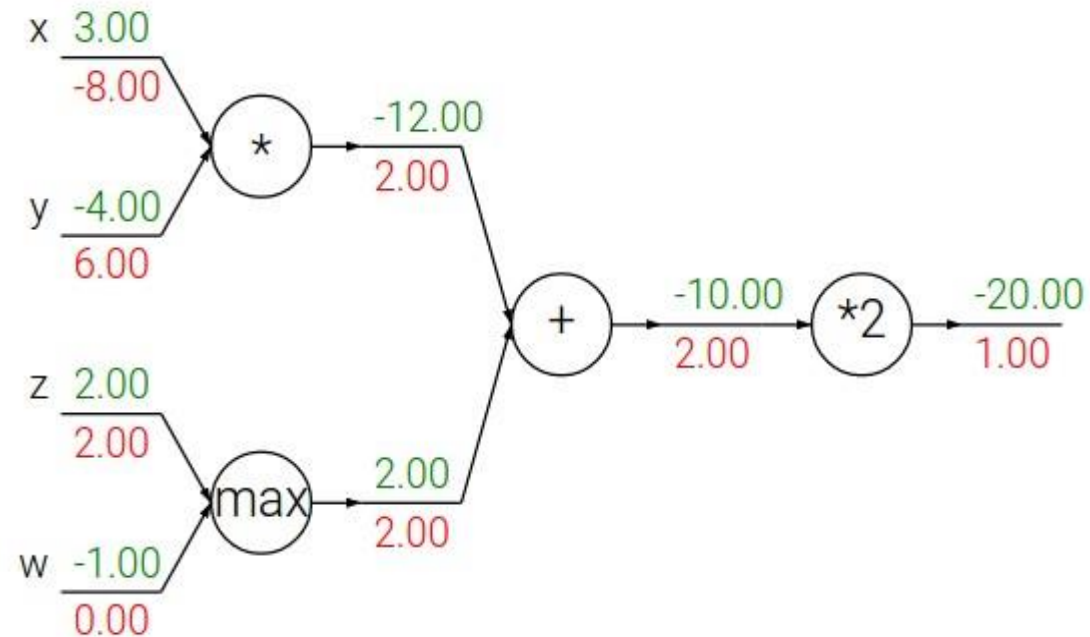
Adição : distribuidor de gradiente



Padrões no Fluxo Reverso

Adição : distribuidor de gradiente

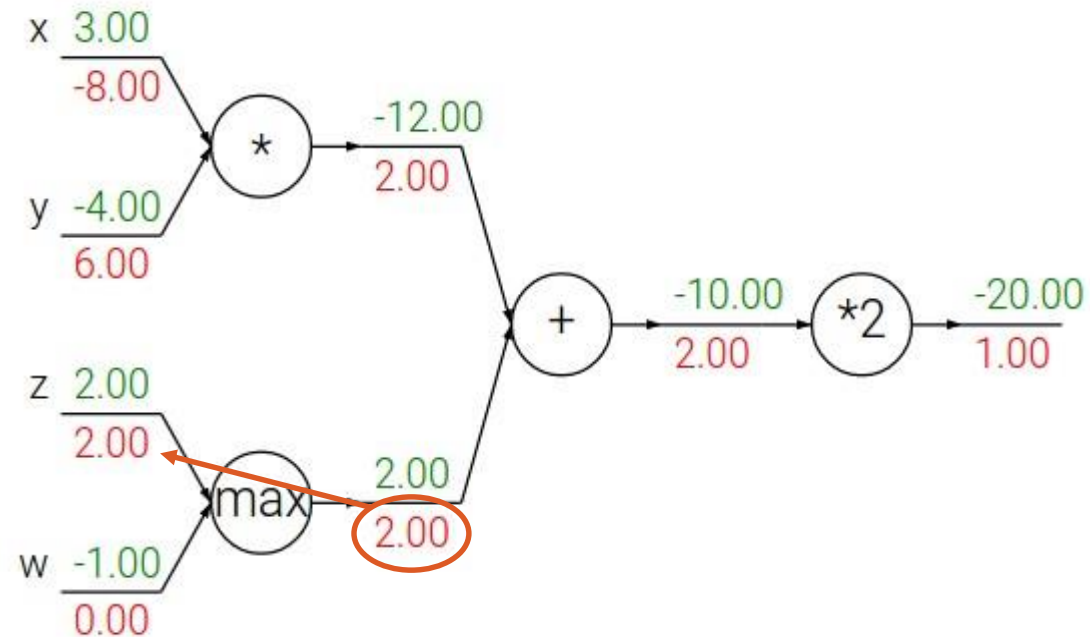
Max : direcionador de gradiente



Padrões no Fluxo Reverso

Adição : distribuidor de gradiente

Max : direcionador de gradiente

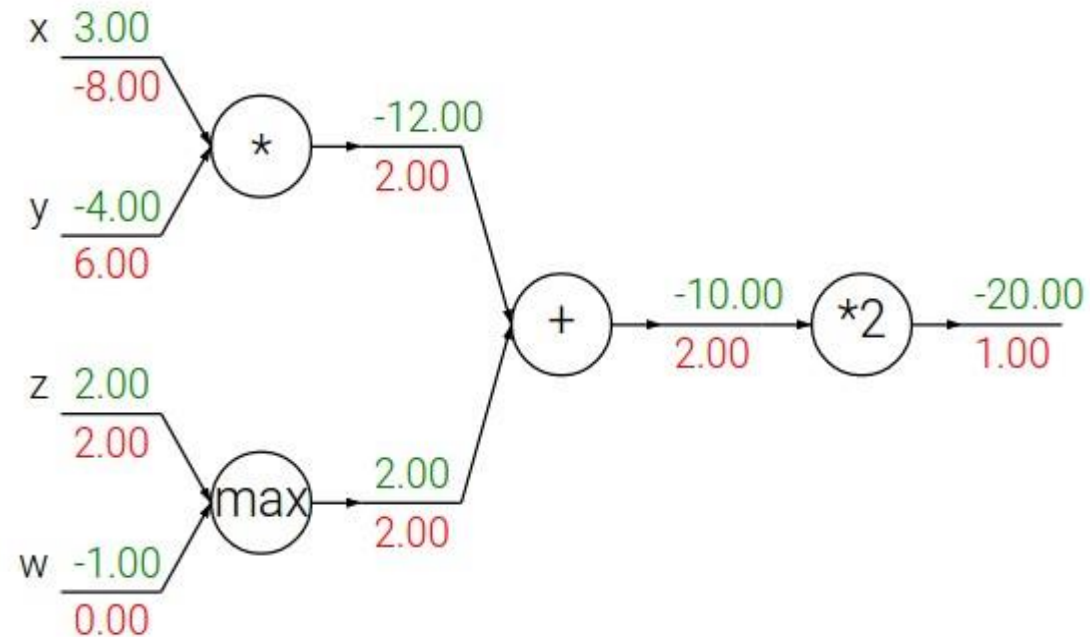


Padrões no Fluxo Reverso

Adição : distribuidor de gradiente

Max : direcionador de gradiente

Produto: “comutador” de gradiente

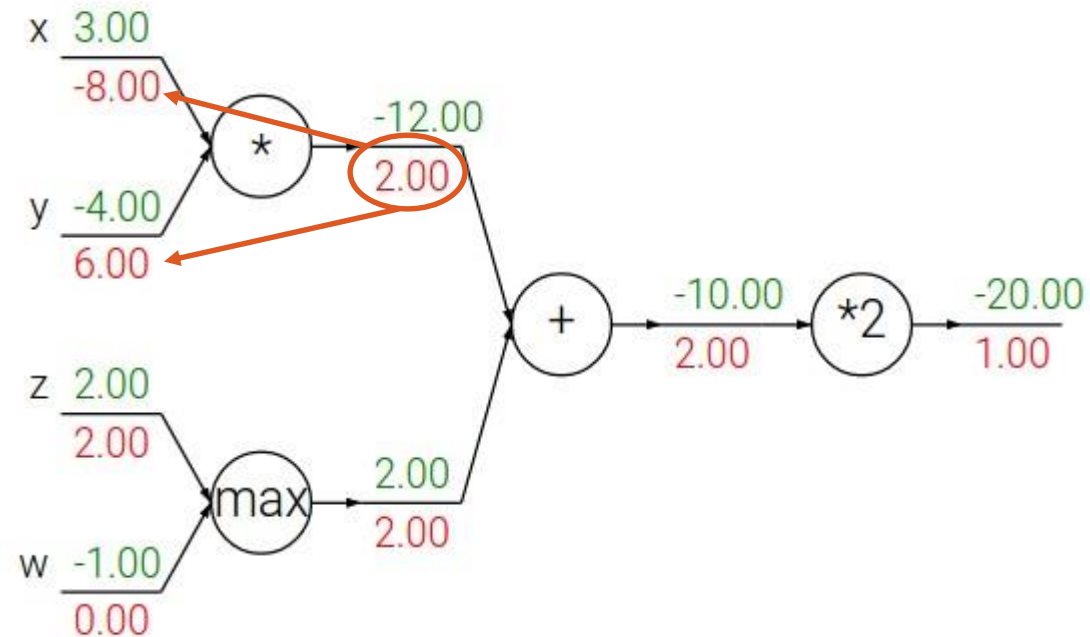


Padrões no Fluxo Reverso

Adição : distribuidor de gradiente

Max : direcionador de gradiente

Produto: “comutador” de gradiente

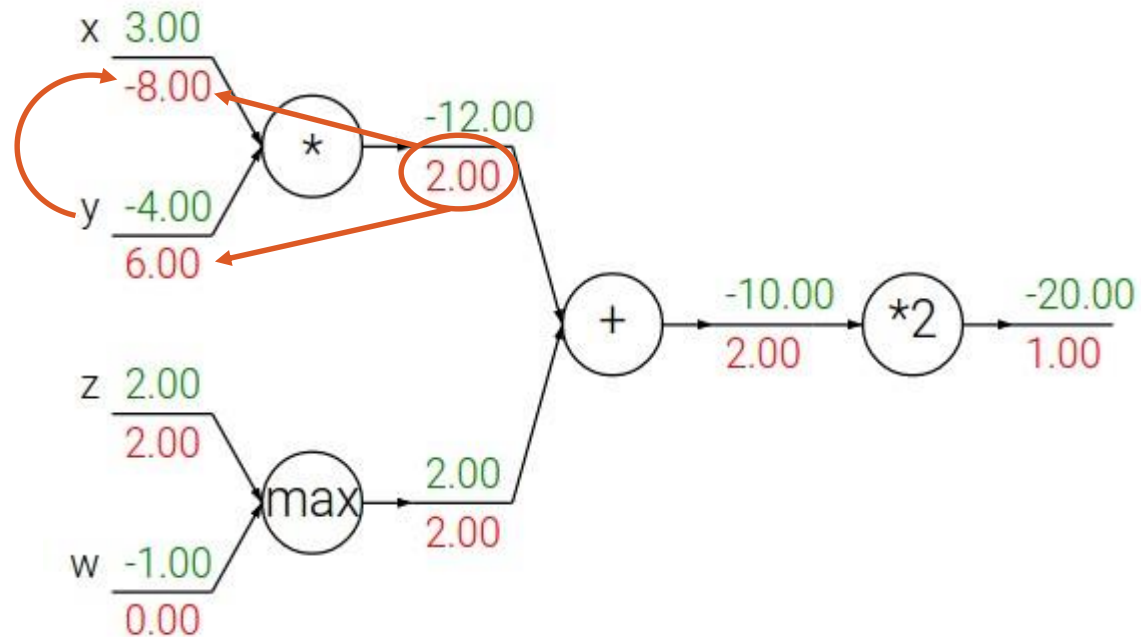


Padrões no Fluxo Reverso

Adição : distribuidor de gradiente

Max : direcionador de gradiente

Produto: “comutador” de gradiente

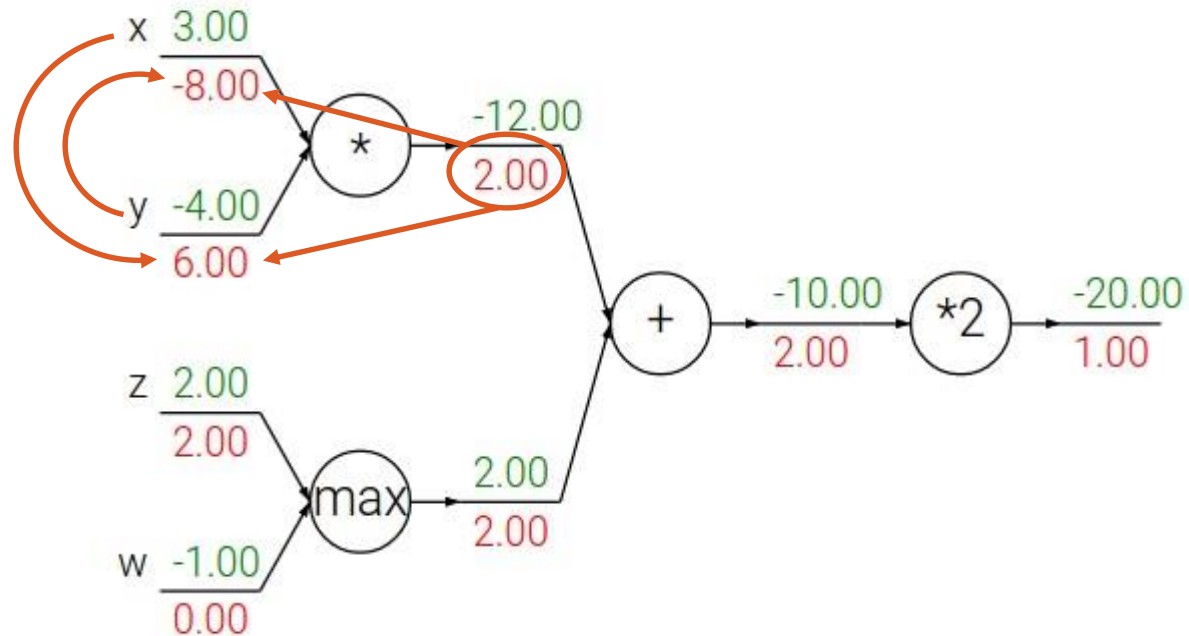


Padrões no Fluxo Reverso

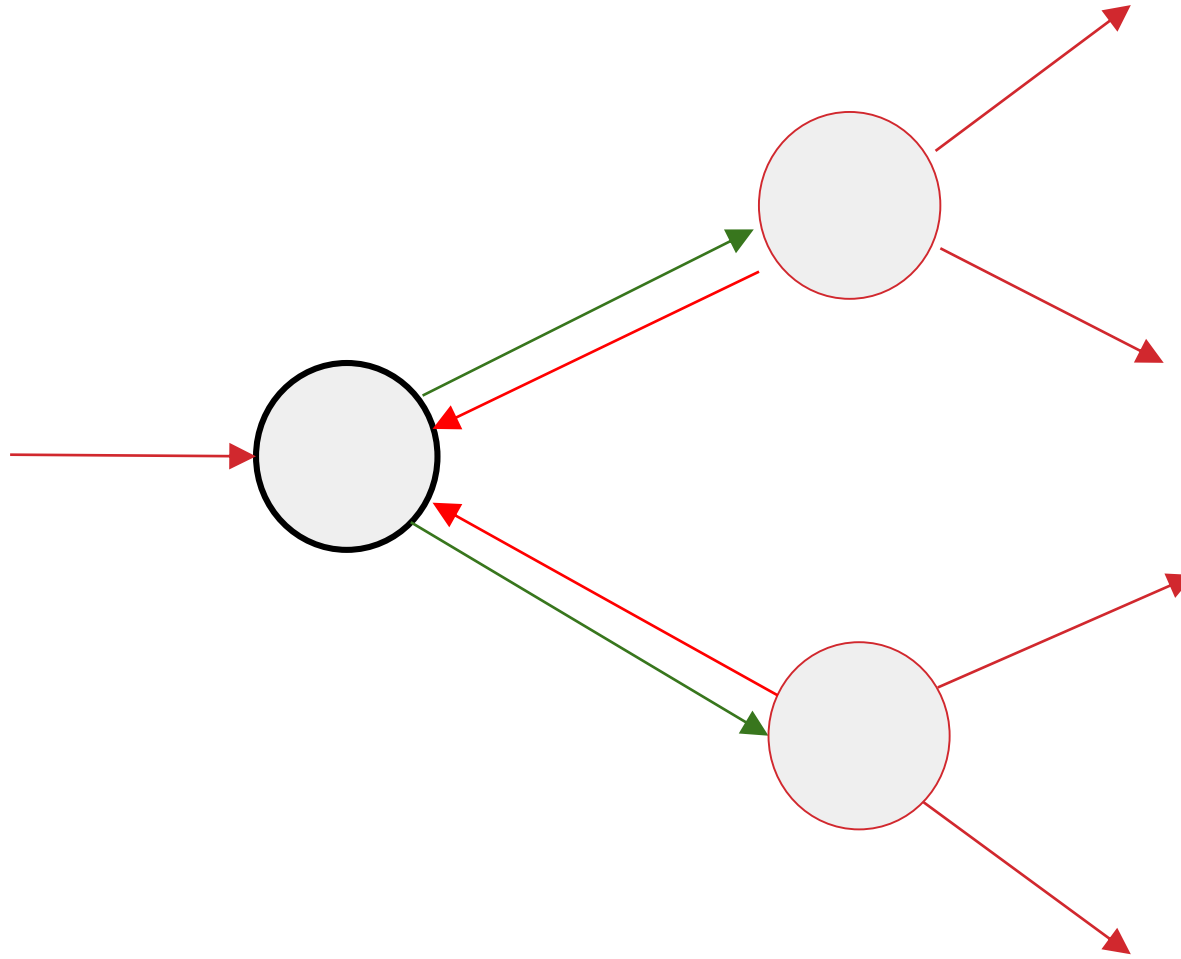
Adição : distribuidor de gradiente

Max : direcionador de gradiente

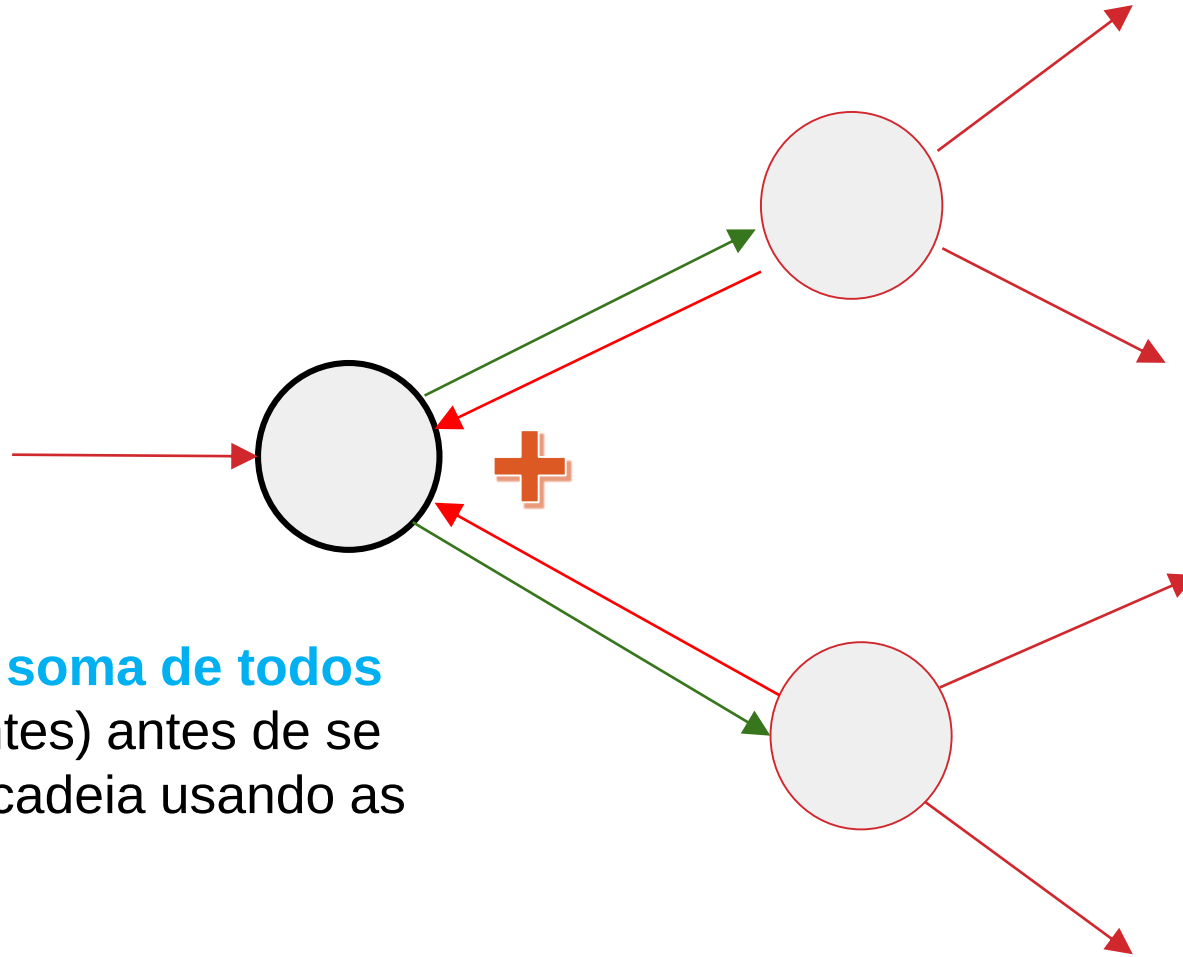
Produto: “comutador” de gradiente



Gradientes nas Ramificações

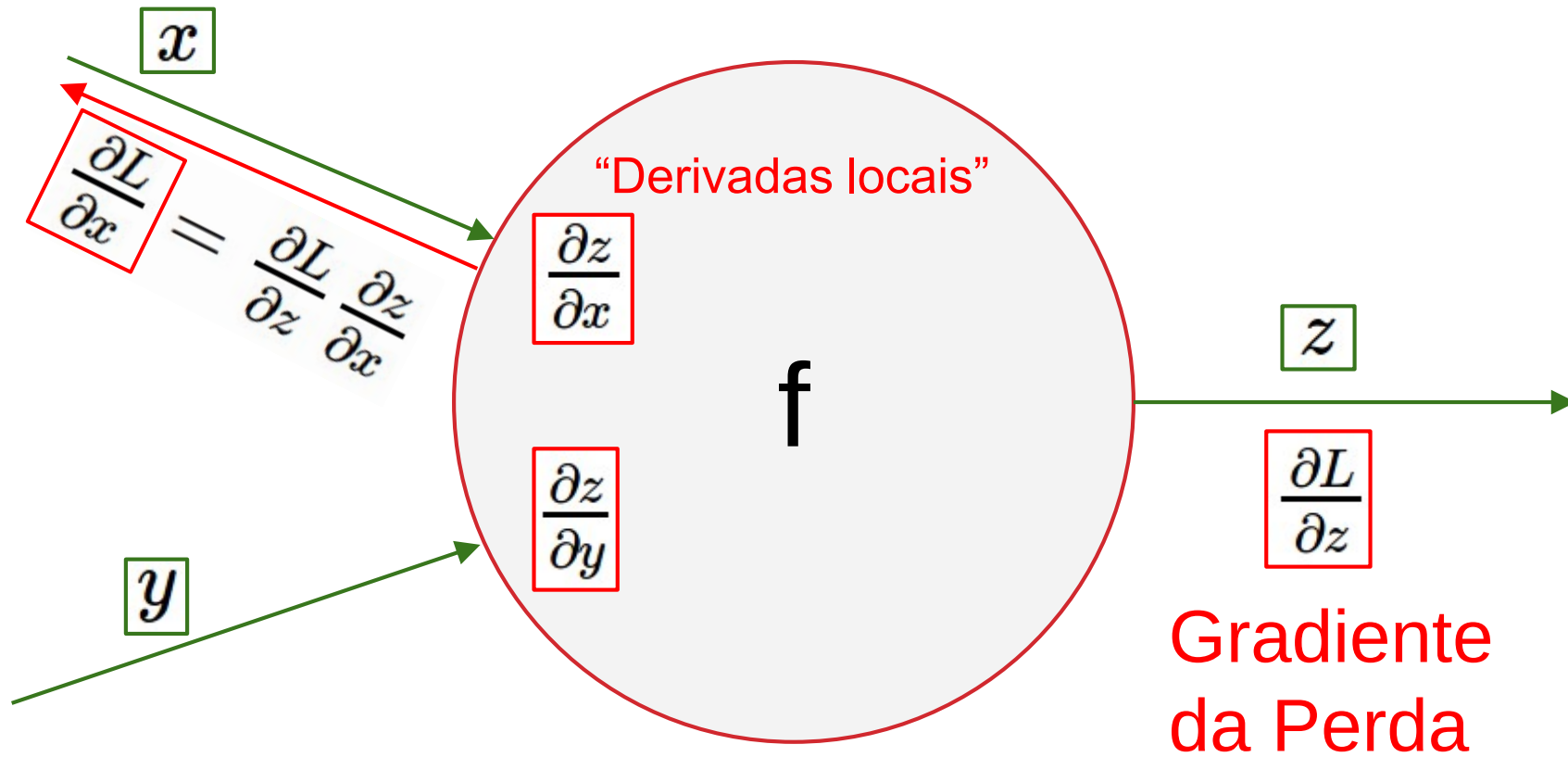


Gradientes nas Ramificações

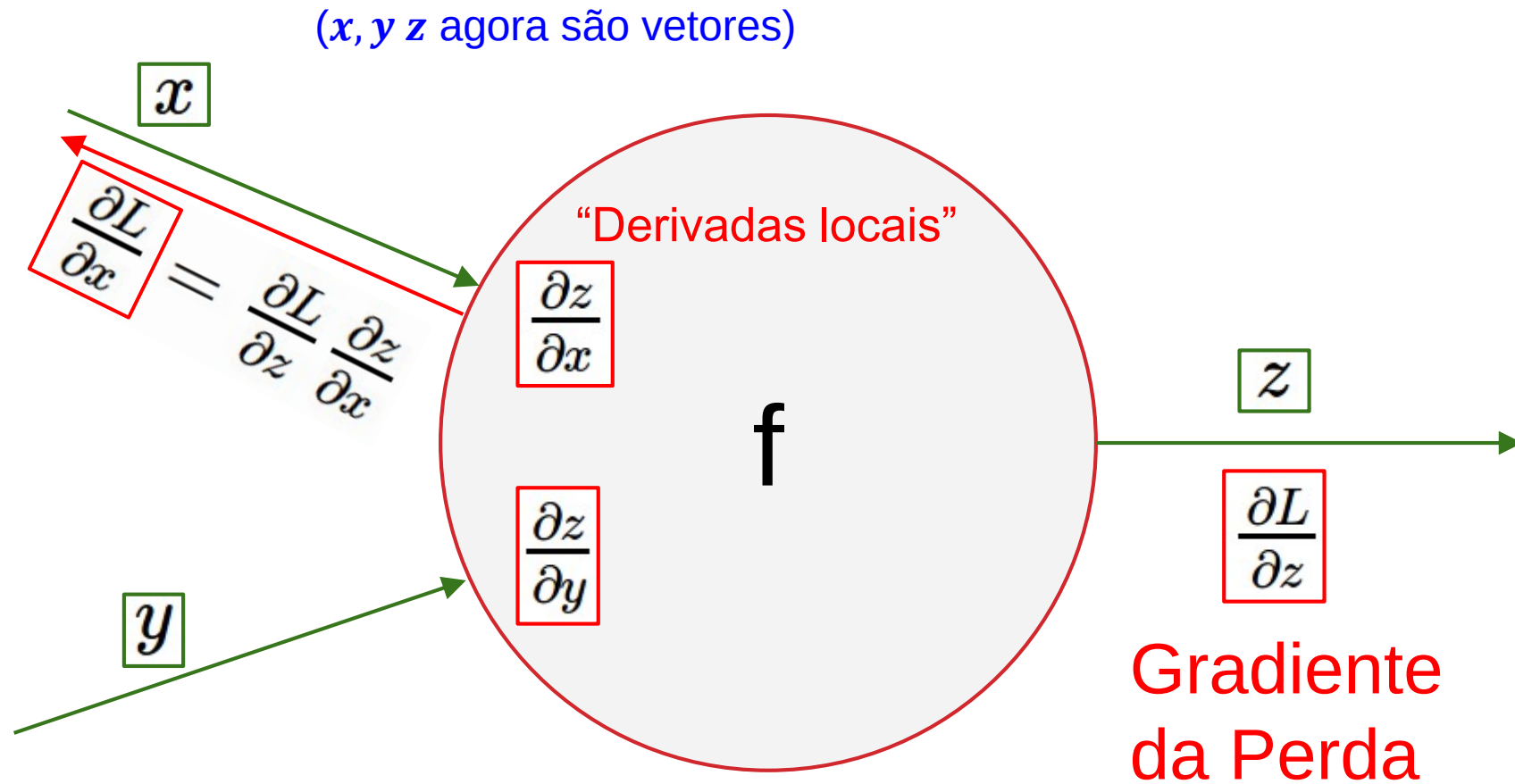


Deve-se realizar a **soma de todos os fluxos** (gradientes) antes de se aplicar a regra da cadeia usando as derivadas locais

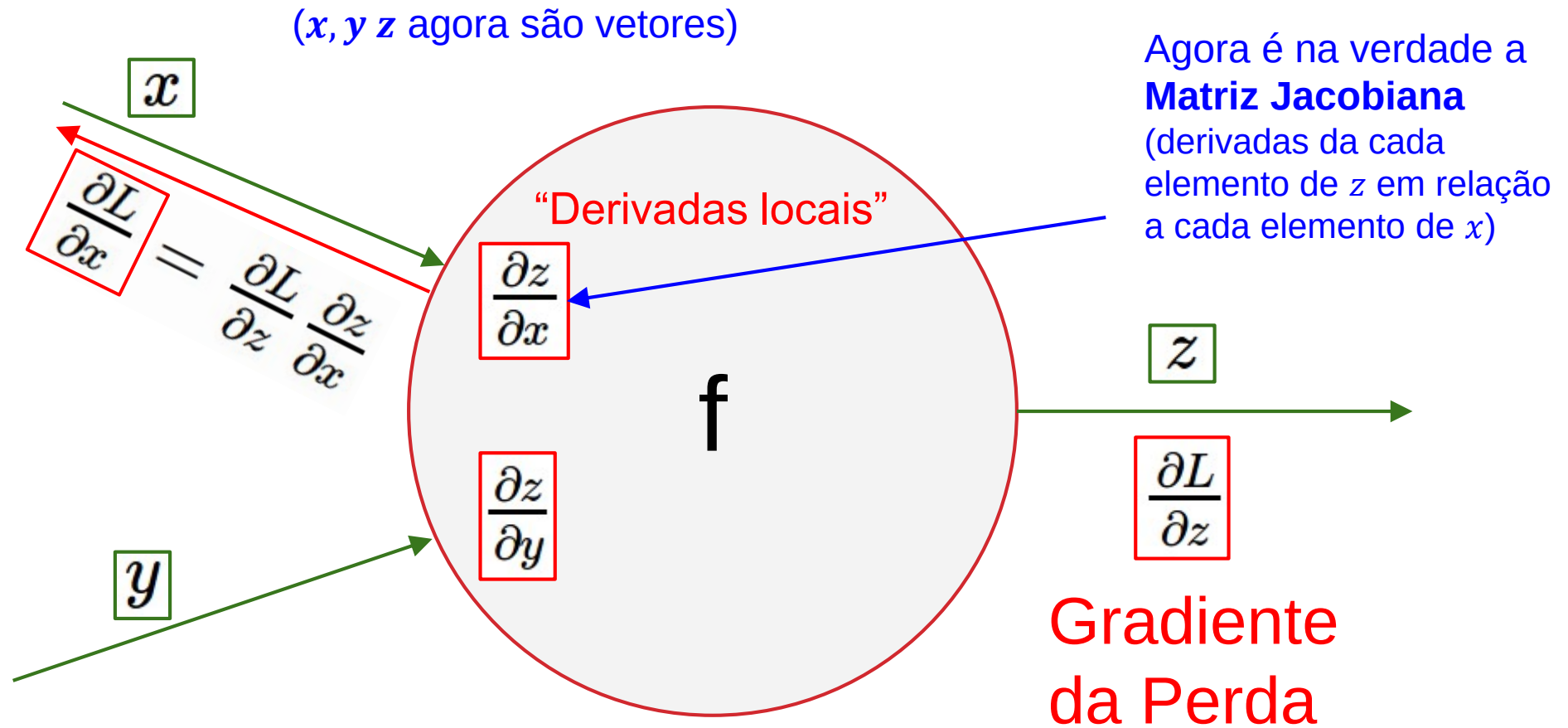
Gradientes para Dados Multidimensionais



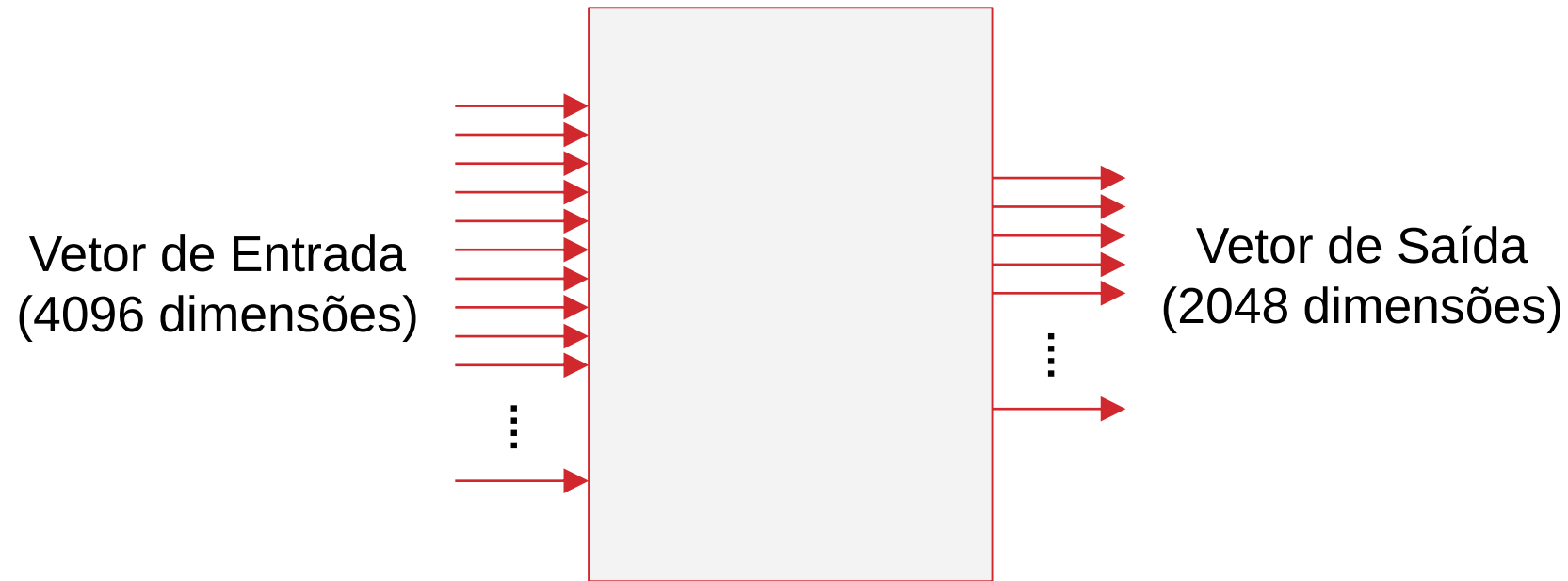
Gradientes para Dados Multidimensionais



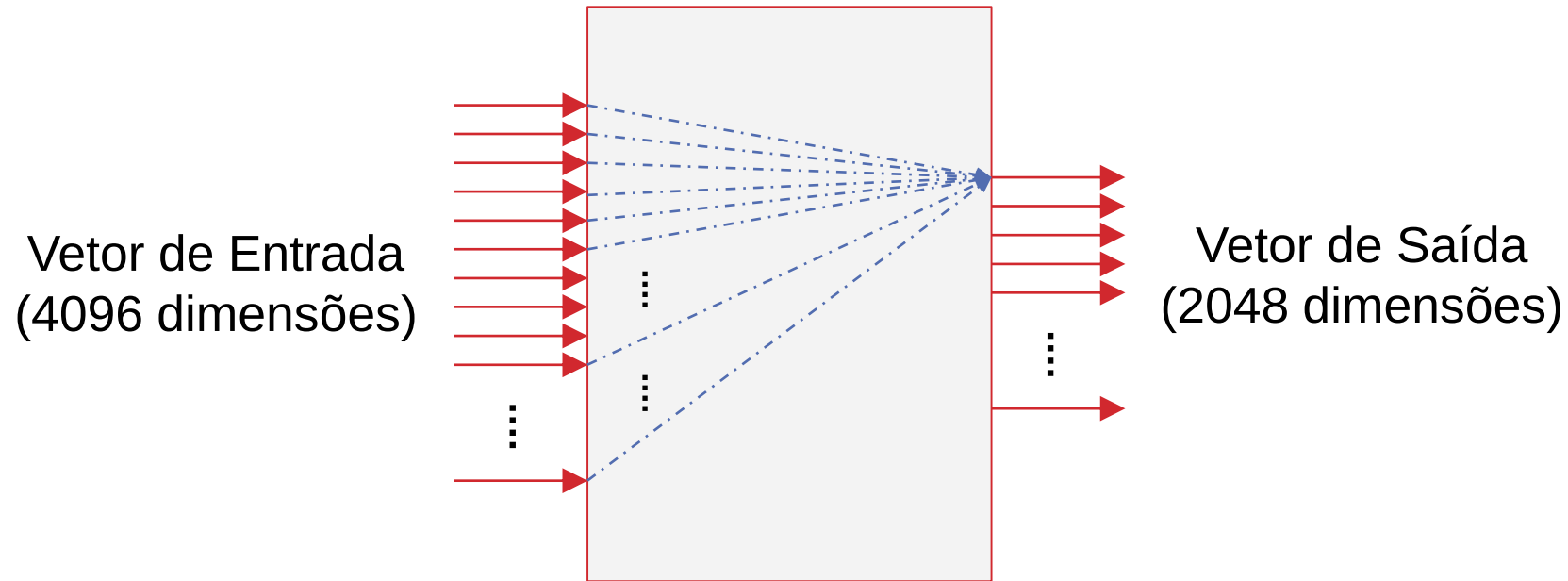
Gradientes para Dados Multidimensionais



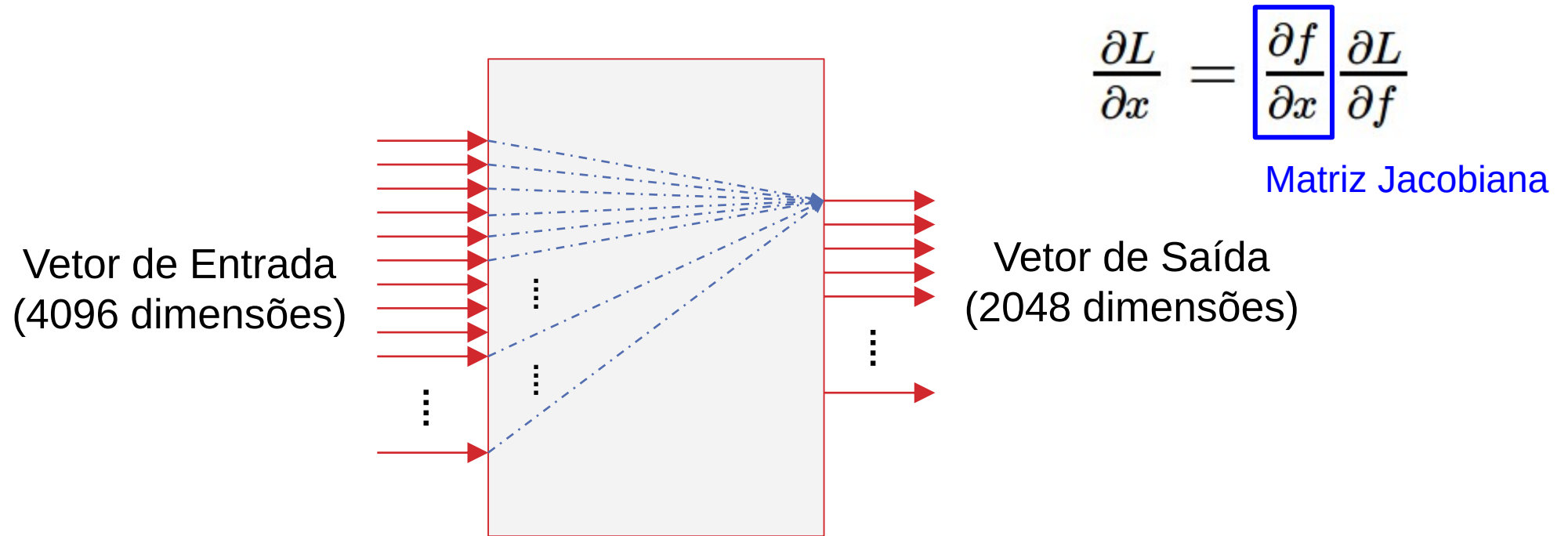
Gradientes para Dados Multidimensionais



Gradientes para Dados Multidimensionais



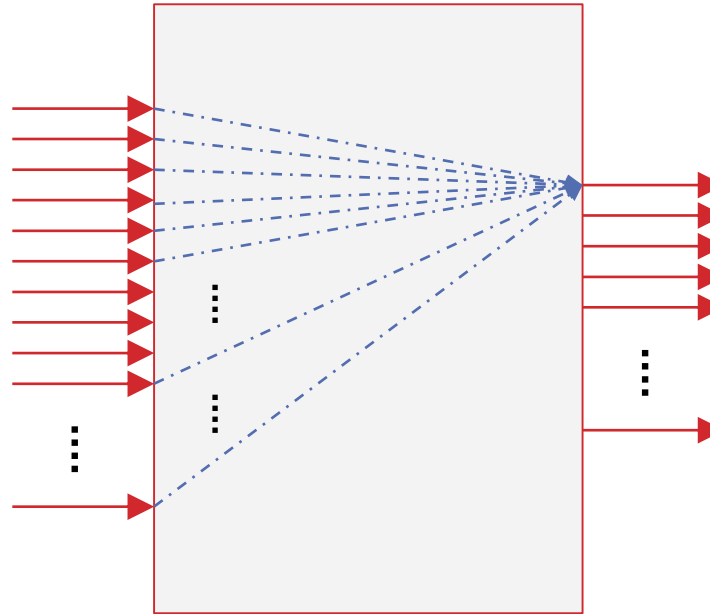
Gradientes para Dados Multidimensionais



Gradientes para Dados Multidimensionais

P1: qual o tamanho
da matriz Jacobiana?

Vetor de Entrada
(4096 dimensões)



$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x}} \frac{\partial L}{\partial f}$$

Matriz Jacobiana

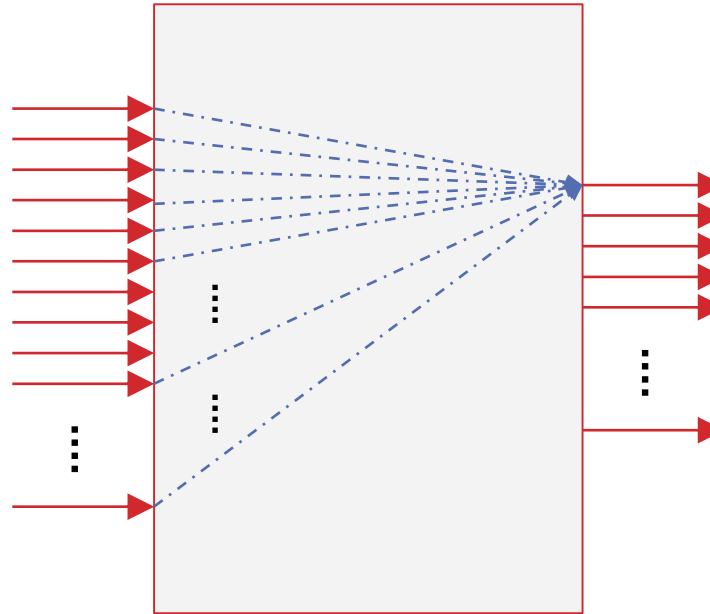
Vetor de Saída
(2048 dimensões)

Gradientes para Dados Multidimensionais

$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x}} \frac{\partial L}{\partial f}$$

Matriz Jacobiana

Vetor de Entrada
(4096 dimensões)



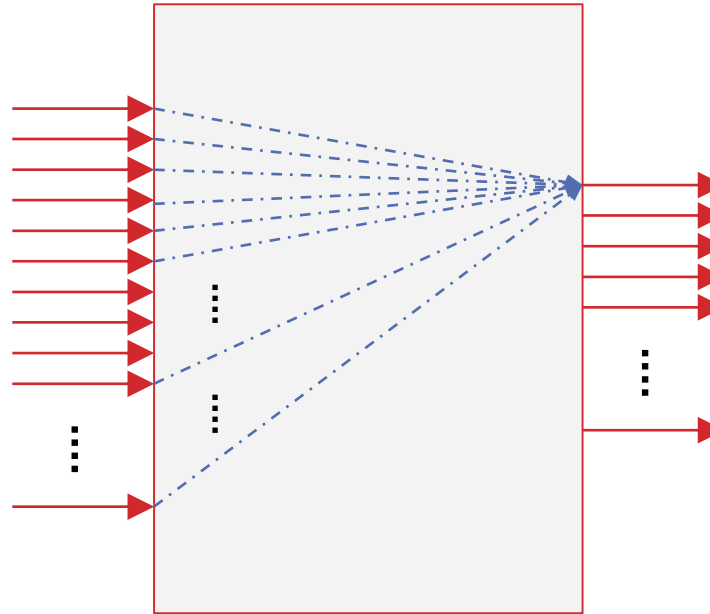
Vetor de Saída
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P1: qual o tamanho
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⇒ 2048 × 4096 !

Gradientes para Dados Multidimensionais

Vetor de Entrada
(4096 dimensões)



$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x}} \frac{\partial L}{\partial f}$$

Matriz Jacobiana

Vetor de Saída
(2048 dimensões)

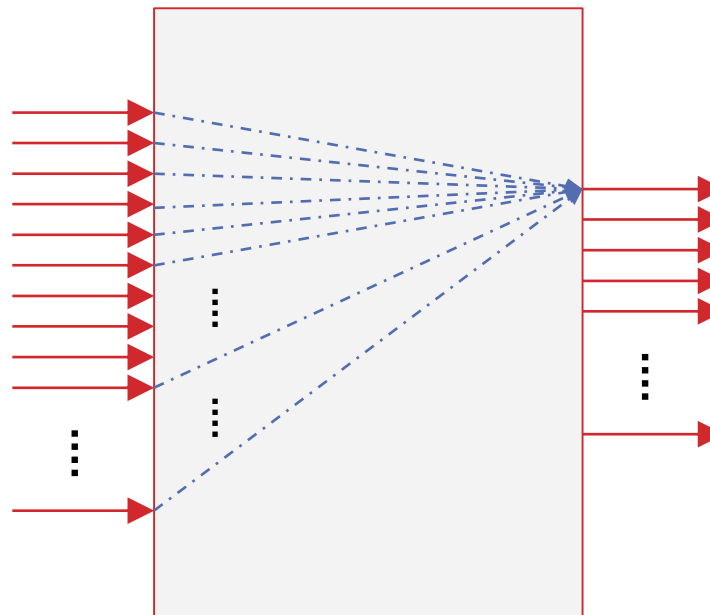
P1: qual o tamanho
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⇒ 2048 × 4096 !

P2: qual a aparência
dessa matriz?

Gradientes para Dados Multidimensionais

Vetor de Entrada
(4096 dimensões)



P1: qual o tamanho
da matriz Jacobiana?

⇒ 2048 × 4096 !

$$\frac{\partial L}{\partial x} = \boxed{\frac{\partial f}{\partial x}} \frac{\partial L}{\partial f}$$

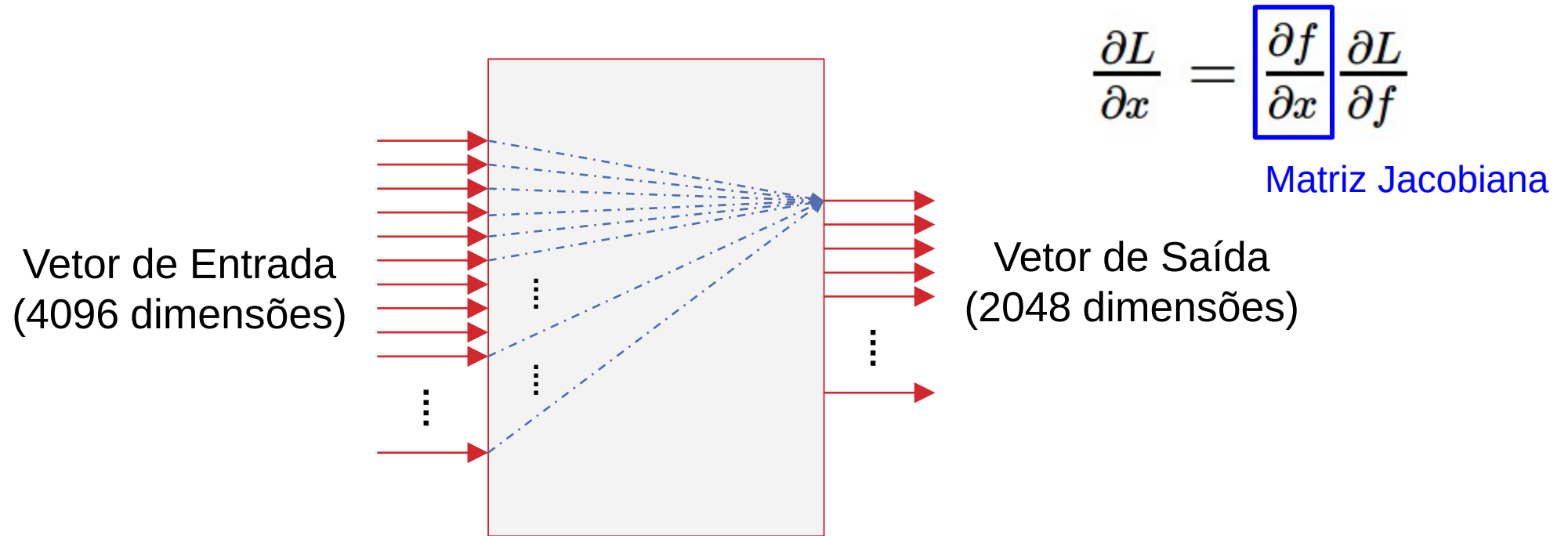
Matriz Jacobiana

Vetor de Saída
(2048 dimensões)

P2: qual a aparência
dessa matriz?

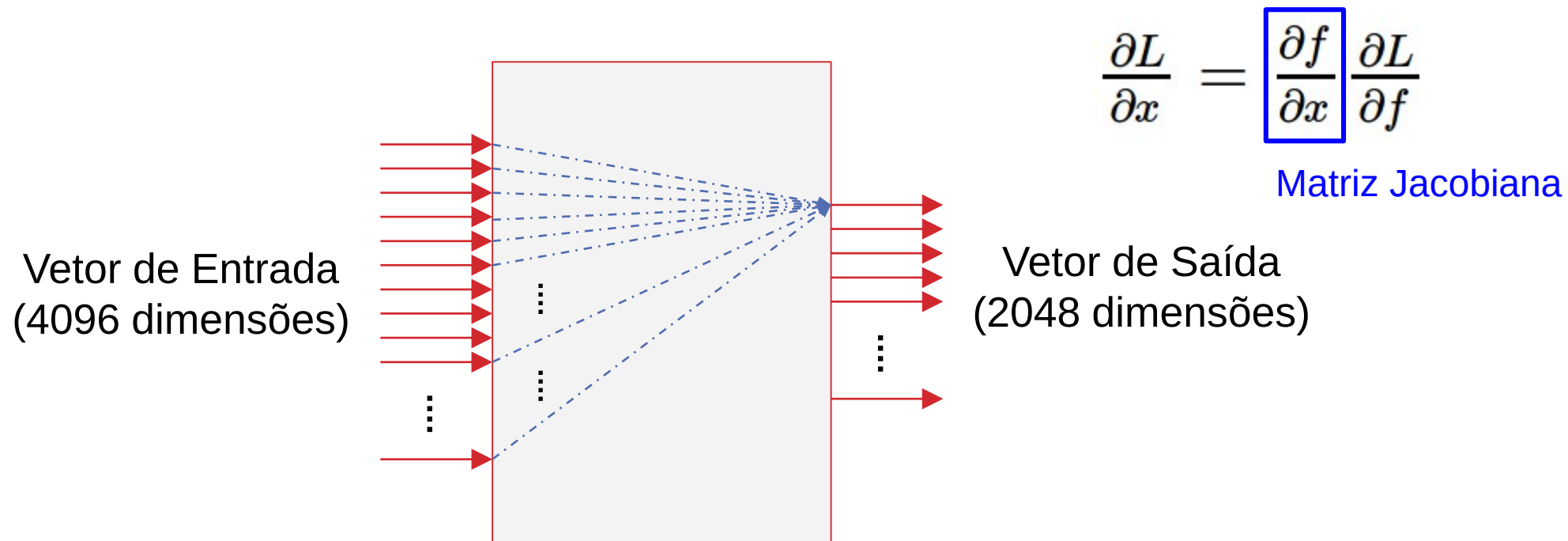
$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_m} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_k}{\partial x_1} & \frac{\partial f_k}{\partial x_2} & \dots & \frac{\partial f_k}{\partial x_m} \end{bmatrix}$$

Gradientes para Dados Multidimensionais



Na prática, processa-se
todo um lote ou “*minibatch*”
(p.ex. 100 amostras) de
uma só vez

Gradientes para Dados Multidimensionais



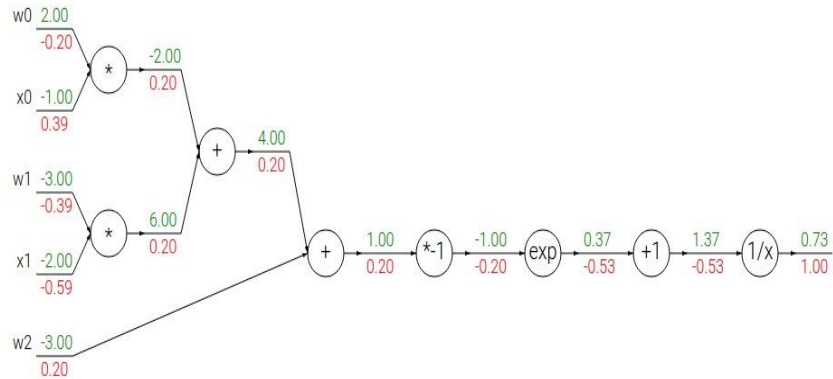
Na prática, processa-se todo um lote ou “*minibatch*” (p.ex. 100 amostras) de uma só vez

Assim, as dimensões da matriz Jacobiana desse exemplo seriam

$$2.048 \times 4.096 \times 100$$

≈ 839 milhões de pesos :(
(Note: The original image has a closing parenthesis at the end of this line)

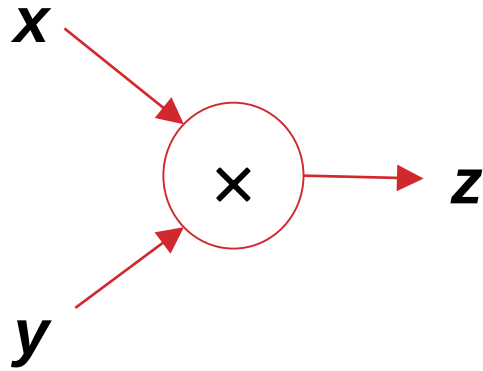
Implementação – Forward / Backward API



Para um grafo ou rede (*pseudocódigo*)

```
class ComputationalGraph(object):  
    #...  
    def forward(inputs):  
        # 1. [pass inputs to input gates...]  
        # 2. forward the computational graph:  
        for gate in self.graph.nodes_topologically_sorted():  
            gate.forward()  
        return loss # the final gate in the graph outputs the loss  
    def backward():  
        for gate in reversed(self.graph.nodes_topologically_sorted()):  
            gate.backward() # little piece of backprop (chain rule applied)  
        return inputs_gradients
```

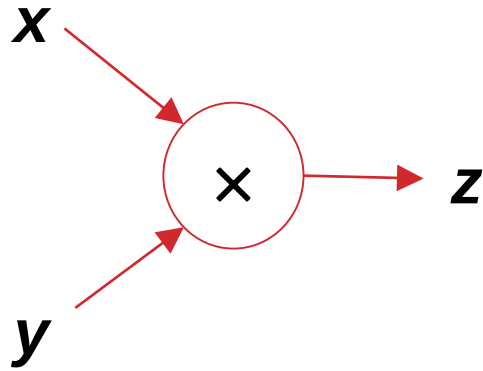
Implementação – Forward / Backward API



(x, y, z escalares)

```
class MultiplyGate(object):  
    def forward(x,y):  
        z = x*y  
        return z  
    def backward(dz):  
        # dx = ... #todo  
        # dy = ... #todo  
        return [dx, dy]
```

Implementação – Forward / Backward API

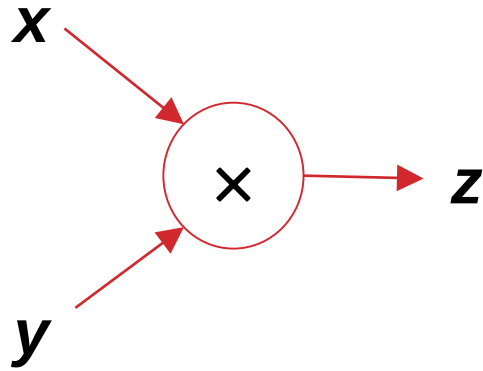


(x, y, z escalares)

```
class MultiplyGate(object):  
    def forward(x,y):  
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        # dx = ... #todo  
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```



Implementação – Forward / Backward API



(x, y, z escalares)

```
class MultiplyGate(object):
```

```
    def forward(x,y):
```

```
        z = x*y
```

```
        return z
```

```
    def backward(dz):
```

```
        # dx = ... #todo
```

```
        # dy = ... #todo
```

```
        return [dx, dy]
```

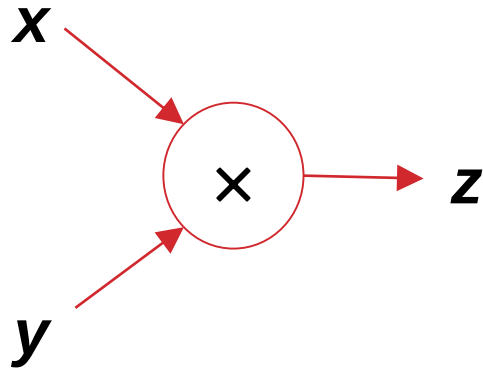
$$\frac{\partial L}{\partial z}$$

An arrow points from this box to the `dz` parameter in the `backward` method definition.

$$\frac{\partial L}{\partial x}$$

An arrow points from this box to the `dx` element in the `return` statement of the `backward` method.

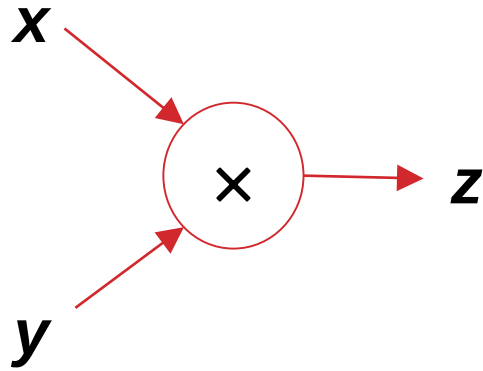
Implementação – Forward / Backward API



(x , y , z escalares)

```
class MultiplyGate(object):  
    def forward(x,y):  
        z = x*y  
        self.x = x # must keep these around!  
        self.y = y  
        return z  
    def backward(dz):  
        dx = self.y * dz # [dz/dx * dL/dz]  
        dy = self.x * dz # [dz/dy * dL/dz]  
        return [dx, dy]
```

Implementação – Forward / Backward API



(x , y , z escalares)

```
class MultiplyGate(object):  
    def forward(x,y):  
        z = x*y  
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        dx = self.y * dz # [dz/dx * dL/dz]  
        dy = self.x * dz # [dz/dy * dL/dz]  
        return [dx, dy]
```



Biblioteca de Componentes / Camadas

torch / nn		Watch		Issues	Pull requests	Wiki	Pulse	Graphs
No description or website provided								
1,839 commits		377 branches		79 releases		88 contributors		
Search master · View pull requests		New file		Find file	HTTPS	https://github.com/torch		Download ZIP
🔍 search Merge pull request #955 from torchresearch/master · 10h						Latest commit 2385176 15 hours ago		
🔍 doc	Fix batch mode in MergeRankingCriterion					4 days ago		
🔍 generic	Improve error message in SpatialConvolutionMM					4 days ago		
🔍 IR	THNN: add missing OpenMP include					2 days ago		
🔍 rocks	Add 'torch' dependency					14 days ago		
🔍 glogfire	let glog to ignore build output					4 months ago		
🔍 hachyderm	Fix TorchView test suite to the top level					4 years ago		
🔍 travis.yml	small fixes for test path					2 months ago		
🔍 Alex.lua	Add THNN conversion of (JLU, LeakyReLU, LogSigmoid, LogSoftmax, Looku...					7 days ago		
🔍 AlexCriterion.lua	Add THNN conversion of (JLU, LeakyReLU, LogSigmoid, LogSoftmax, Looku...					7 days ago		
🔍 Add.lua	fix Add with multi-dim bias					10 months ago		
🔍 AddConstant.lua	Adding in-place AddConstant and MulConstant					5 months ago		
🔍 BatchCriterion.lua	Remove unnecessary modules from BatchCriterion					3 months ago		
🔍 BatchNormalization.lua	fix batchnorm reset					3 months ago		
🔍 BatchTable.lua	fixing table modules to return correct number of gradients					6 months ago		
🔍 CO-Table.lua	fixing table modules to return correct number of gradients					6 months ago		
🔍 CMakeLists.txt	Add C implementation of SpatialBatchNormalization					7 days ago		
🔍 CModule.lua	in-Module preserve type sharing semantics (P187) add in-Module apply					4 months ago		
🔍 CModuleTable.lua	fixing table modules to return correct number of gradients					6 months ago		
🔍 COPYRIGHTING.txt	add copyright file					3 months ago		
🔍 CUPYRIGHT.txt	add copyright file					2 years ago		
🔍 CudaTable.lua	fixing table modules to return correct number of gradients					6 months ago		
🔍 Clamp.lua	Use custom range in torch.nn and use it as Clamp					3 months ago		
🔍 ClassNLLCriterion.lua	Add functional conversion of ClassNLLCriterion					13 days ago		
🔍 Concat.lua	fix a bug in conditional expression					4 months ago		
🔍 ConcatTable.lua	fixing bug in ConcatTable variable length					4 months ago		
🔍 Container.lua	Adding applyTableModules() to Container, which is like apply(), but ...					3 months ago		
🔍 Copy.lua	in-Module preserve type sharing semantics (P187) add in-Module apply					4 months ago		
🔍 Coarse.lua	Fix types() in Coarse					4 months ago		
🔍 CoarseDistance.lua	Do not change state variables in CoarseDistanceCoarseEmbeddingCriterion					2 months ago		
🔍 CoarseEmbeddingCriterion.lua	Do not change state variables in CoarseDistanceCoarseEmbeddingCriterion					2 months ago		
🔍 Criterion.lua	in-Module preserve type sharing semantics (P187) add in-Module apply					4 months ago		
🔍 CriterionTable.lua	Rename unpack to table.unpack for Lua 5.2					6 months ago		
🔍 CrossEntropyCriterion.lua	Check for 'in-Module' and 'in-Criterion' in recursiveType					6 months ago		
🔍 DeepConv.lua	adding direct backward to Conv2d, DepthConv2d, Sequential					6 months ago		
🔍 DistL2Criterion.lua	Use tensor for THNN functions even for single element outputs					10 days ago		
🔍 DotProduct.lua	Add batch mode in DotProduct + unit test					2 months ago		
🔍 Dropout.lua	in-place dropout					4 months ago		
🔍 ELU.lua	Add THNN conversion of (JLU, LeakyReLU, LogSigmoid, LogSoftmax, Looku...					7 days ago		
🔍 ErrorMessage.lua	Give better error messages when trying to use the wrong kind of Tensor					4 years ago		
🔍 Evolve.lua	in-Module preserve type sharing semantics (P187) add in-Module apply					4 months ago		
🔍 Exp.lua	Exp made lua only					6 months ago		
🔍 FastTable.lua	in-Module preserve type sharing semantics (P187) add in-Module apply					4 months ago		
🔍 FastRecurrent.lua	Add GradientRecurrent layer					4 months ago		
🔍 FastThink.lua	Add functional conversion of FastThink					10 days ago		
🔍 FastThink.lua	Add functional conversion of FastThink					10 days ago		
🔍 HingeEmbeddingCriterion.lua	rewrite HingeEmbeddingCriterion to support batch mode					6 months ago		
🔍 Identity.lua	Revert to previous Identity.lua implementation					2 months ago		
🔍 Index.lua	Simplifying and more efficient in-Index					2 months ago		
🔍 Jacobian.lua	Add unit tests for jacobian.lua, fix bugs detected by the tests					6 months ago		
🔍 JoinTable.lua	in-Module preserve type sharing semantics (P187) add in-Module apply					4 months ago		
🔍 L1Cost.lua	Use tensor for THNN functions even for single element outputs					10 days ago		
🔍 L1HingeEmbeddingCriteria...	Make types() truly recursive					6 months ago		
🔍 L1Penalty.lua	fixed L1Penalty constructor arguments					4 years ago		
🔍 LeakyReLU.lua	Add THNN conversion of (JLU, LeakyReLU, LogSigmoid, LogSoftmax, Looku...					7 days ago		
🔍 Linear.lua	Remove spurious matrices from in-Linear					4 months ago		

🔍 LogSigmoid.lua	Add THNN conversion of (JLU, LeakyReLU, LogSigmoid, LogSoftmax, Looku...	7 days ago
🔍 LogSoftmax.lua	Add THNN conversion of (JLU, LeakyReLU, LogSigmoid, LogSoftmax, Looku...	7 days ago
🔍 LookupTable.lua	Harmonize LookupTable signature with torch.nn	5 days ago
🔍 MM.lua	Rename unpack to table.unpack for Lua 5.2	6 months ago
🔍 MSECriterion.lua	Add SoftAverage to criterion in the constructor	2 months ago
🔍 MergeCriterion.lua	modernized MergeCriterion	4 years ago
🔍 MergeRankingCriterion.lua	Fix batch mode in MergeRankingCriterion	4 days ago
🔍 Mix.lua	Merge pull request #944 from vighnesh	2 months ago
🔍 Mix.lua	Add support for negative dimension and both batch and non-batch input...	2 months ago
🔍 Mix.lua	Merge pull request #944 from vighnesh	2 months ago
🔍 ModuleTable.lua	cannot uncast variable and updates expression	26 days ago
🔍 Module.lua	Reset "Can't re-define parameters if they are already defined"	15 hours ago
🔍 Mul.lua	removing the requirement for providing ops in nn.Module	4 years ago
🔍 MulConstant.lua	Ignore updateOutInput if self.gradInput is nil	3 months ago
🔍 MulCriterion.lua	asserts in MulCriterion and ParallelCriterion add	2 months ago
🔍 MulTableBatchingCriterion.lua	initial rewrap of torch7 tree	4 years ago
🔍 MulTableBatchingCriterion.lua	multiversion supports grad	11 months ago
🔍 Narrow.lua	typo in Narrow not done in place	6 months ago
🔍 Norm.lua	NormTable	6 months ago
🔍 NormTable.lua	Remove inner and backward from NormTable, because they allocate memory ...	26 days ago
🔍 PReLU.lua	Buffer for PReLU code implementation	6 months ago
🔍 Padding.lua	fixed bug in Padding: input was returned in backward	5 months ago
🔍 ParameterCriterion.lua	Merge pull request #932 from vighnesh	28 days ago
🔍 Parallel.lua	fix a bug in conditional expression	4 months ago
🔍 ParallelCriterion.lua	asserts in ParallelCriterion and ParallelCriterion add	2 months ago
🔍 ParallelTable.lua	Parallel optimization: ParallelTable inherits Container: unit tests	4 years ago
🔍 Power.lua	Use UNO line endings	7 months ago
🔍 REAME.md	docs restructured	5 months ago
🔍 ReLU.lua	Add randomized leaky modified linear unit (ReLU)	3 months ago
🔍 ReLU.lua	adds in-place ReLU and fixes a potential divide-by-zero in nn.Soft	6 months ago
🔍 Replicate.lua	Replicate batch mode	6 months ago
🔍 Reshape.lua	Add more informative pretty-printing	4 years ago
🔍 Select.lua	initial rewrap of torch7 tree	4 years ago
🔍 SelectTable.lua	in-Module preserve type sharing semantics (P187) add in-Module apply	4 months ago
🔍 Sequential.lua	Being Sequential removes corner cases	6 months ago
🔍 Sigmoid.lua	initial rewrap of torch7 tree	4 years ago
🔍 SmoothL1Criterion.lua	Add SoftAverage to criterion in the constructor	2 months ago
🔍 Soft.lua	Fix various unused variables in nn	4 years ago
🔍 SoftMax.lua	Fix various unused variables in nn	4 years ago
🔍 SoftPlus.lua	fixed a numerical issue in the SoftPlus module (it breaks for input g...	2 years ago
🔍 SoftShrink.lua	initial rewrap of torch7 tree	4 years ago
🔍 SoftSign.lua	initial rewrap of torch7 tree	4 years ago
🔍 SparseAnchor.lua	Fix various unused variables in nn	4 years ago
🔍 SparseLinear.lua	Using sparse implementation of paramGradParameters for SparseLinear	4 months ago
🔍 SpatialAdaptivePooling.lua	Added SpatialAdaptivePooling	4 years ago
🔍 SpatialAveragePooling.lua	SpatialAveragePooling supports padding, cell mode and exclude_pad de...	26 days ago
🔍 SpatialBatchNormalization.lua	Add C implementation of SpatialBatchNormalization	7 days ago
🔍 SpatialConvolutional.lua	Make types() truly recursive	6 months ago
🔍 SpatialConvolution.lua	Fix types() in SpatialConvolution	3 months ago
🔍 SpatialConvolutionMM.lua	Fix types() in SpatialConvolution	3 months ago
🔍 SpatialConvolutionMM.lua	Remove unused and expensive initialization logic from in.SpatialConv...	6 months ago
🔍 SpatialConvolutionMM.lua	code consistency	18 days ago
🔍 SpatialConvolutional.lua	SpatialConvolutional: Divisive Subtractive Normalization work with bat...	6 months ago
🔍 SpatialDropout.lua	small fix on error message	6 months ago
🔍 SpatialFractionalMaxPool...	Adding Fractional Max Pooling	3 months ago
🔍 SpatialFullConvolution.lua	Add adjustment term to SpatialFullConvolution to control the size of ...	5 days ago
🔍 SpatialFullConvolution.lua	New NN classes	3 years ago
🔍 SpatialFPooling.lua	SpatialAveragePooling divides by NN^H	10 months ago
🔍 SpatialMapPooling.lua	SpatialMapPooling supports padding and cell mode	6 months ago
🔍 SpatialMapPooling.lua	Add SpatialMapUnpooling	26 days ago
🔍 SpatialSoftmax.lua	Update Softmax to work in spatial mode	4 months ago
🔍 SpatialSubSampling.lua	Merge branch 'nn_test_fixes'	3 years ago
🔍 SpatialSubtractiveNormaliz...	SpatialConvolutional: Divisive Subtractive Normalization work with bat...	6 months ago
🔍 SpatialSubSampling.lua	Use UNO line endings	7 months ago
🔍 SpatialZeroPadding.lua	Add more informative pretty-printing	4 years ago
🔍 SplitTable.lua	Add support for negative indices in nn.SplitTable	7 months ago

Biblioteca de Componentes / Camadas

[illegible][illegible]

Biblioteca de Componentes / Camadas

[illegible]

AddNorm.lua	Add THNN converson of Lelu, LuayLuo, LuigiPong, LuigiStella, Luks...	7 days ago
LuigiStella.lua	Add THNN converson of Lelu, LuayLuo, LuigiPong, LuigiStella, Luks...	7 days ago
LookupTable.lua	Hemizon LookupTable signature with sum input	5 days ago
MM.lua	Rename output to table lookup for Lua 5.2	6 months ago
MSECriterion.lua	Add SaeZawaga to criterion in the constructor	2 months ago
MarginCriterion.lua	modernized MarginCriterion	a year ago
MergeRankingCriterion.lua	Fix batch mode in MergeRankingCriterion	4 days ago
Min.lua	Merge pool request HARA from gisharener	2 months ago
Mean.lua	Add support for negative dimension and both odd and non-odd inputs	2 months ago
Min.lua	Merge pool request HARA from gisharener	2 months ago
MinMaxTable.lua	cannot uncast variable and useless assignment	6 months ago
Module.lua	Revert "Don't re-factor patterns if they are already factored"	15 hours ago
Mul.lua	removing the requirement for providing size on Mul	a year ago
MulConstant.lua	Ignore updates because I feel gradcheck is nil	2 months ago
MuxCriterion.lua	asserts in MuxCriterion and PatchCriterion add	2 months ago
MLLibUtilMergeCriterion.lua	initial saving of torch7 tree	4 years ago
MLMergeCriterion.lua	multimerge supports p=2	11 months ago
Name.lua	typo in Nameve not done in place.	6 months ago
NormalizeTable.lua	NameTable	6 months ago
Normaliza.lua	Remove item and badlines from Normalize, because they allocate memory ...	20 days ago
PRNUC.lua	Buffer for PRNUC code implementation.	6 months ago
Padding.lua	Fixed broken in Padding: input was returned in backward	6 months ago
PairwiseDistance.lua	Merge pool request H22 from wagnermarter	28 days ago
Permute.lua	Is a bug in conditional expression	a month ago
PerseuCriterion.lua	asserts in MuxCriterion and PatchCriterion add	2 months ago
PerseuTable.lua	Parallel optimization: ParallelTable inherits Context, unit tests	2 months ago
Power.lua	Use UNIX file endings	7 months ago
README.md	doc modification	6 months ago
PRNUC.lua	Add embedded huck modified base code (PRNUC)	1 months ago
ReLU.lua	adds n-by-k ReLU and fixes a potential divide-by-zero in Set	3 months ago
Replicate.lua	Replicate batchSize	6 months ago
Reshape.lua	Add new informative pretty printing	a year ago
Select.lua	Initial saving of torch7 tree	4 years ago
SequentialTable.lua	In Module preserve type sharing semantics (H27), add In Module apply	6 months ago
Sequential.lua	Using Sequential remove corner case	6 months ago
Signoid.lua	Initial saving of torch7 tree	4 years ago
SmoothL1Criterion.lua	Add SaeZawaga to criterion in the constructor	2 months ago
SoftMax.lua	Fix various unused variables in m	a year ago
SoftMin.lua	Fix various unused variables in m	a year ago
SoftPlus.lua	fixed a numerical issue in the SoftPlus module (it breaks for large inputs)	2 years ago
SoftSigmoid.lua	Initial saving of torch7 tree	4 years ago
SoftSig.lua	Initial saving of torch7 tree	4 years ago
SparseJacobian.lua	Fix various unused variables in m	a year ago
SparseLinear.lua	Using sparse implementation of zeroOnParameters for SparseLinear	a month ago
SpatialAveragePooling.lua	Add SpatialAveragePooling	a year ago
SpatialWarpingPooling.lua	SpatialWarpingPooling supports padding, cell mode and exclude pad cells	a year ago
SpatialBatchNormalization.lua	Add C Implementation of SpatialBatchNormalization	7 days ago
SpatialContrastiveNormalized.lua	Make typoe truly recursive.	6 months ago
SpatialConvolution.lua	Fix typo in SpatialConvolution	1 months ago
SpatialConvolutionMM.lua	Fix typo in SpatialConvolution	1 months ago
SpatialConvolutionMap.lua	Remove unused and expensive information from m in SpatialConvo...	6 months ago
SpatialCrossMapPool.lua	code consistency	18 days ago
SpatialConvolutionMax.lua	SpatialContrastive Distance.SubtractionNormalization work with bat...	6 months ago
SpatialDropout.lua	small fix in error message	6 months ago
SpatialFractionalMaxPooling.lua	Adding Fractional Max Pooling	6 months ago
SpatialFullConvolution.lua	Add adjustment term in SpatialFullConvolution to control the size of ...	5 days ago
SpatialFullConvolutionMap.lua	New class	3 years ago
SpatialHardwiringPooling.lua	SpatialWarpingPooling differs by NIMRA	16 months ago
SpatialMaxPooling.lua	SpatialMaxPooling supports padding and cell mode	6 months ago
SpatialMaxUpooling.lua	Add SpatialMaxUpooling	26 days ago
SpatialSoftMax.lua	Update SoftMax to work in spatial mode	6 months ago
SpatialSoftmaxSampling.lua	Merge branch 'in_dev_wmf'	3 years ago
SpatialSubtractiveNormalized.lua	SpatialContrastive Distance.SubtractionNormalization work with bat...	6 months ago
SpatialTernarySampling.lua	Use UNIX file endings	7 months ago
SpatialTopkPooling.lua	Add more informative pretty-printing	6 months ago
Soft.lua	Add support for negative indices in m.SoftTable	7 months ago

